

MAHMOUD TALAAT

Architecture | BIM | Working Drawings | Automation

Alexandria, Egypt

mahmoud.talaat605@gmail.com · +20 115 674 1997 · linkedin.com/in/mahmoud-talaat605

SECTIONS

- 01

Working Drawings

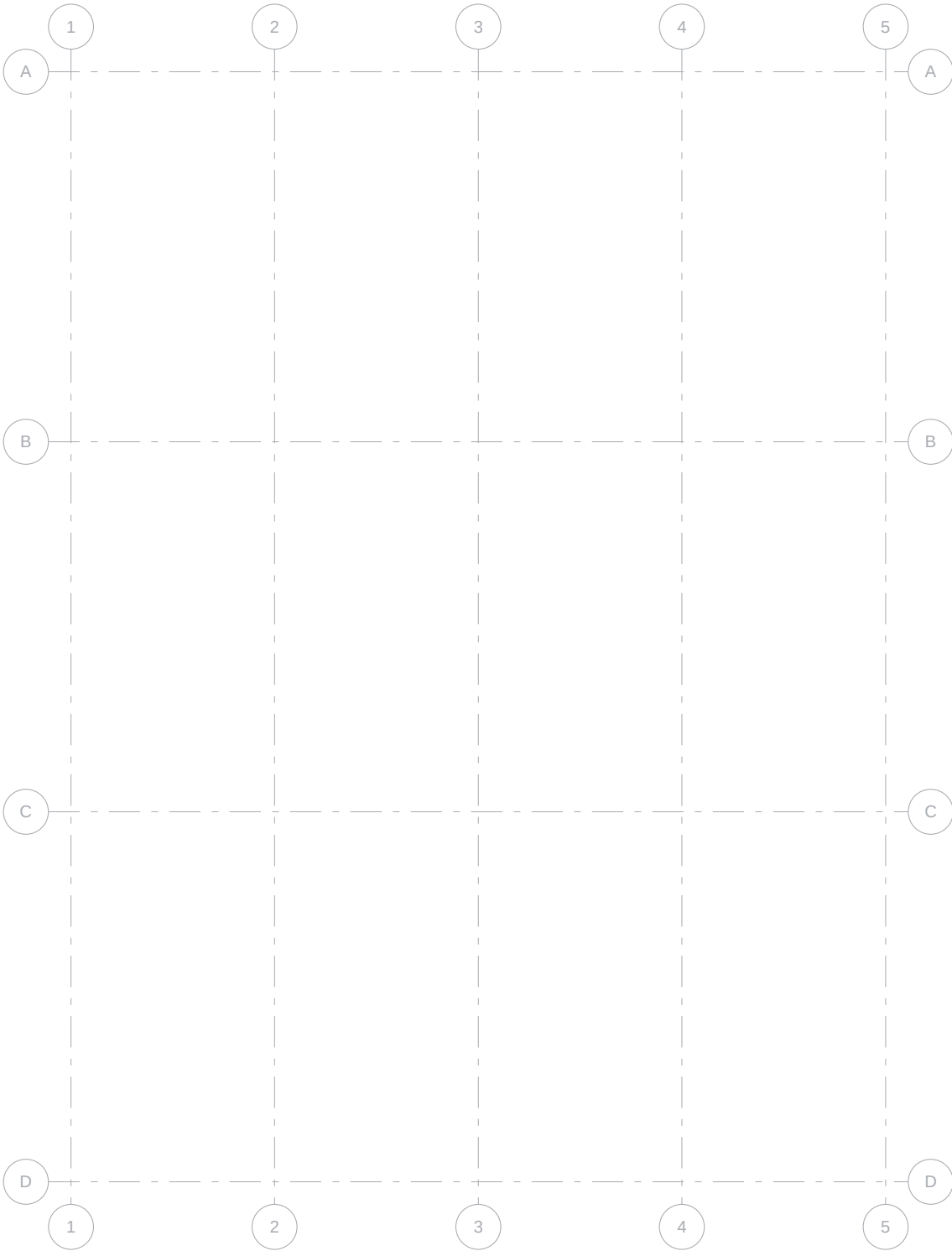
Residential Villa · Multipurpose Hall · Car Showroom
- 02

BIM Automation & Families

Dynamo scripts · Revit plugin · Parametric family
- 03

Architectural Design

Five design projects · Graduation project



PROFILE

ARCHITECTURE, BIM & AUTOMATION

BIM Architect | Revit | Dynamo | BIM Automation

Architecture graduate and ITI graduate with hands-on experience in Revit-based BIM modeling and architectural documentation, including parametric families, working drawings, and multidisciplinary project documentation. Developed Dynamo workflows and a C# Revit API plugin to automate practical BIM tasks, with a programming background that supports BIM automation and workflow development.

3

WORKING-DRAWING
PROJECTS

35

DRAWING
SHEETS

4

TOOLS & PARAMETRIC
FAMILIES

6

DESIGN PROJECTS
INCL. GRADUATION PROJECT

CONTACT

EMAIL mahmoud.talaat605@gmail.com

PHONE +20 115 674 1997

LINKEDIN linkedin.com/in/mahmoud-talaat605

AT A GLANCE

FOCUS	BIM Architect · BIM Automation
EDUCATION	B.Sc. Architecture, Alexandria University
GRADUATED	2024
TRAINING	ITI Code Camp, .NET & Generative AI (2025)
LANGUAGES	Arabic (native) · English (B2)
BASED IN	Alexandria, Egypt

BIM & ARCHITECTURE

Revit · Dynamo · BIM Modeling · Revit API

Architectural Documentation · Parametric Families

ARCHITECTURAL TOOLS

Rhino · Grasshopper · Enscape · Twinmotion

SOFTWARE ENGINEERING

C# · .NET / ASP.NET Core · Entity Framework Core

SQL Server · REST APIs · Git

CONTENTS

Working drawings, BIM automation and architectural design

SECTIONS	Working drawings · BIM automation & families · Architectural design
PROJECTS	3 working-drawing sets + 4 automation items + 6 design projects
SHEETS	35 drawing sheets + 8 automation pages + 21 design pages
SOFTWARE	Revit, Dynamo, Rhino, Grasshopper, Twinmotion, Enscape

01 WORKING DRAWINGS			4 – 42
W1	Residential Villa	10 sheets	4
W2	Multipurpose Hall	14 sheets	15
W3	Car Showroom	11 sheets	30
02 BIM AUTOMATION & FAMILIES			43 – 51
A1	Clash-Driven Wall Openings		44
A2	Face-Hosted Family Placement		46
A3	Beam Automation Pipeline		49
A4	Parametric Door Family		51
03 ARCHITECTURAL DESIGN			52 – 74
D1	The Sun That Sets in the South	Train station	53
D2	Audi Showroom	Car showroom	57
D3	Boutique Hotel of the Arabian Nights	Boutique hotel	60
D4	Siwa Discovery Center	Discovery center	63
D5	Alexandria Community Center	Community center	65
D6	Oblivion: The Sin of Time	Graduation project	68

W1

RESIDENTIAL VILLA

Architectural Working Drawings

TYPE	Residential villa
LEVELS	Ground + first floor
SITE	40.00 x 30.00 m plot with swimming pool
SOFTWARE	Autodesk Revit 2022
YEAR	2022

PLANS

A-001	Site Plan	5
A-101	Ground Floor Plan	6
A-102	First Floor Plan	7
A-111	Ground Floor Finishes	8
A-112	First Floor Finishes	9

SECTIONS

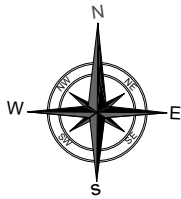
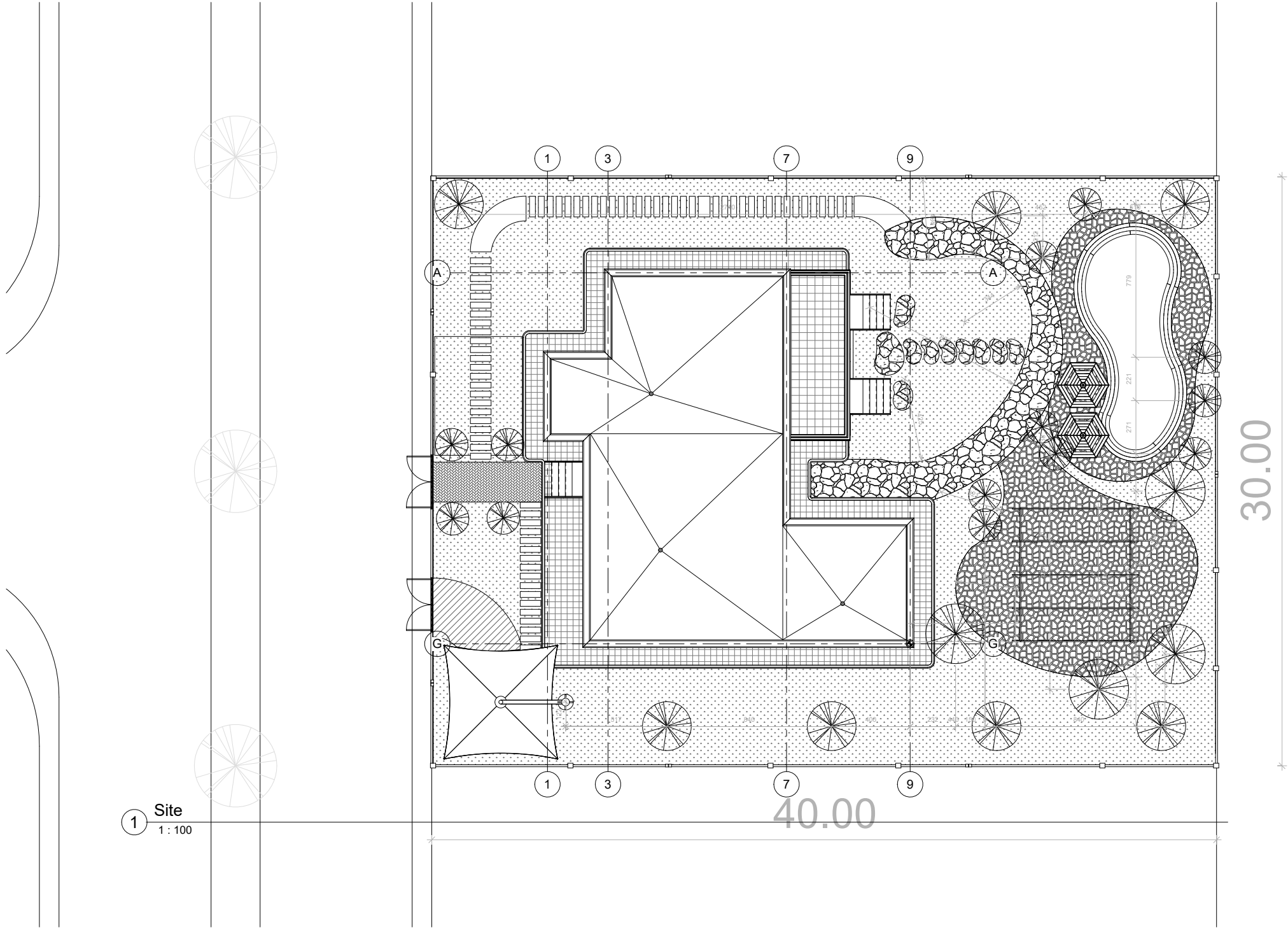
A-201	Section 1	10
A-202	Section 2	11

DETAILS

A-301	Door & Window Details	12
A-302	Swimming Pool	13

3D VIEWS

A-401	3D Views	14
-------	----------	----



Trees

notes	height	diameter	name	sympol
shading	3m	3m	japanese maple	
privacy	6m	5m	decidious golden chain	

Hardscape

Stone Dressed Fieldstone	
herbe-grass	
brick pavers	

Symbols

Description	Sympol
Project start point	

MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

RESIDENTIAL VILLA

Architectural Working Drawings

SHEET TITLE

SITE PLAN

Landscape, hardscape & planting

SCALE

1:100

DRAWN BY

M. Talaat

DATE

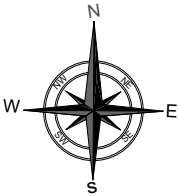
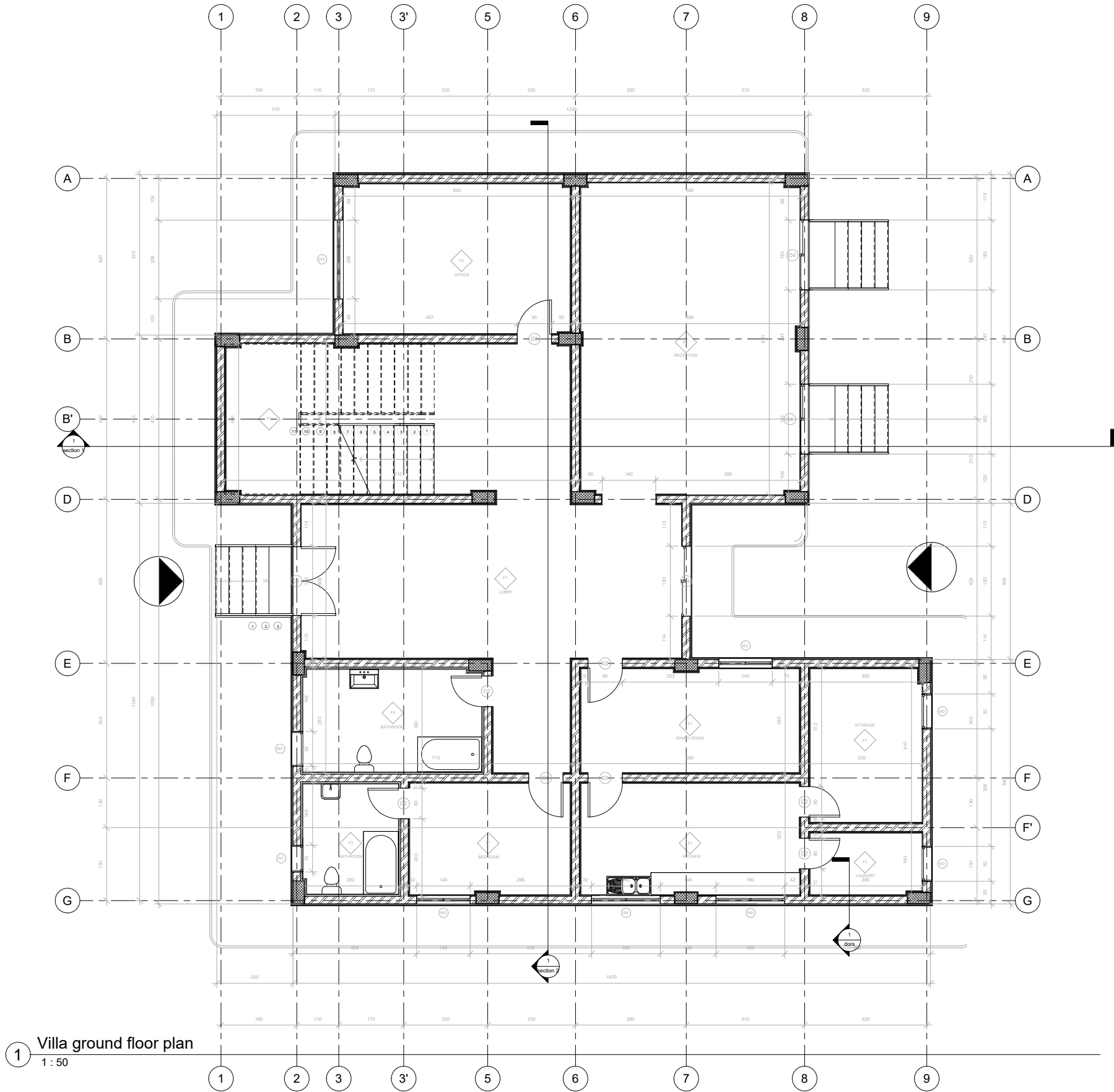
2022

DISCIPLINE

Architectural

SHEET NO.

A-001



FINISHES

wazara	ceilings	Walls	Floors	Room name	Mark
Ceramic 3x10cm			Ceramic 40x40 cm		F1
Parquet 10cm			Parquet		F2
Parquet 10cm			Ceramic 40x40cm		F3

Floors

Type	Description	Floors	Room name	Mark
		Ceramic 40x40 cm		F1
		Parquet		F2
		Ceramic 40x40cm		F3

Doors schedule

Type	Description	Dimensions		Number	mark
		height	width		
Flush Door	Entrance	220CM	180CM		D1
Flush Door	W.C	220CM	80CM		D2
Flush Door	Rooms	220CM	90CM		D3
Flush Door	Sliding Door	220CM	180CM		D4

Windows Schedule

Type	Description	Dimensions			Number	Mark
		Stool height	Height	Width		
		100cm	120cm	70cm		W1
		100cm	120cm	90cm		W2
		100cm	120cm	140cm		W3
		100cm	120cm	180cm		W4
		100cm	120cm	200cm		W5

MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

RESIDENTIAL VILLA

Architectural Working Drawings

SHEET TITLE

GROUND FLOOR PLAN

Door & window schedules

SCALE

1:50

DRAWN BY

M. Talaat

DATE

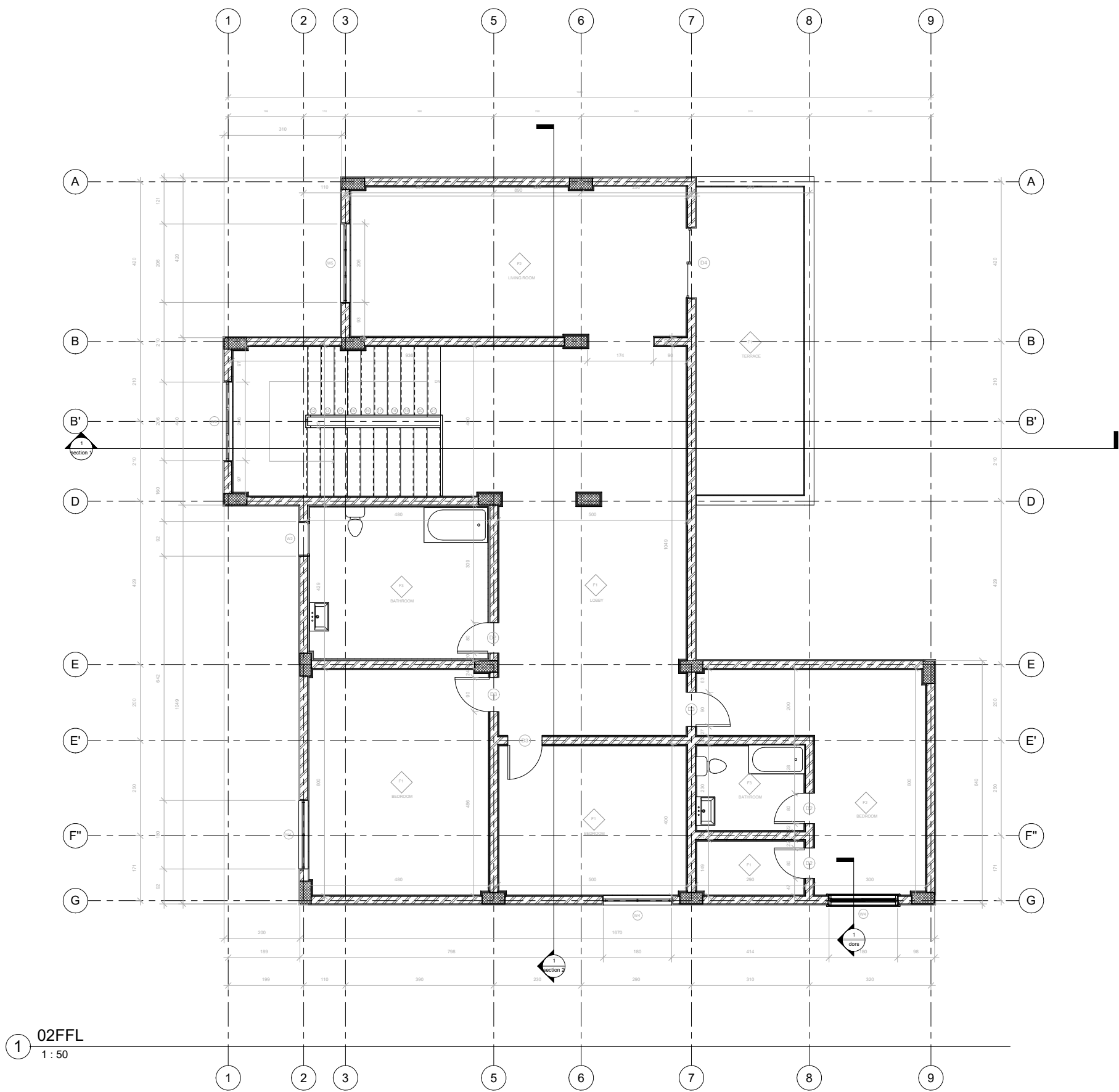
2022

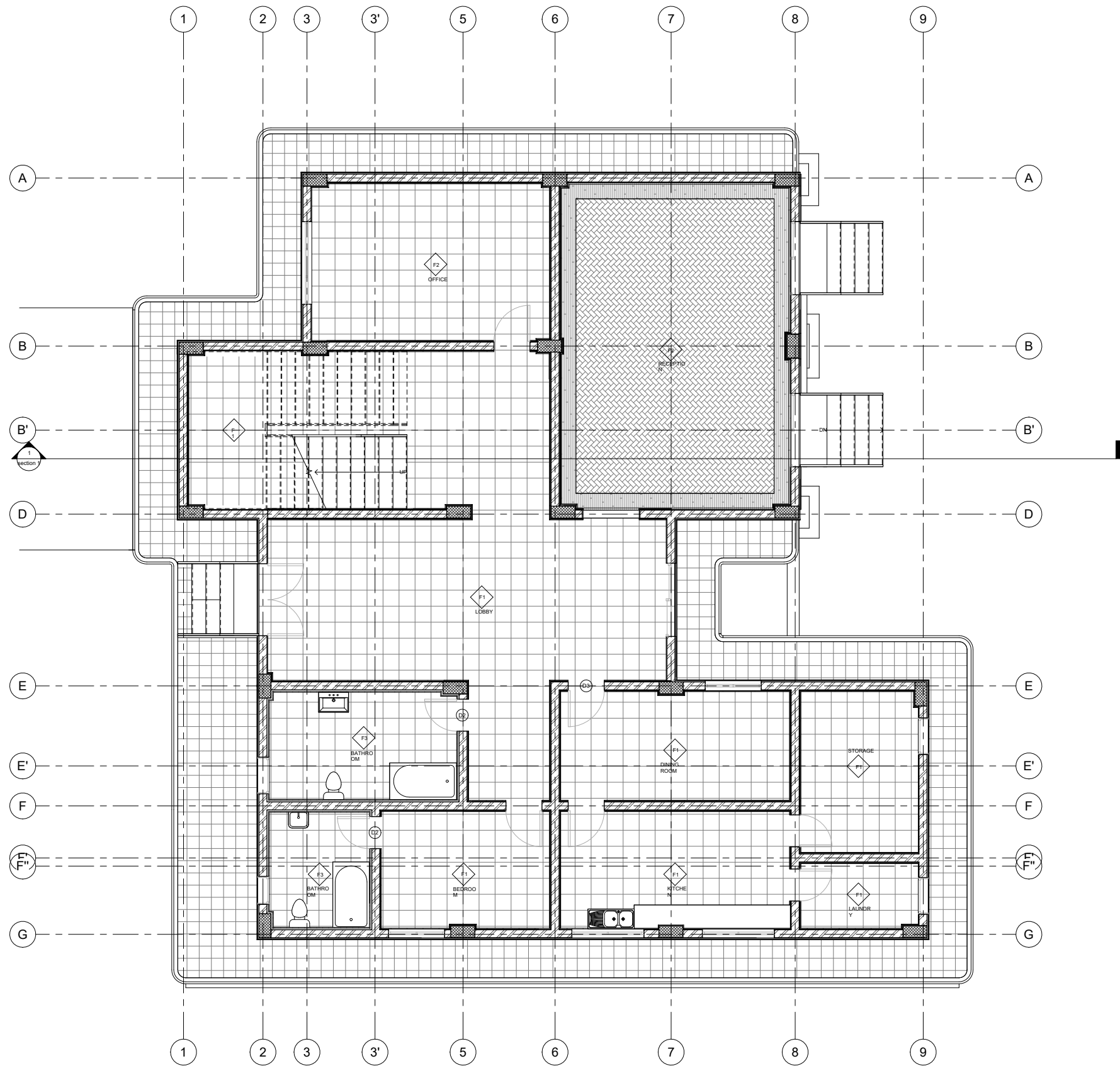
DISCIPLINE

Architectural

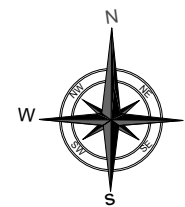
SHEET NO.

A-101

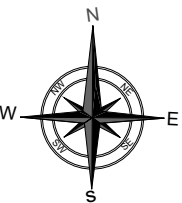
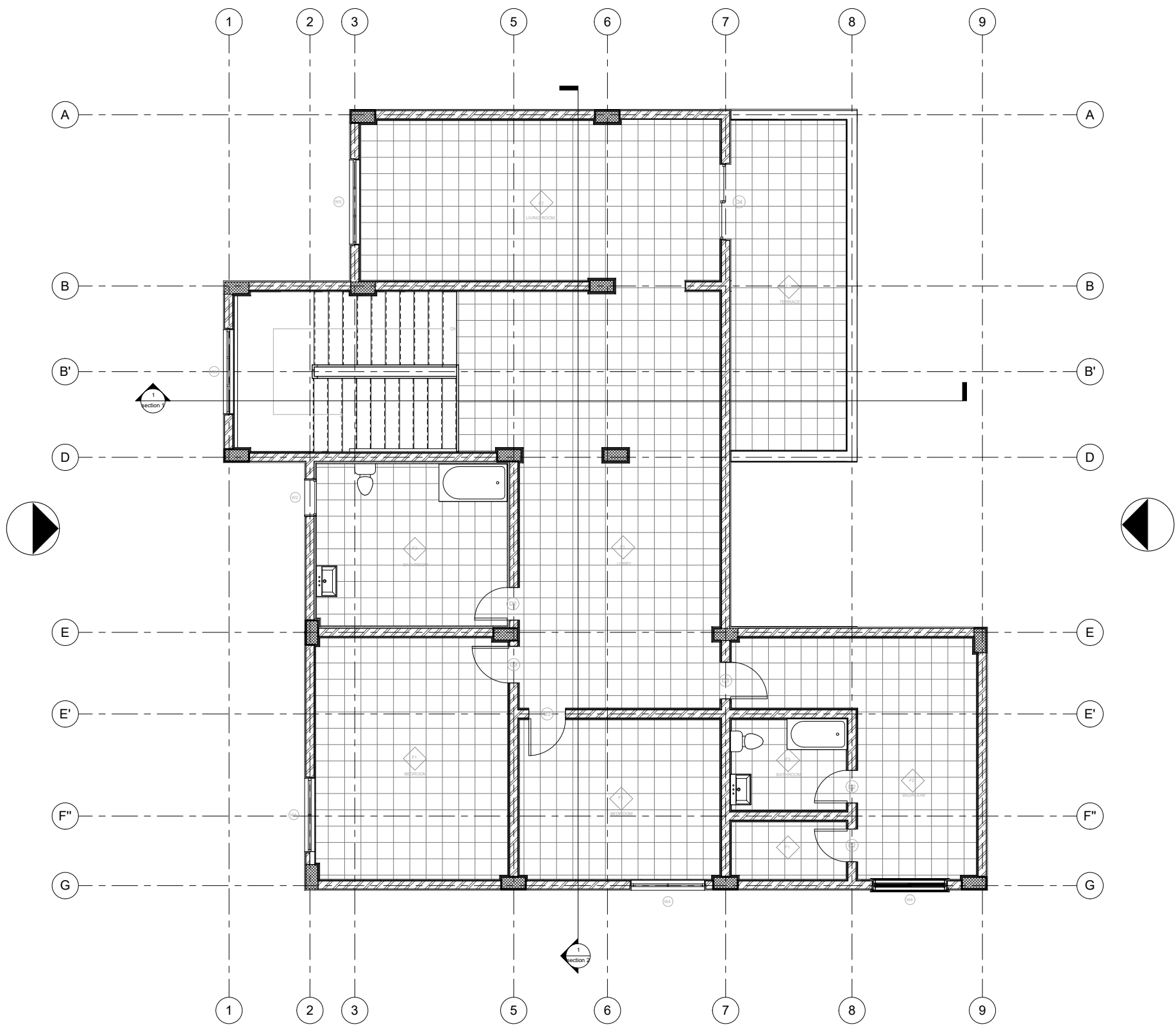




1 ground floor finishes
1 : 50



MAHMOUD TALAAT		
ARCHITECTURE BIM		
PROJECT		
RESIDENTIAL VILLA		
Architectural Working Drawings		
SHEET TITLE		
GROUND FLOOR FINISHES		
Floor finishes plan		
SCALE	DATE	SHEET NO.
1:50	2022	A-111
DRAWN BY	DISCIPLINE	
M. Talaat	Architectural	



1 first floor finishes
1 : 50

MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

RESIDENTIAL VILLA

Architectural Working Drawings

SHEET TITLE

FIRST FLOOR FINISHES

Floor finishes plan

SCALE

1:50

DATE

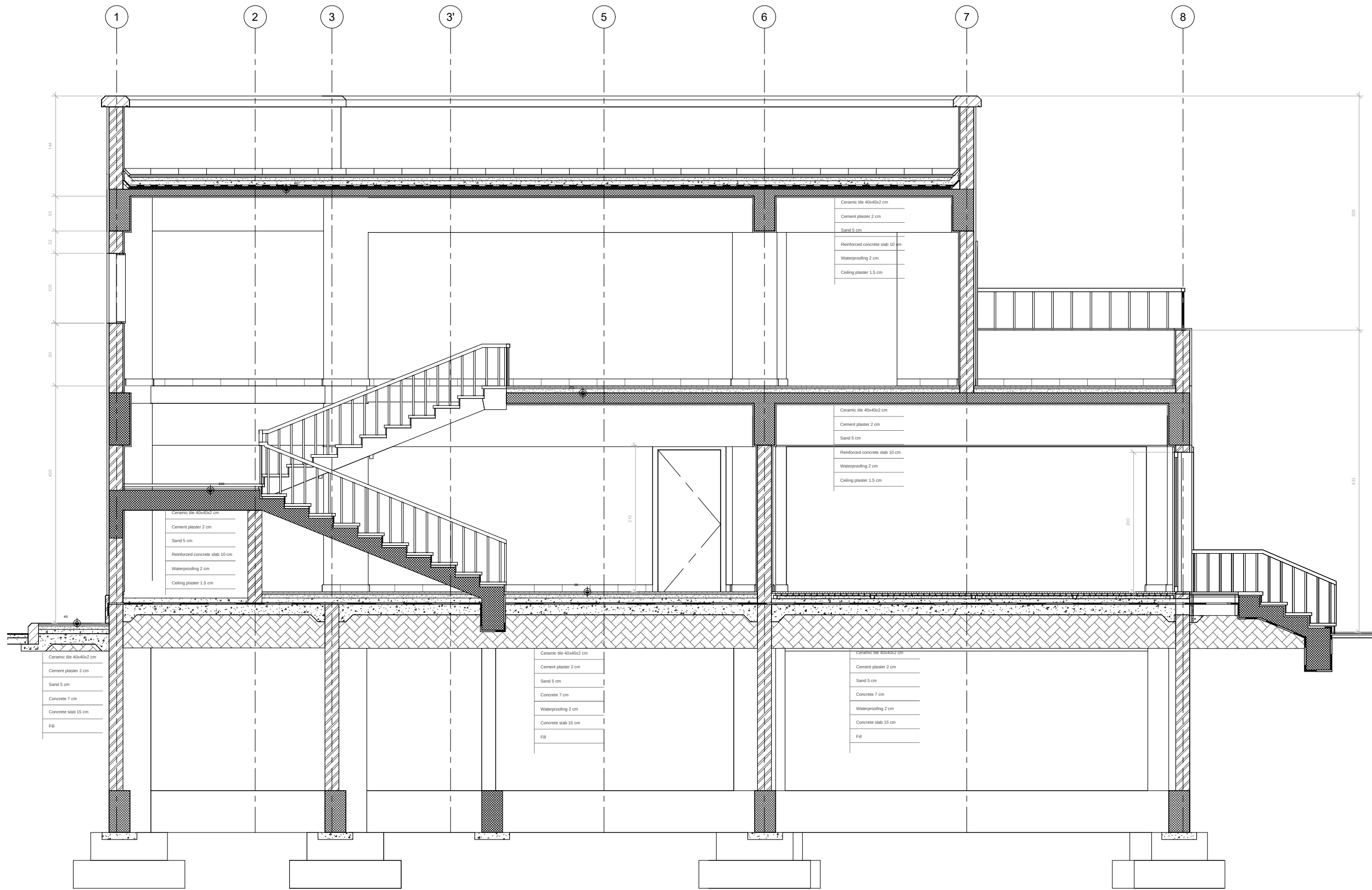
2022

DRAWN BY
M. Talaat

DISCIPLINE
Architectural

SHEET NO.

A-112



1 Section 1
1 : 25

MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

RESIDENTIAL VILLA

Architectural Working Drawings

SHEET TITLE

SECTION 1

Stair section, axes 1-8

SCALE

1:25

DATE

2022

DRAWN BY

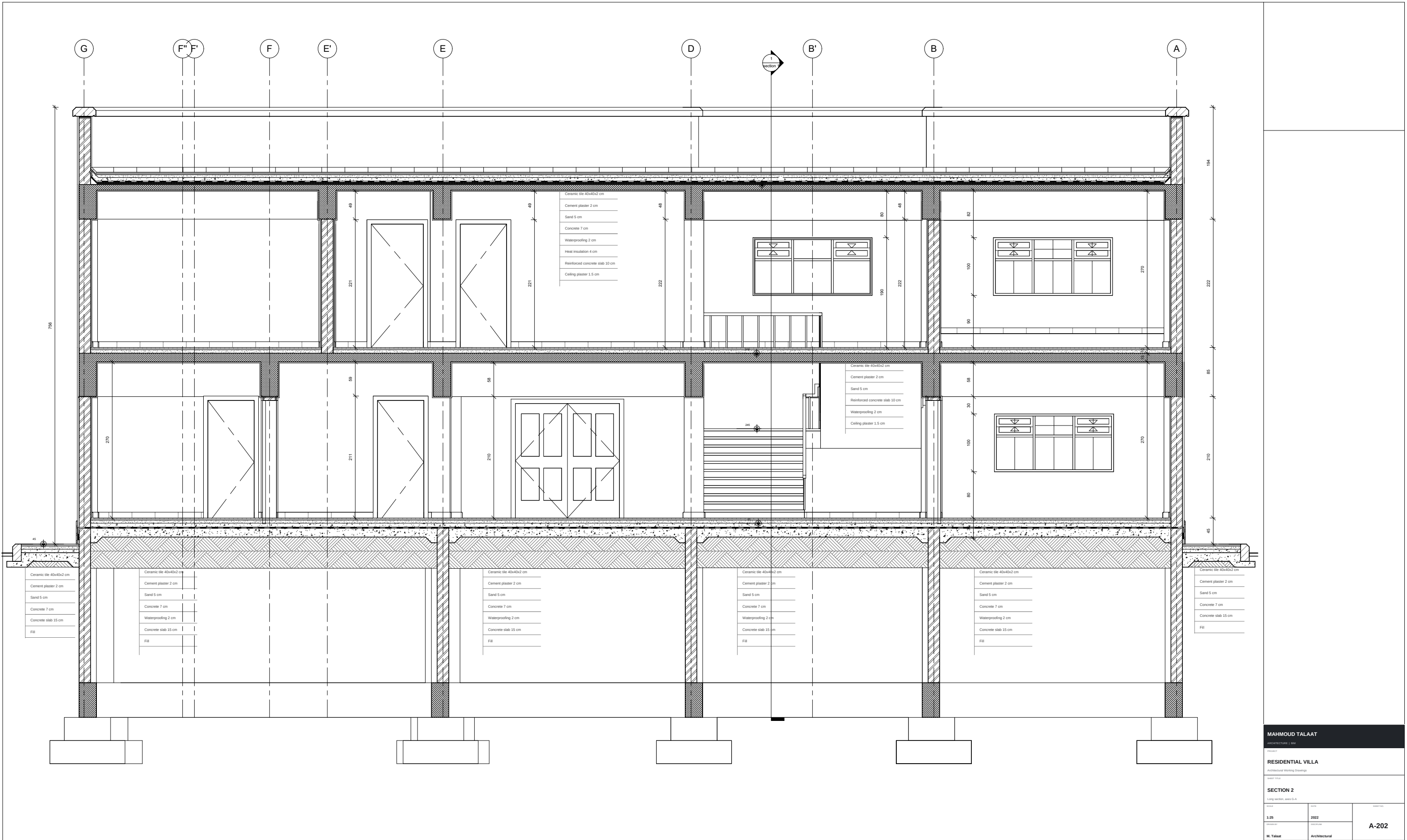
M. Talaat

DISCIPLINE

Architectural

SHEET NO.

A-201



MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

RESIDENTIAL VILLA

Architectural Working Drawings

SHEET TITLE

SECTION 2

Long section, level G.A.

SCALE

1:25

DATE

2022

DRAWN BY

M. Talaat

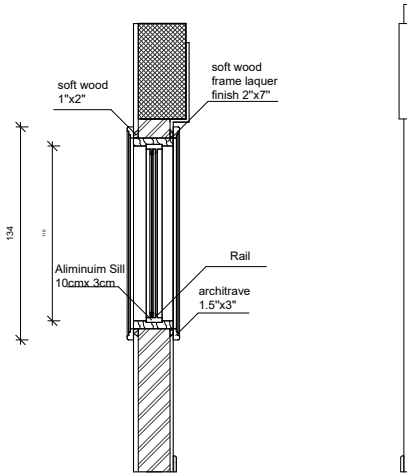
CHECKED BY

Architectural

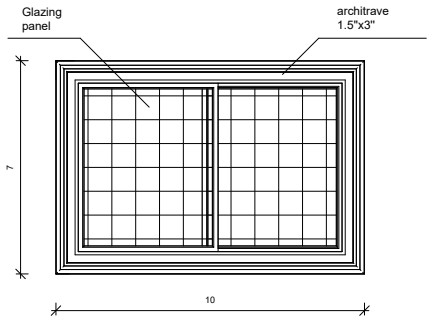
SHEET NO.

A-202

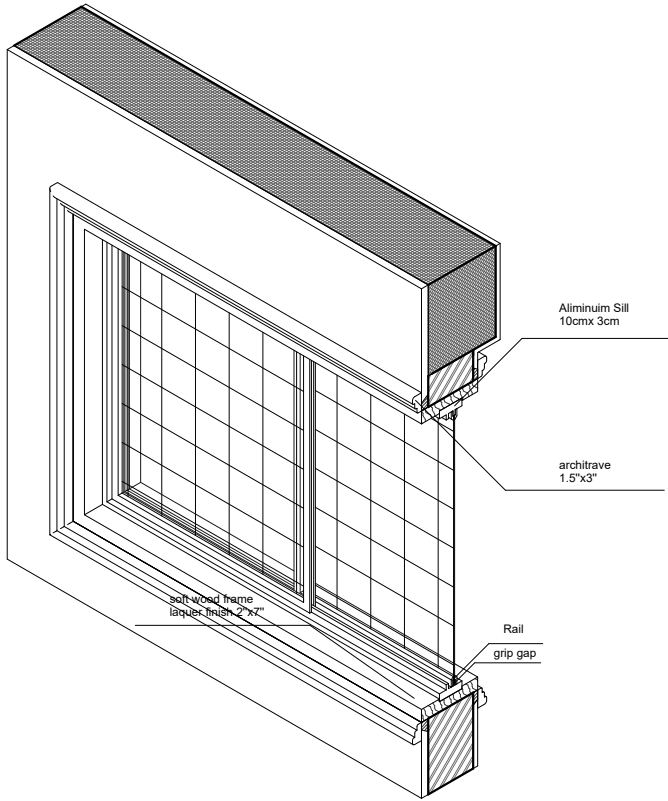
1 Window - vertical section



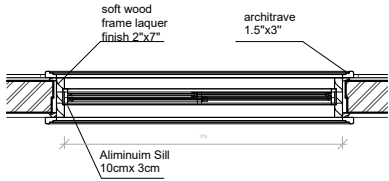
2 Window - elevation



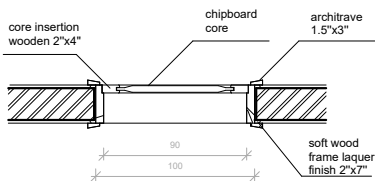
3 Window - 3D view



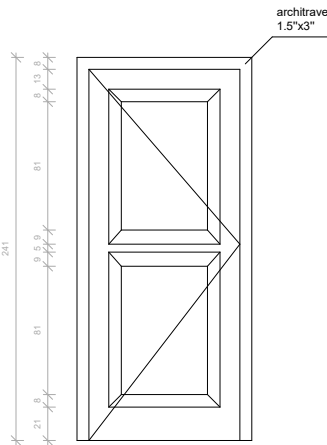
4 Window - horizontal section



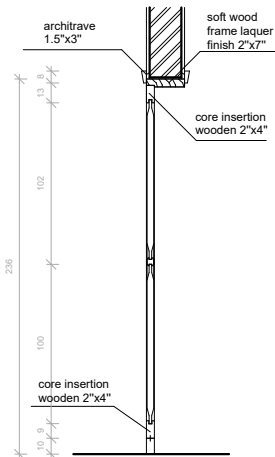
5 Door - horizontal section



6 Door - elevation



7 Door - vertical section



MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

RESIDENTIAL VILLA

Architectural Working Drawings

SHEET TITLE

DOOR & WINDOW DETAILS

SCALE

As indicated

DATE

2022

DRAWN BY

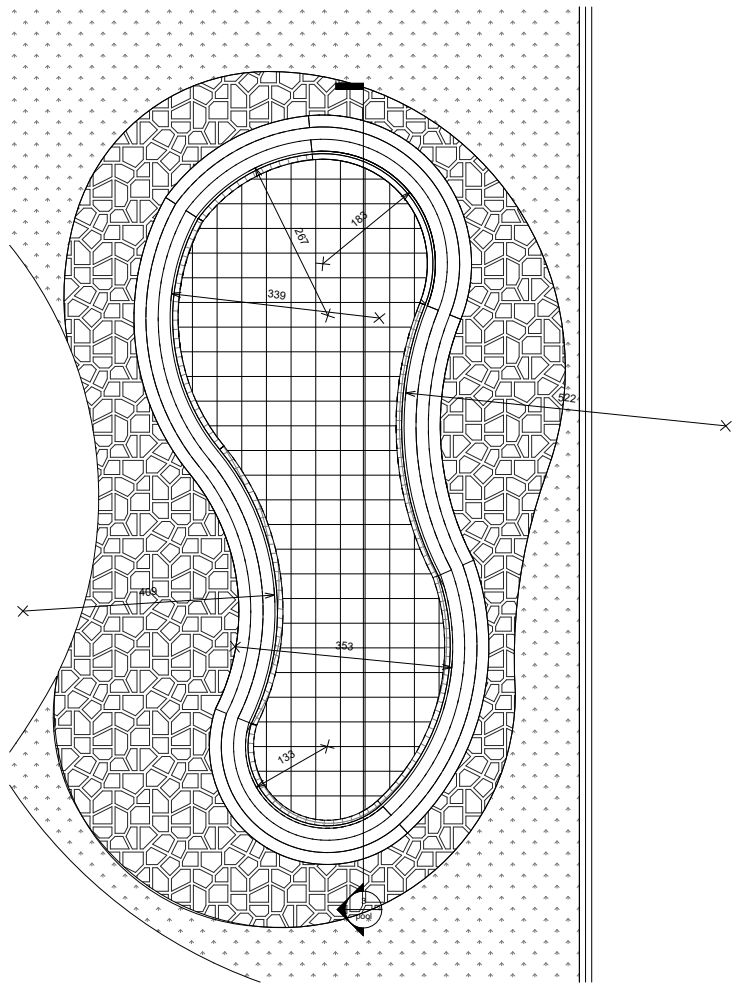
M. Talaat

DISCIPLINE

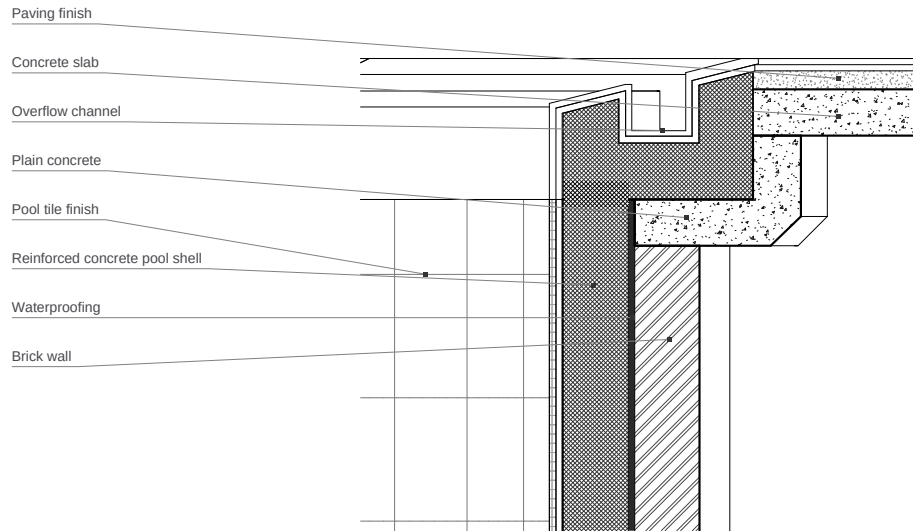
Architectural

SHEET NO.

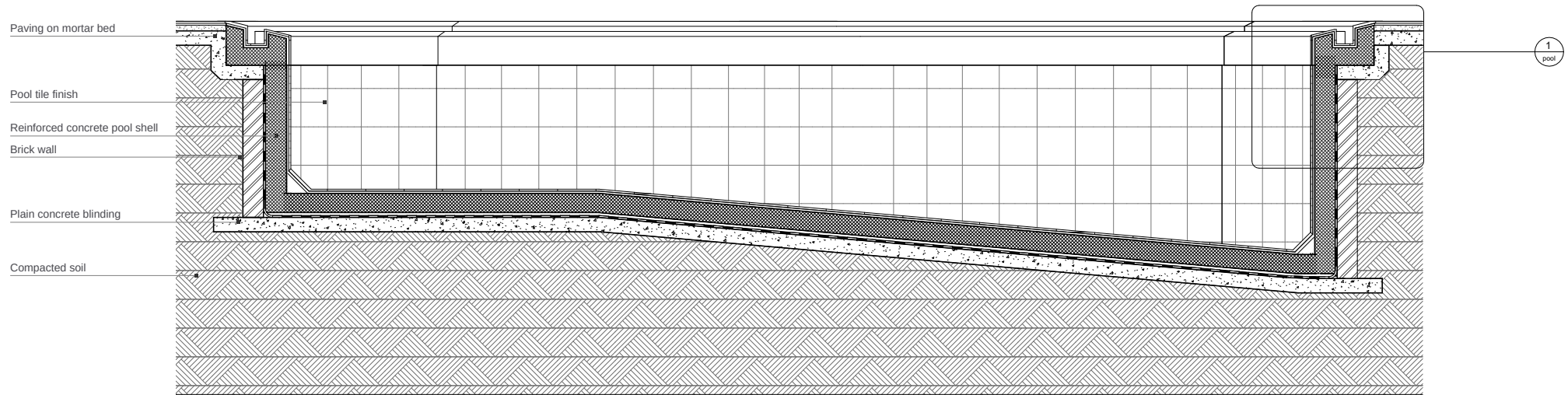
A-301



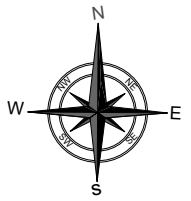
2 Pool plan
1 : 50



1 Pool edge detail
1 : 10



3 Pool section
1 : 25



MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

RESIDENTIAL VILLA

Architectural Working Drawings

SHEET TITLE

SWIMMING POOL

Plan, section & details

SCALE

As indicated

DRAWN BY

M. Talaat

DATE

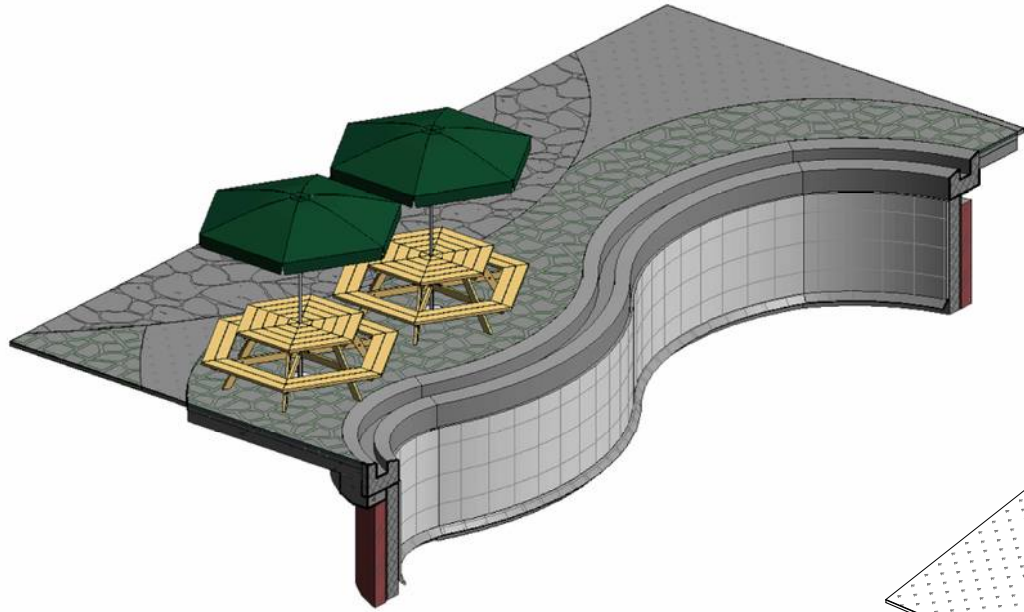
2022

DISCIPLINE

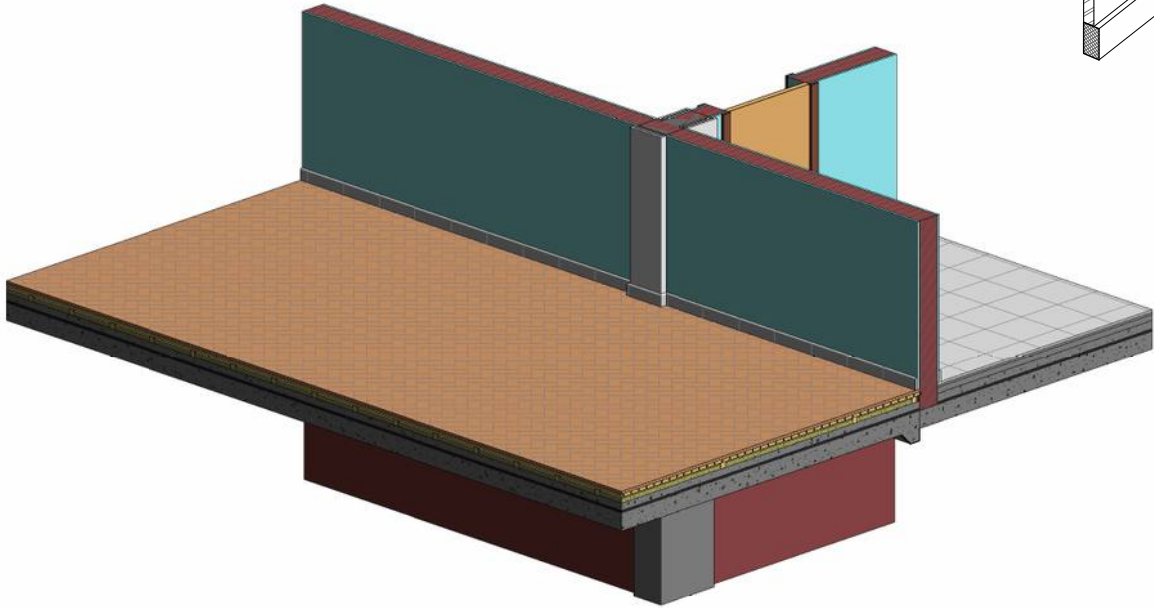
Architectural

SHEET NO.

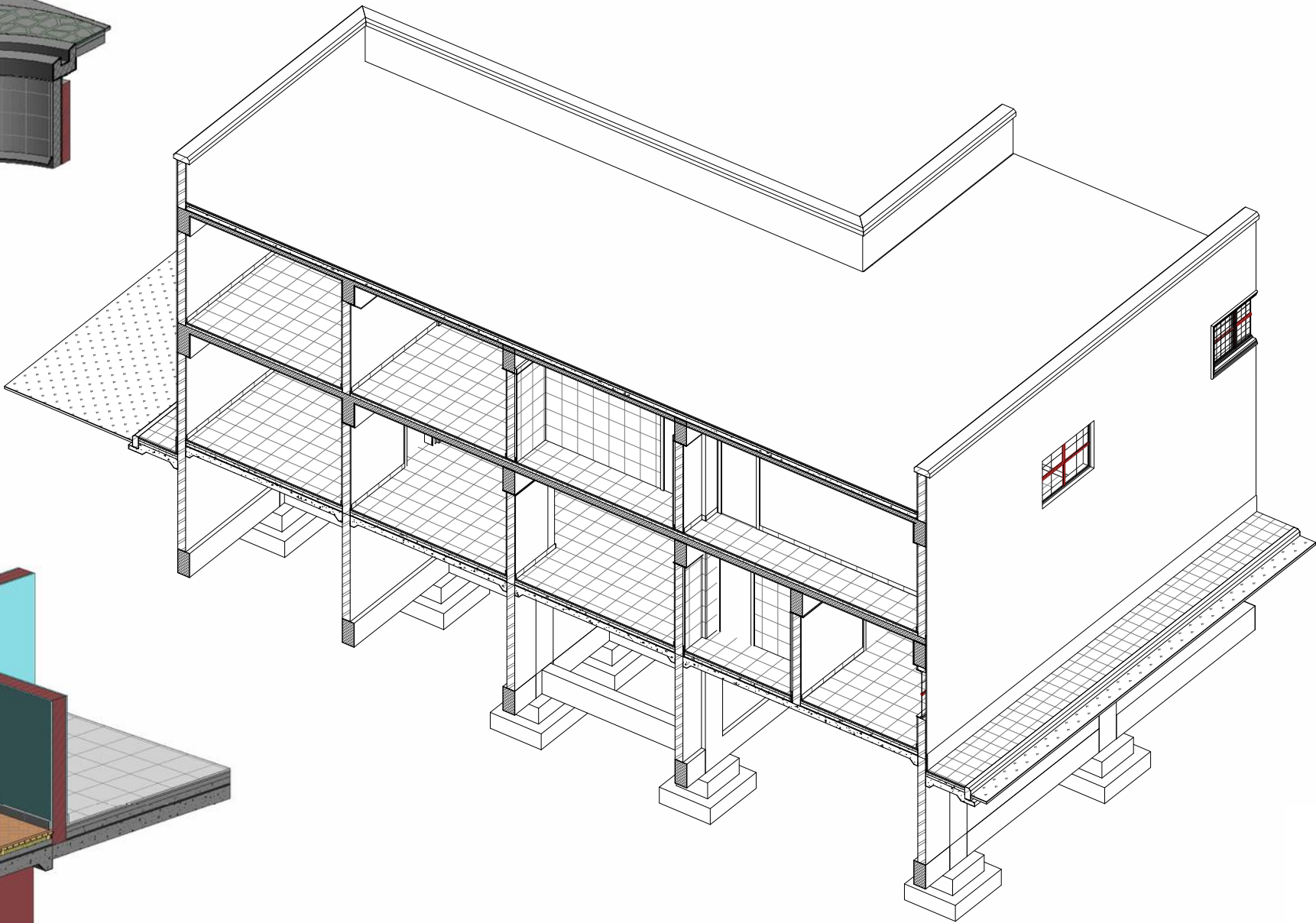
A-302



Pool Terrace - Paving & Seating



Parquet Floor



3D Section - Foundations, Columns & Slabs

MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

RESIDENTIAL VILLA

Architectural Working Drawings

SHEET TITLE

3D VIEWS

Sectional axonometric & assemblies

SCALE

N.T.S.

DRAWN BY

M. Talaat

DATE

2022

DISCIPLINE

Architectural

SHEET NO.

A-401

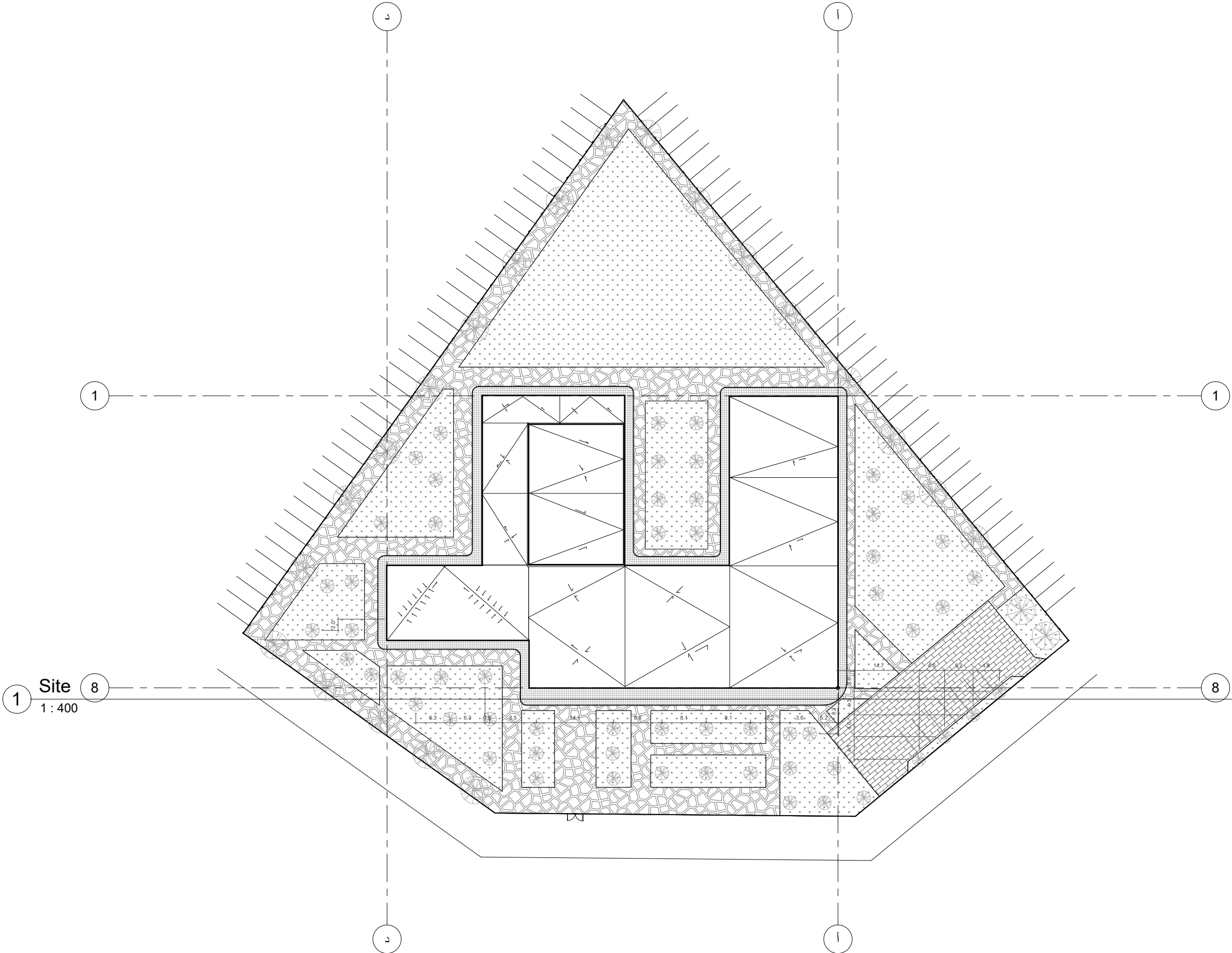
W2

MULTIPURPOSE HALL

Architectural, Structural & MEP Working Drawings

TYPE	Multipurpose hall with administration wing
LEVELS	Ground floor + partial first floor
SITE	Irregular triangular plot with landscaping
SOFTWARE	Autodesk Revit
YEAR	2022

PLANS		
A-001	Site Plan	16
A-101	Ground Floor Plan	17
A-102	First Floor Plan	18
S-101	Foundation Plan	19
A-121	Reflected Ceiling Plan	20
E-101	Electrical Plan	21
M-101	HVAC Plan	22
FP-101	Fire Alarm & Fire Fighting Plan	23
P-101	Drainage Plan	24
SECTIONS		
A-201	Section 1	25
A-202	Section 2	26
DETAILS		
A-301	Section Details	27
A-302	False Ceiling Details	28
S-301	Steel Connection & Cladding Details	29

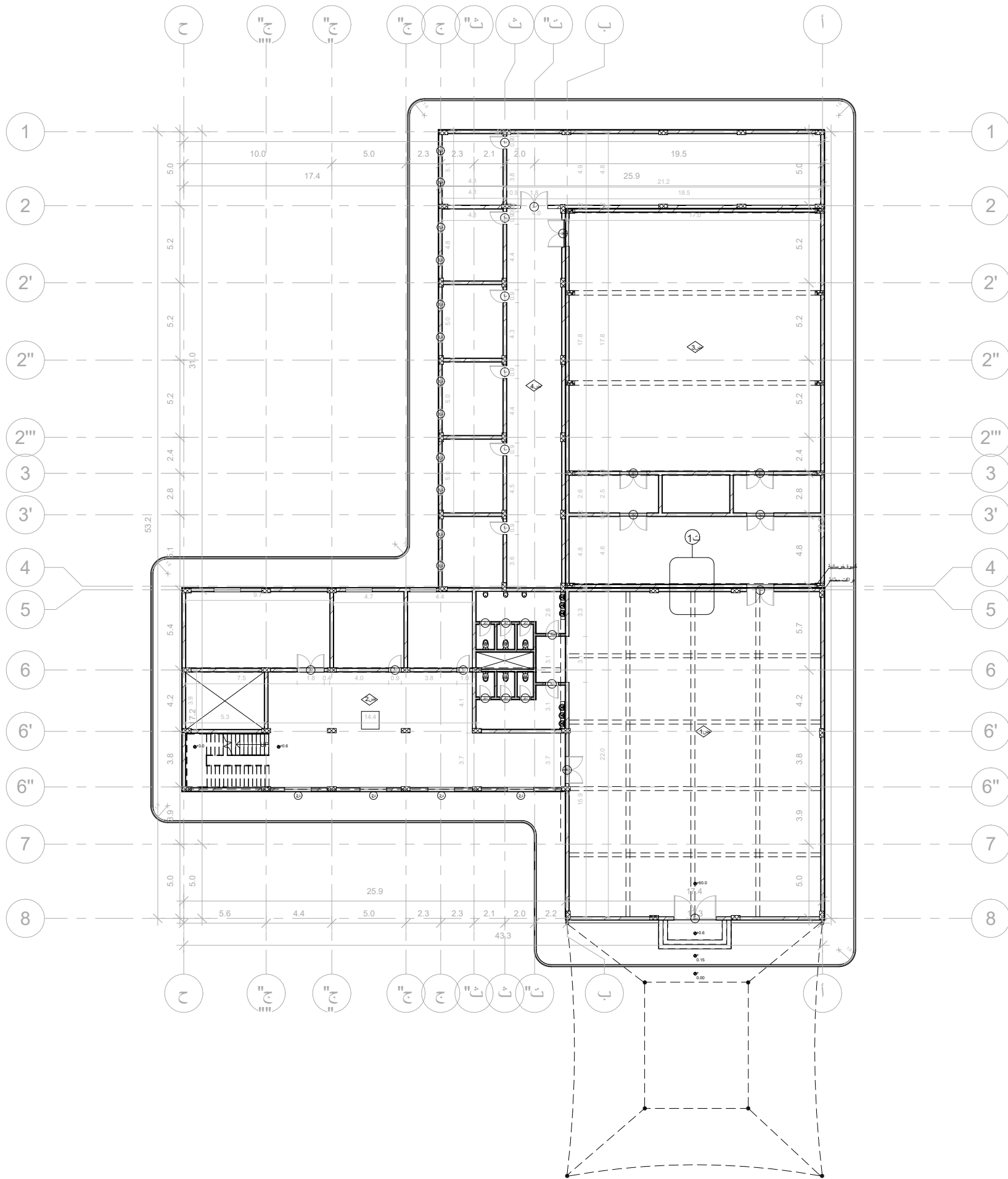


الأشجار				
الرمز	اسم	نصف قطر	ارتفاع	ملاحظات
	شجرة الماعنونا	3م	3م	تظليل
	اشجار اللبخ	5م	15	خصوصية

الارضيات	
	قوالب خرسائية
	عشب اخضر
	بلاط حجري

الرموز	
	الوصف
	نقطة بداية المشروع

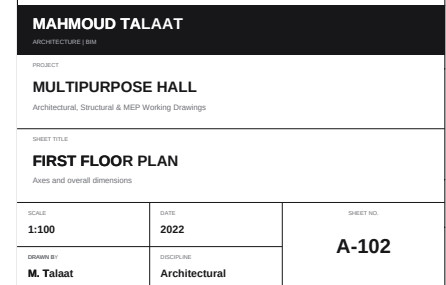
MAHMOUD TALAAT ARCHITECTURAL & MEP		
PROJECT MULTIPURPOSE HALL Architectural, Structural & MEP Working Drawings		
SHEET TITLE SITE PLAN Landscape and hardscape layout		
SCALE 1:400	DATE 2022	SHEET NO. A-001
DRAWN BY M. Talaat	DISCIPLINE Architectural	

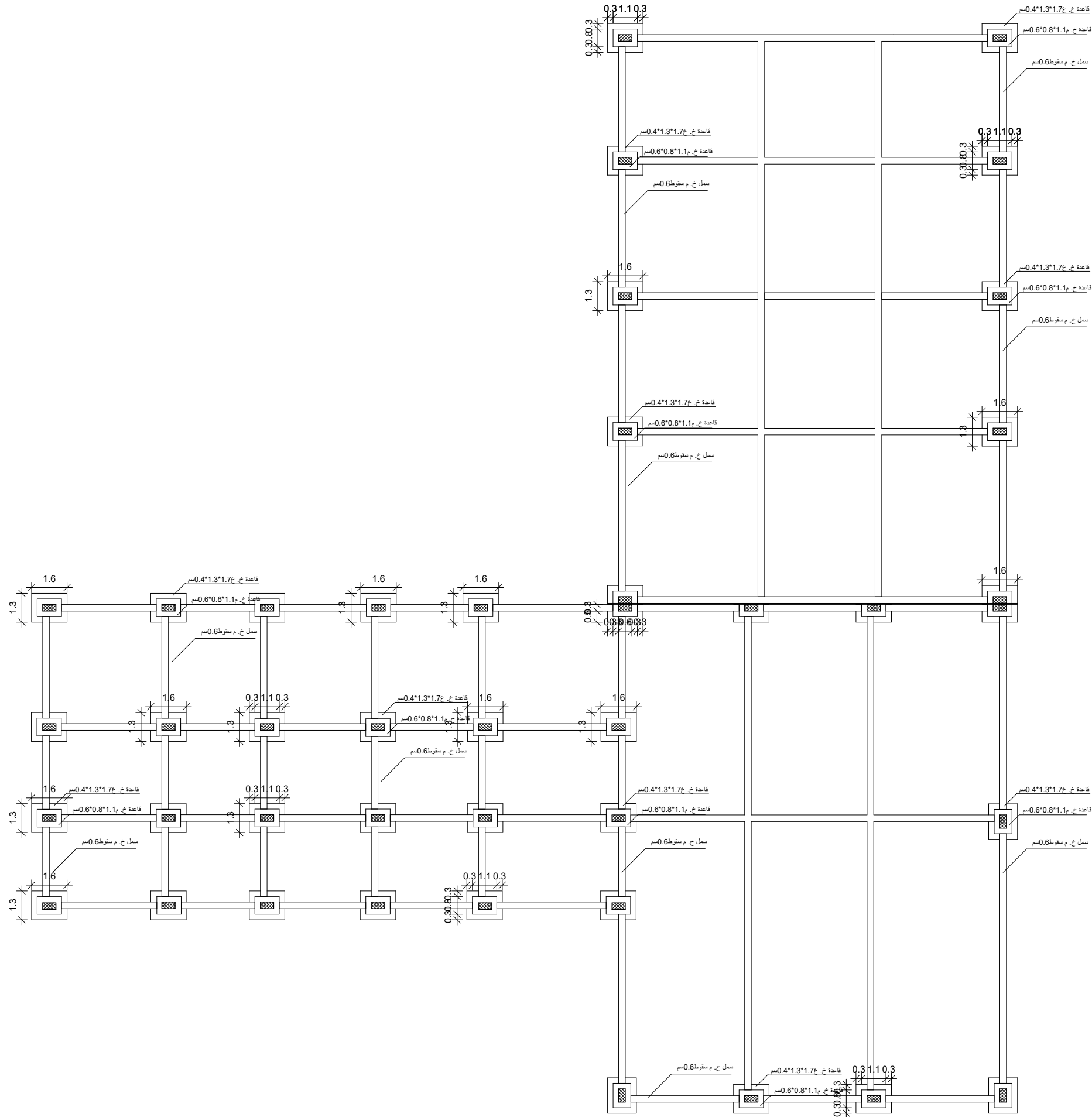


جدول الابواب				
رمز	العدد	مقاسات عرض الارتفاع	نوع	المواصفات
ب1	8	2.20 1.20	مفصلي	باب تجليد البكاج خلفه
ب2	6	2.20 0.90	مفصلي	باب ضلقتين حشو خشب القرو
ب3	8	3.20 1.90	مفصلي	باب حشو خشب القرو خلفه
ب4	2	2.20 1.20	مفصلي	باب تجليد البكاج خلفه
ب5	1	4.20 3.20	مفصلي	باب ضلقتين حشو خشب زان بفتح للخارج

جدول الشبائيك				
رمز	العدد	مقاسات عرض الارتفاع	نوع	المواصفات
ش1	9	1.20 1.00 70	مفصلي	شباك المونيوم زجاج فيه خلفه واحدة
ش2	10	1.20 1.60 1.8	مفصلي	شباك ضلقتين خشب فراغ زجاج

جدول التشطيبات				
رمز	الاسقف	الارضيات	الحوائط	الوزرة
ض1		سويدي 320*160	ترابيع رخام ذهبي 120*260سم	ترابيع بورسلين معرق
ض2		ترابيع سيراميك 30*60سم	دهان بلاستيك	
ض3		سويدي 120*100سم	الواح خشب تجليد 40*40*2 pvc خشب	



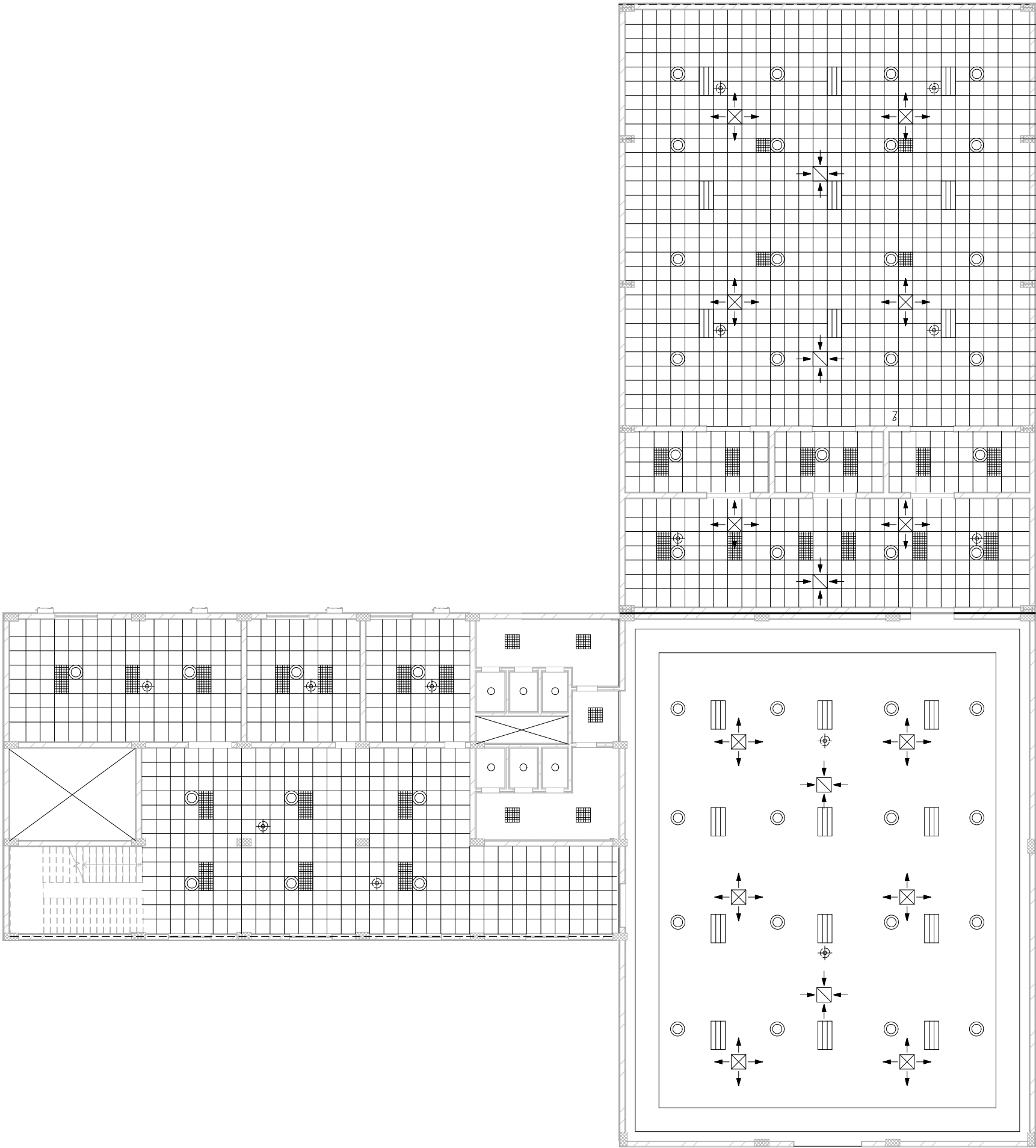


MAHMOUD TALAAT
ARCHITECTURE L&M

MULTIPURPOSE HALL
Architectural, Structural & MEP Working Drawings

FOUNDATION PLAN
Footings layout and dimensions

SCALE 1:100	DATE 2022	SHEET NO. S-101
DRAWN BY M. Talaat	DISCIPLINE Structural	

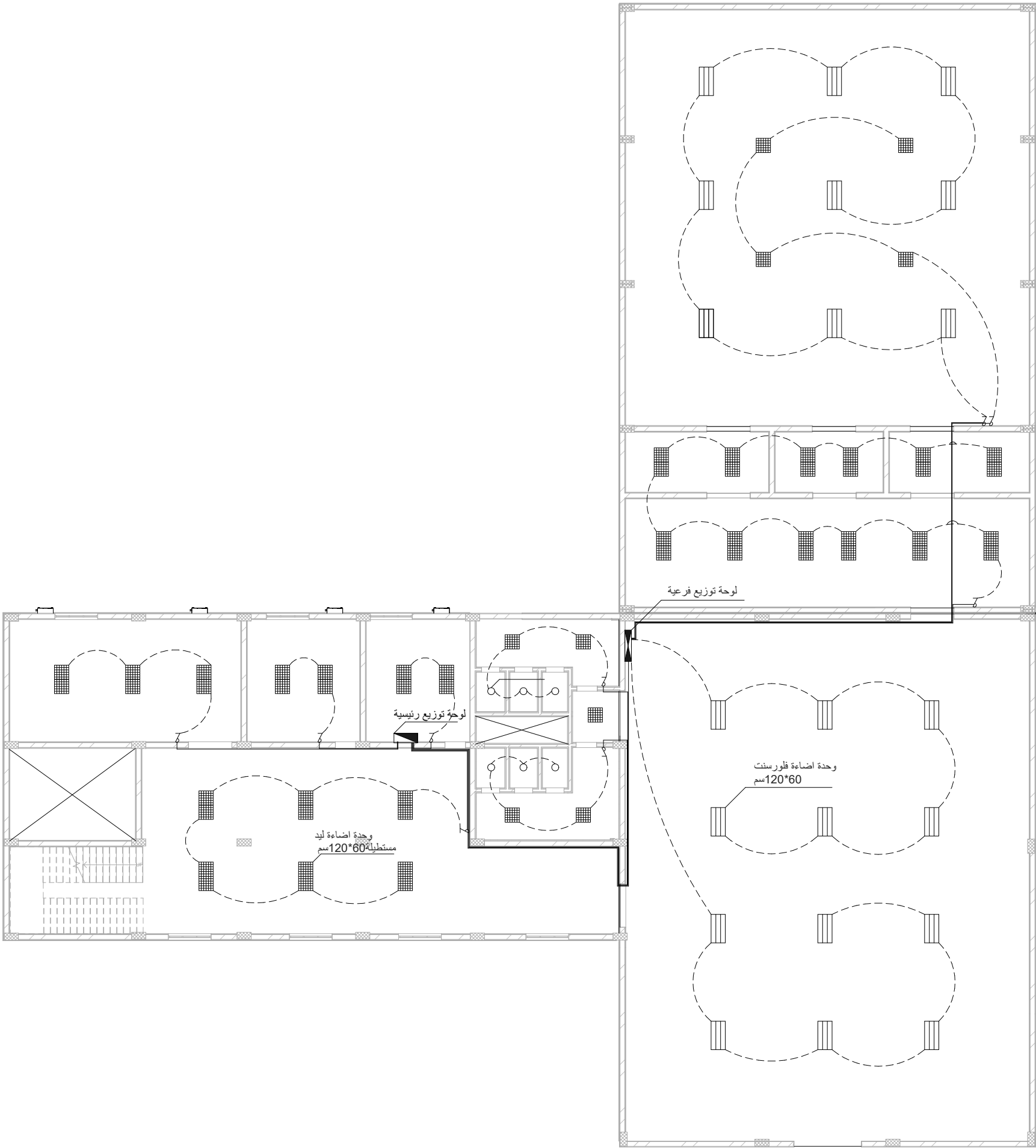


الرمز	عدد	البيان	ملاحظات النوع
7	٢٢	مفتاح أضواء سكة واحدة	سائي علي ارتفاع سم ١٦٠
7	٤	مفتاح أضواء سكتين	سائي علي ارتفاع سم ١٦٠
	١٢	وحدة أضواء فلور سنت مزدوجة بالسقف	
	١٨	وحدة أضواء ليد بالسقف سم ٦٠*١٢٠	
	١٤	وحدة أضواء ليد بالحائط	
	٥٤	سوتلت للأضواء	
	٥٢	وحدة أضواء ليد مربعة سم ٣٠*٣٠ سم لانتارة الممرات	
	٤٦	وحدة أضواء ليد بالسقف	

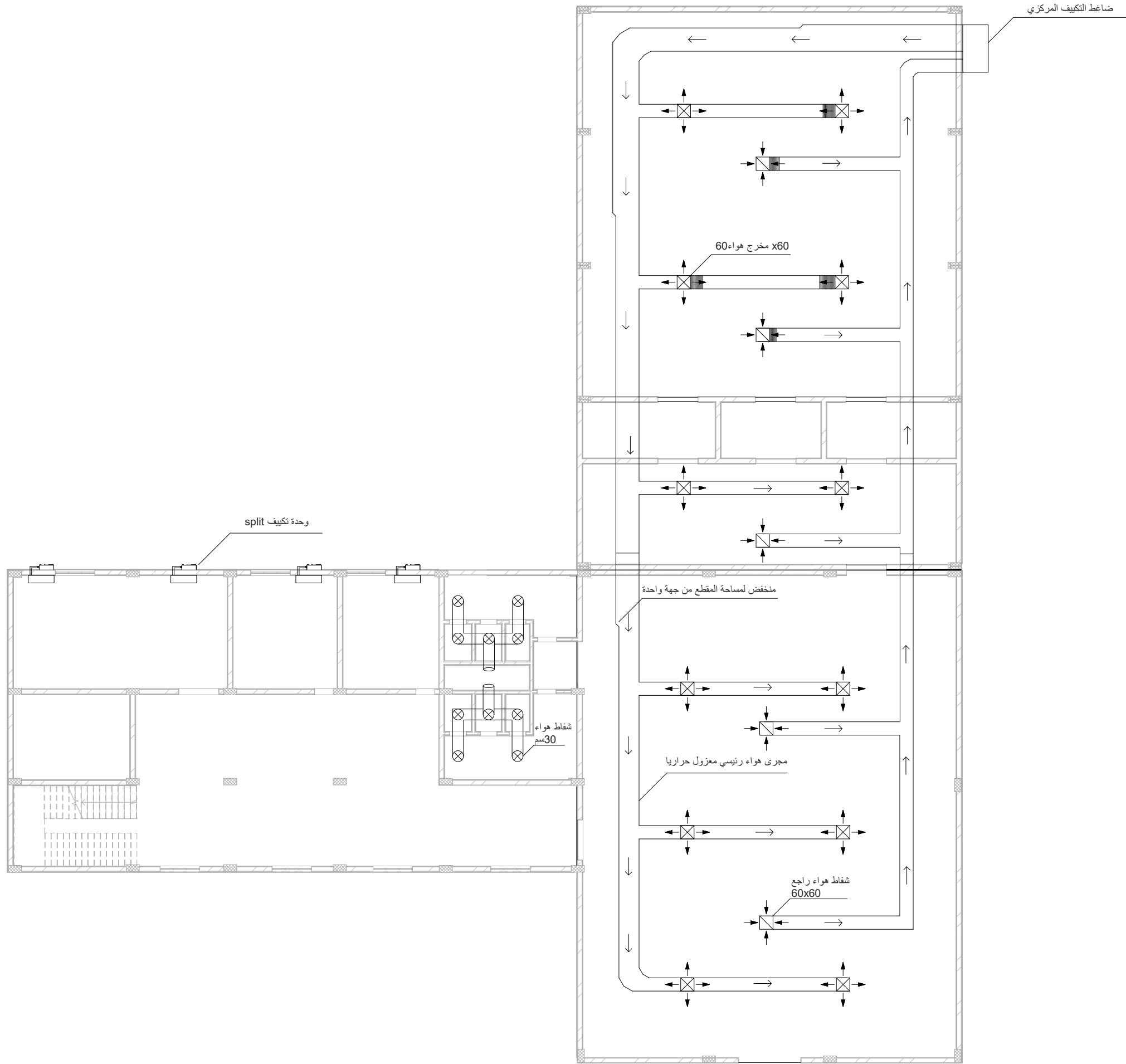
الرمز	التوصيف
	انذار ضوئي باللون الاحمر
	انذار صوتي (سماعات)
	كاشف حراري
	كاشف للدخاد
	كاشف للغاز
	اللوحة الرئيسية للانذار

الرمز	البيان
	مخرج للهواء بارد
	راجع للهواء بارد
	كوع ٩٠
	وصله تي
	شفاط هواء

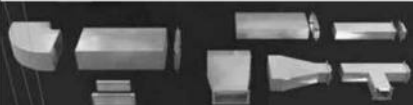
MAHMOUD TALAAT ARCHITECTURE LLM		
PROJECT MULTIPURPOSE HALL Architecture, Structural & MEP Working Drawings		
SHEET TITLE REFLECTED CEILING PLAN Lighting and suspended ceiling layout		
SCALE 1:100	DATE 2022	SHEET NO. A-121
DRAWN BY M. Talaat	DISCIPLINE Architectural	



الرمز	عدد	البيان	ملاحظات النوع
7	٢٢	مفتاح اضاءة سكة واحدة	علي ارتفاع سم ١٦٠ ساتني
7	٤	مفتاح اضاءة سكتين	علي ارتفاع سم ١٦٠ ساتني
	١٢	وحدة اضاءة فلورسنت مزدوجة بالسقف	
	١٨	وحدة اضاءة ليد بالسقف ٦٠*١٢٠ سم	
	١٤	وحدة اضاءة ليد بالحائط	
	٥٤	سويتات للاضاءة	
	٥٢	وحدة اضاءة ليد مربعة ٣٠ سم ٣٠ سم لانتارة الممرات	
	٤٦	وحدة اضاءة ليد بالسقف	
	٣٥	مأخذ تيار (بريز ١٠ أمبير)	علي ارتفاع سم ٥٠ ساتني
	٦	مأخذ تيار فوي (بريز ٣٢ أمبير)	علي ارتفاع سم ٥٠ ساتني
	٤	مخرج تليفون (بريز ٢ تليفون)	علي ارتفاع سم ١٠٠ ساتني
	٥	مخرج تلفزيون (بريز ٥ تلفزيون)	علي ارتفاع سم ٢٤٠ ساتني
		دائرة ثانوية (خط فرعي للاضاءة سلك ٣ مللي شعر نحاسي)	
		دائرة رئيسية (خط هومي) للاضاءة و برايز القوي داخل الحوائط متصلة بلوح للتوزيع	
	١	لوحة توزيع رئيسية للمبني (داخل الحائط)	علي ارتفاع سم ١٨٠
	١	لوحة توزيع فرعية للمبني (داخل الحائط)	علي ارتفاع سم ١٨٠
	٤	سخن مياه	



البيان	الرمز
مخرج للهواء بارد	
راجع للهواء بارد	
كوع ٩٠	
وصلته تي	
شفاط هواء	

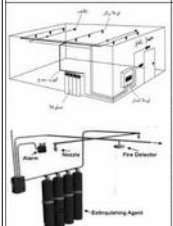


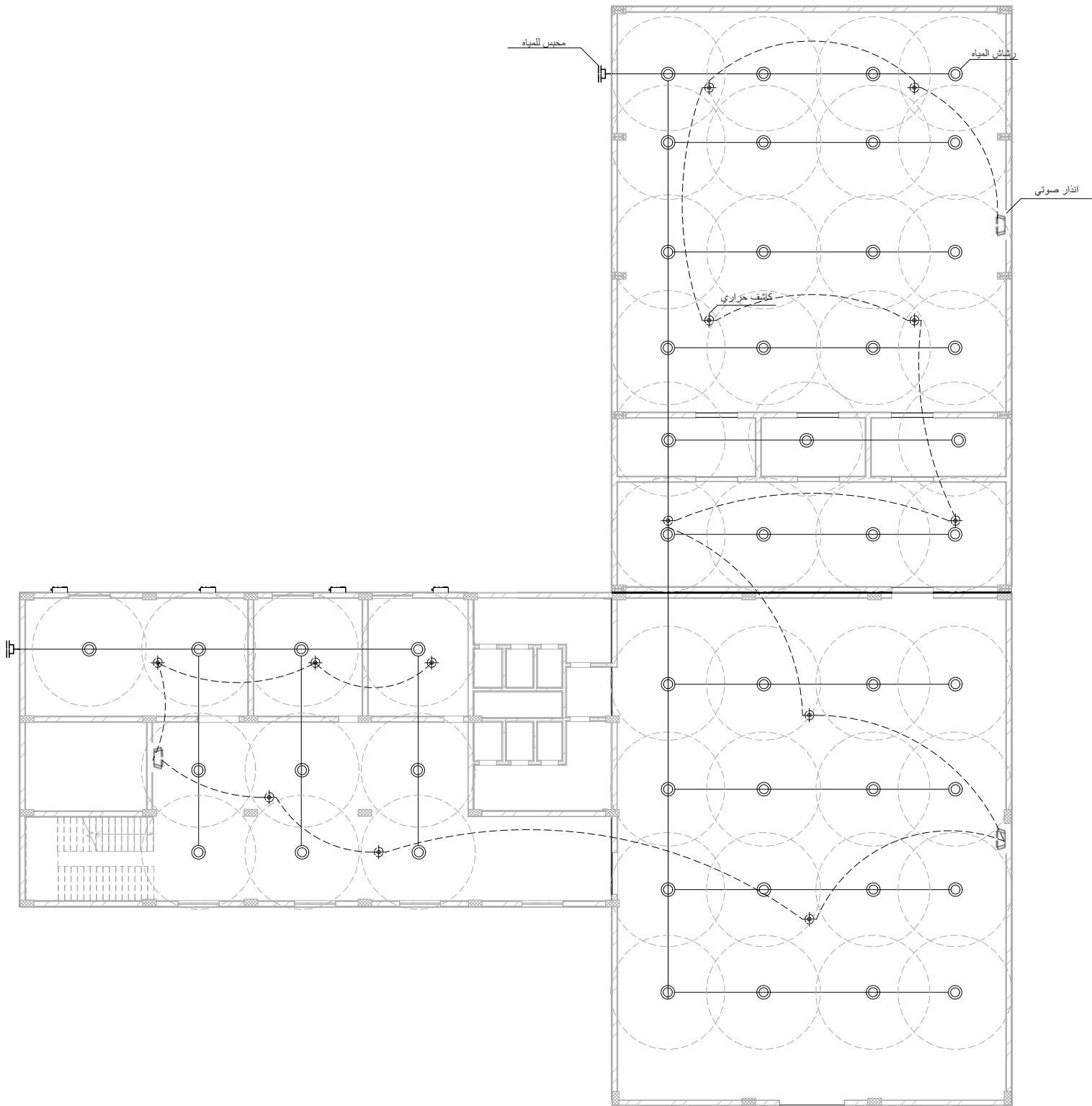
MAHMOUD TALAAT
ARCHITECTURE LDM

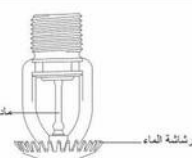
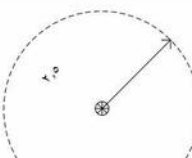
PROJECT
MULTIPURPOSE HALL
Architecture, Structural & MEP Working Drawings

SHEET TITLE
HVAC PLAN
Air distribution and duct layout

SCALE 1:100	DATE 2022	SHEET NO. M-101
DRAWN BY M. Talaat	DISCIPLINE Mechanical	

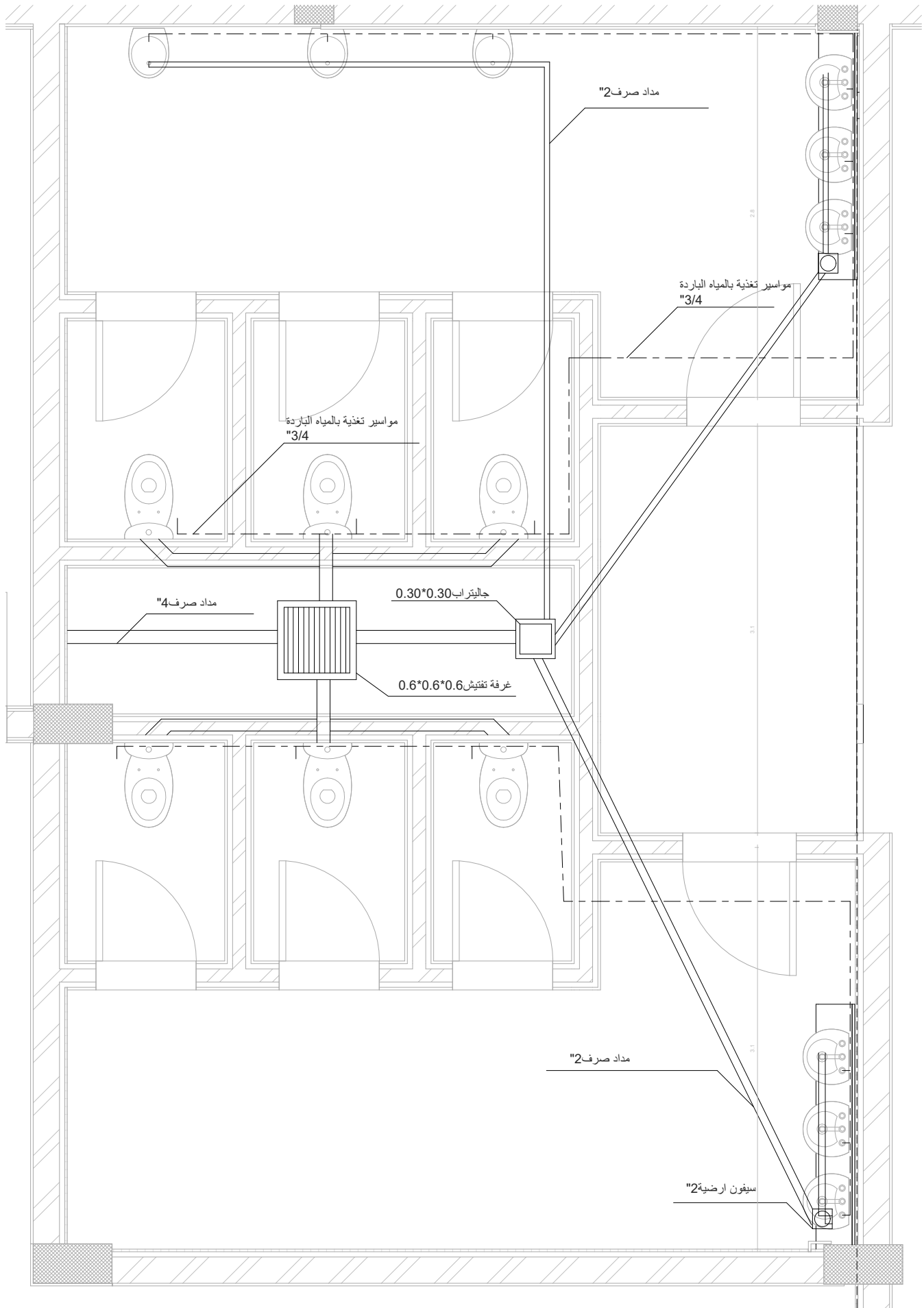
انظمة الاطفاء المستخدمة		
الاطفاء بالماء	الاطفاء بغاز CO2	شكل الرشاش
		
عن طريق ماسورة تغذية رئيسية قطر ها ٢ بوصة ١- في الجزء الإداري ٢- في منطقة الاستقبال المساحة المطلوبة	يستخدم نظام ال CO2 في اخماد الحريق الناتج من الكهربية ١- في الجزء الإداري ٢- في منطقة الاستقبال	نظام الاطفاء
يتم التشغيل تلقائي عن طريق كاشف الحريق	يتم التشغيل تلقائي عن طريق كاشف الحريق	
		



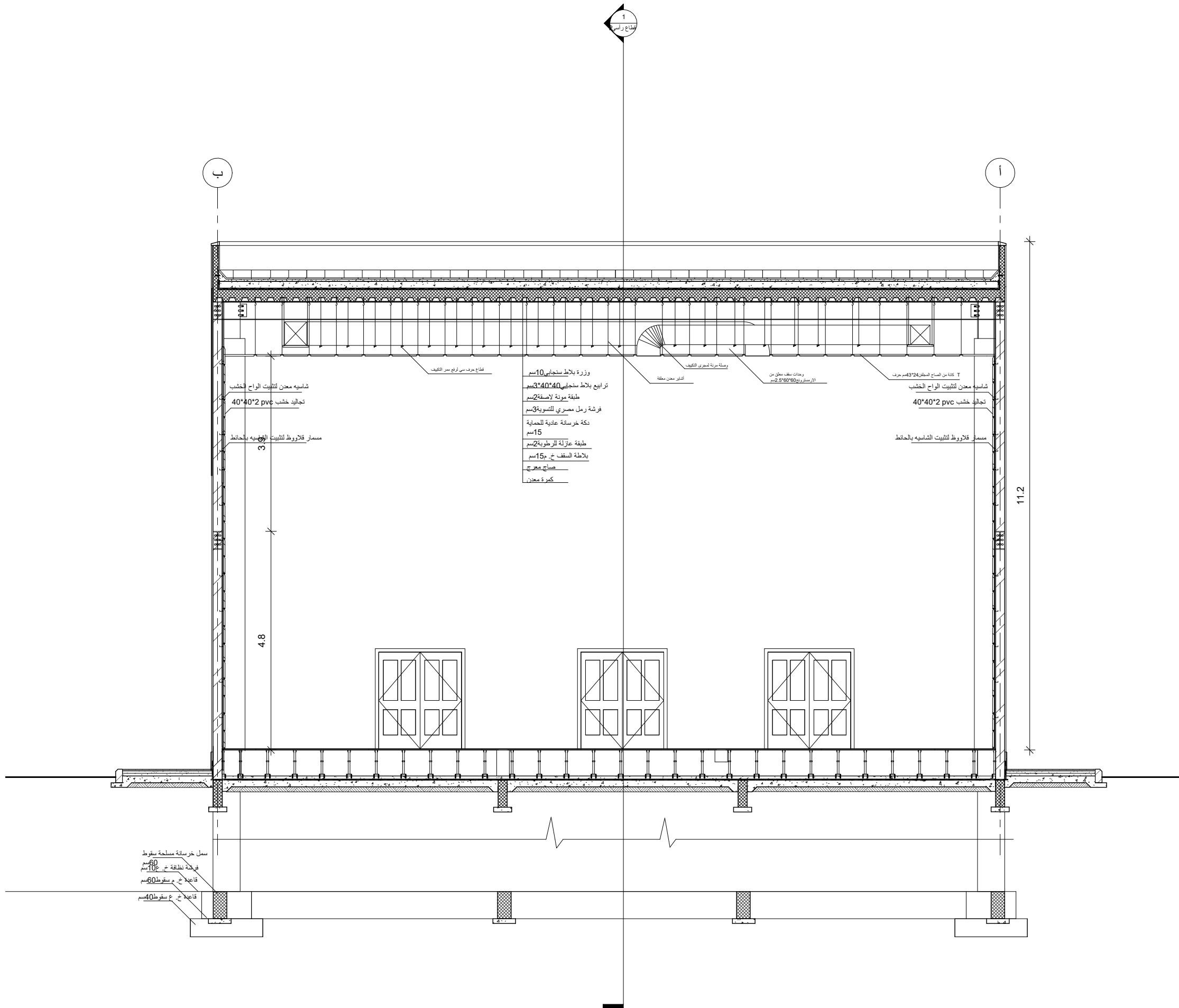
أنواع رشاش المياه	النوع المستخدم هو رأس الرشاش لأسفل	نصف قطر مدى وحدات الرش ٢,٥ م	المواصفات
			١-النظام يستخدم الماء العادي الذي يتم تحويله الى رزاز بقوة ضغط ٢٠٠-٨٠ بار ٢-يشتمل كل نظام صحن على محرك كهربائي بدير مضخات موصل بمصدر تغذية احتياطي ٣-يشتمل كل نظام على مضخة مساعدة للحفاظ على مستوى ثابت في الصحن
رأس الرشاش لأعلى	رأس الرشاش لأسفل		

الرمز	التوصيف
	مدى الرشاش
	كوع ٩٠
	وصلة صليبية
	وصلة تية
	مواسير تغذية ب المياه للحريق بولي بروبيلين
	محبس للمياه
	رشاش المياه

MAHMOUD TALAAT		
ARCHITECTURE / E&M		
PROJECT		
MULTIPURPOSE HALL		
Architectural, Structural & MEP Working Drawings		
SHEET TITLE		
FIRE ALARM & FIRE FIGHTING PLAN		
Detection, alarm and sprinkler layout		
SCALE	DATE	SHEET NO.
1:100	2022	
DRAWN BY	DISCIPLINE	FP-101
M. Talaat	Fire Protection	



جدول الرموز والمصطلحات	
الرمز	المواصفات
ع ٤ بوصة	عمود عمل ٤ بوصة بلاستيك p.v.c
ع ٣ بوصة	عمود صرف ٣ بوصة بلاستيك p.v.c
ع ٢ بوصة	عمود تصوية ٢ بوصة بلاستيك p.v.c
سيفون ارضية مخرج ٢ بوصة بلاستيك p.v.c	
كوع زاوية ٤٥ صرف ٢ بوصة بلاستيك p.v.c	
وصلة تيه ٣ بوصة بلاستيك p.v.c	
وصلة تيه ٢ بوصة بلاستيك p.v.c	
وصلة حرف ٣*٧ بوصة بلاستيك p.v.c	
كوع زاوية ٩٠ صرف ٢ بوصة بلاستيك p.v.c	
مداد صرف ٢ بوصة بلاستيك p.v.c	
جلية ٣ و ٢ بوصة لا عمال الصرف بلاستيك p.v.c	
مرحاض فريين بصندوق طرد واطى على شكل حرف p	
حوض غسيل ايدي	
ع ٤ ت ب	عمود تغذية من البلاستيك ع ٢ ب ٤ مم
محبس ٣ بوصة	
مواسير تغذية بالمياه الباردة داخل الحمام ٣ بوصة	
مواسير تغذية بالمياه الساخنة ٣ بوصة	
جاليتراب بلاستيك ٣٠٠*٣٠٠*٣٠٠	
غرفة تقطيش ٦٠٠*٦٠٠*٦٠٠	
سخان مياه	
حوض غسيل اواني وخضروات	
حوض استحمام (بانير) ٧٠٠*٧٠٠*١٧٠٠	



MAHMOUD TALAAT

ARCHITECTURE 188X

PROJECT

MULTIPURPOSE HALL

Architecture, Structural & MEP Working Drawings

SHEET TITLE

SECTION 1

Vertical section

SCALE

1:50

DRAWN BY

M. Talaat

DATE

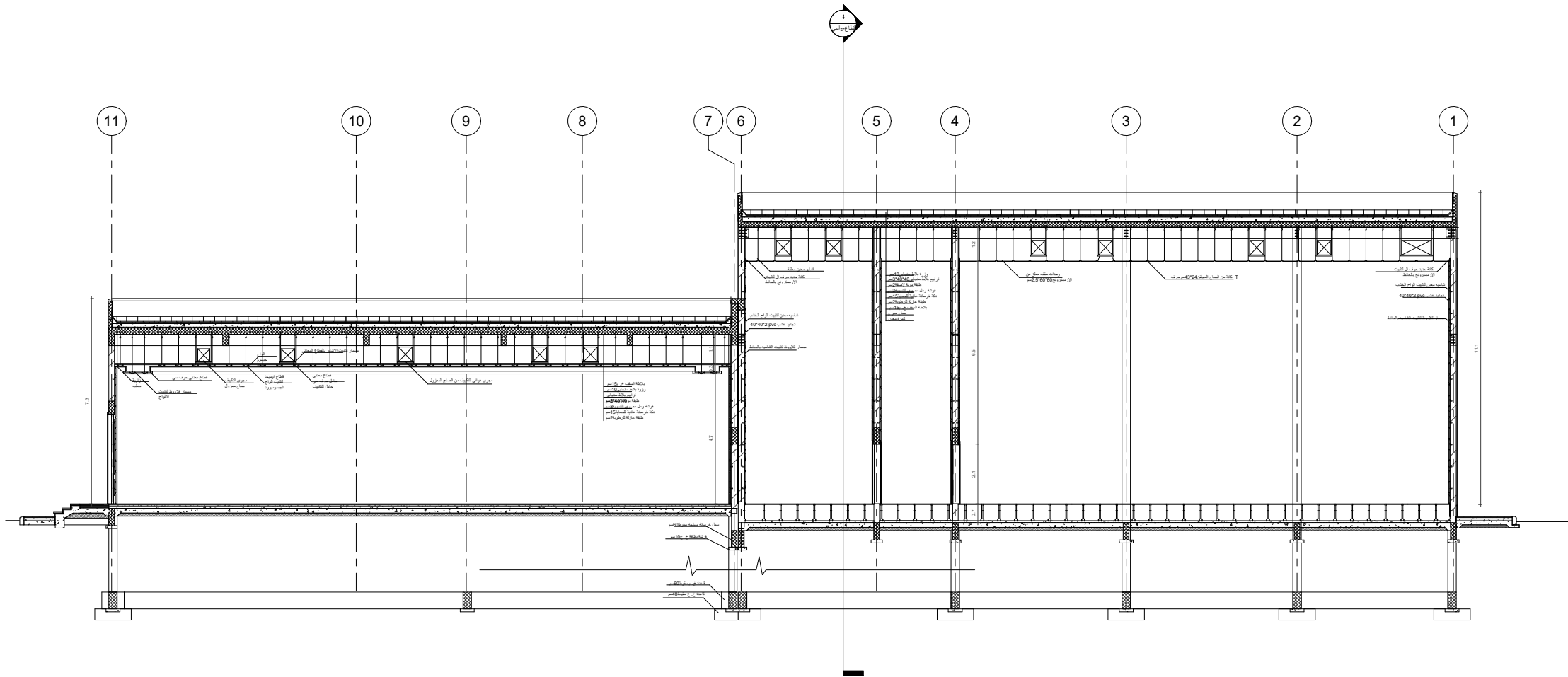
2022

DISCIPLINE

Architectural

SHEET NO.

A-201



MAHMOUD TALAAT

ARCHITECTURE / BIM

MULTIPURPOSE HALL

Architectural, Structural & MEP Working Drawings

SHEET TITLE

SECTION 2

Vertical section

SCALE

1:100

DATE

2022

DRAWN BY

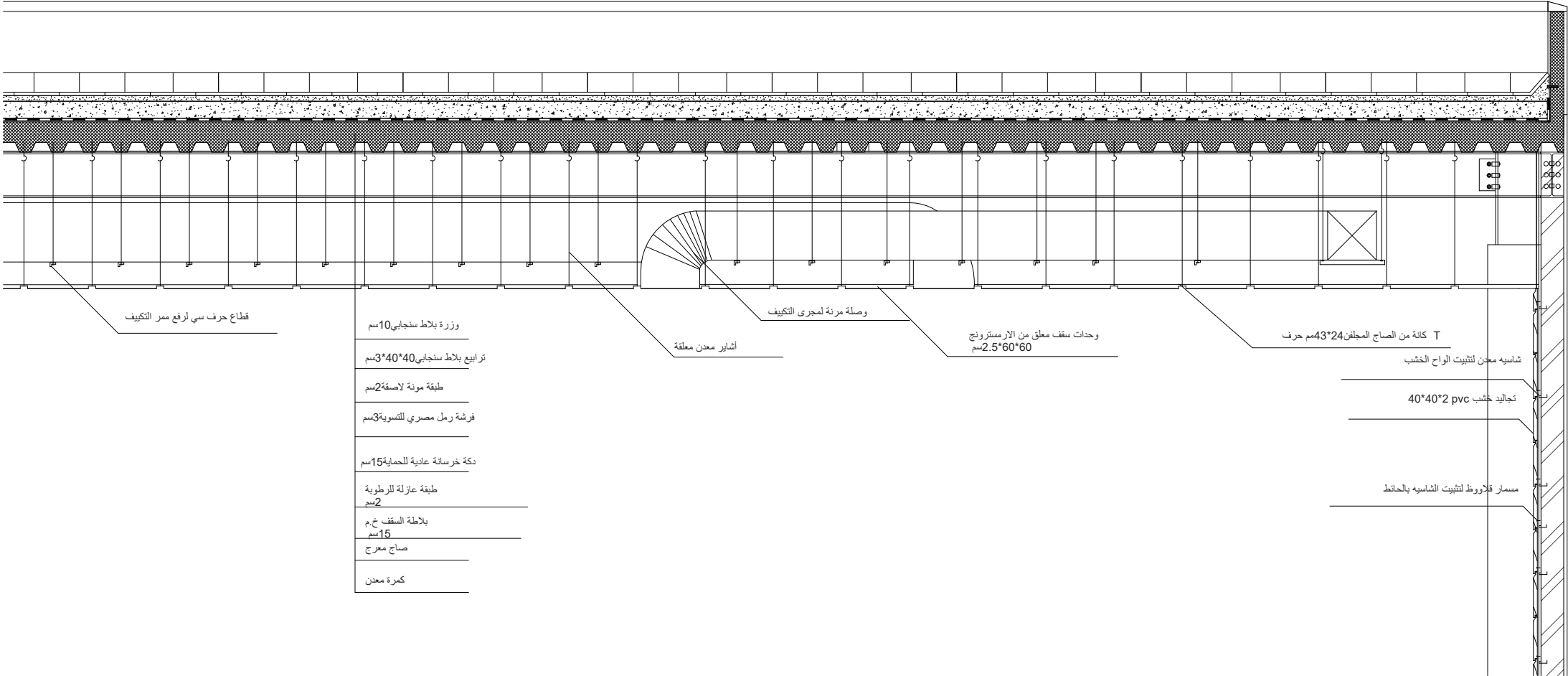
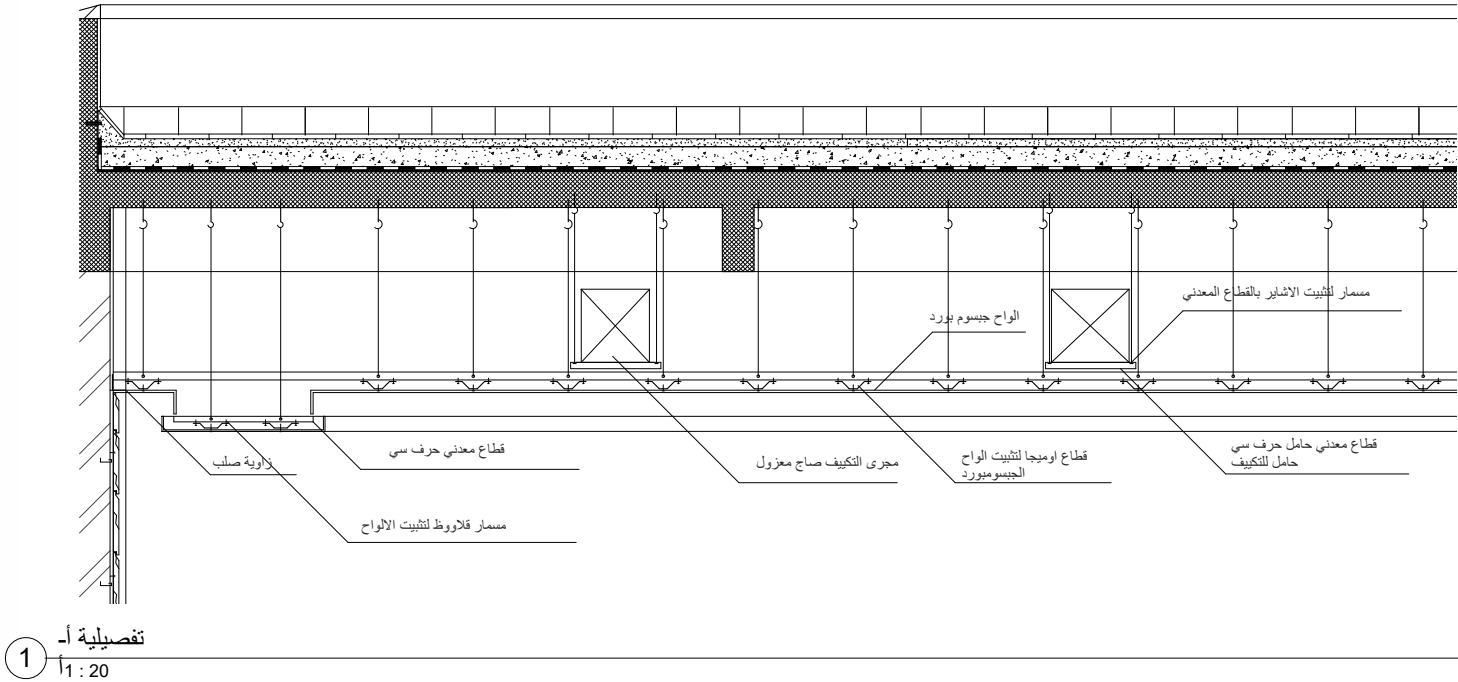
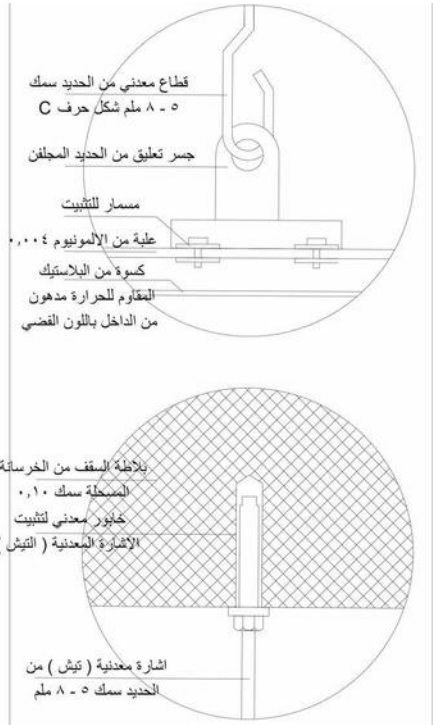
M. Talaat

DISCIPLINE

Architectural

SHEET NO.

A-202



2

تفصيلية ب-

20

MAHMOUD TALAAT

ARCHITECTURE / BIM

PROJECT

MULTIPURPOSE HALL

Architectural, Structural & MEP Working Drawings

SHEET TITLE

SECTION DETAILS

Suspended ceiling and floor build-up

SCALE

As indicated

DATE

2022

DRAWN BY

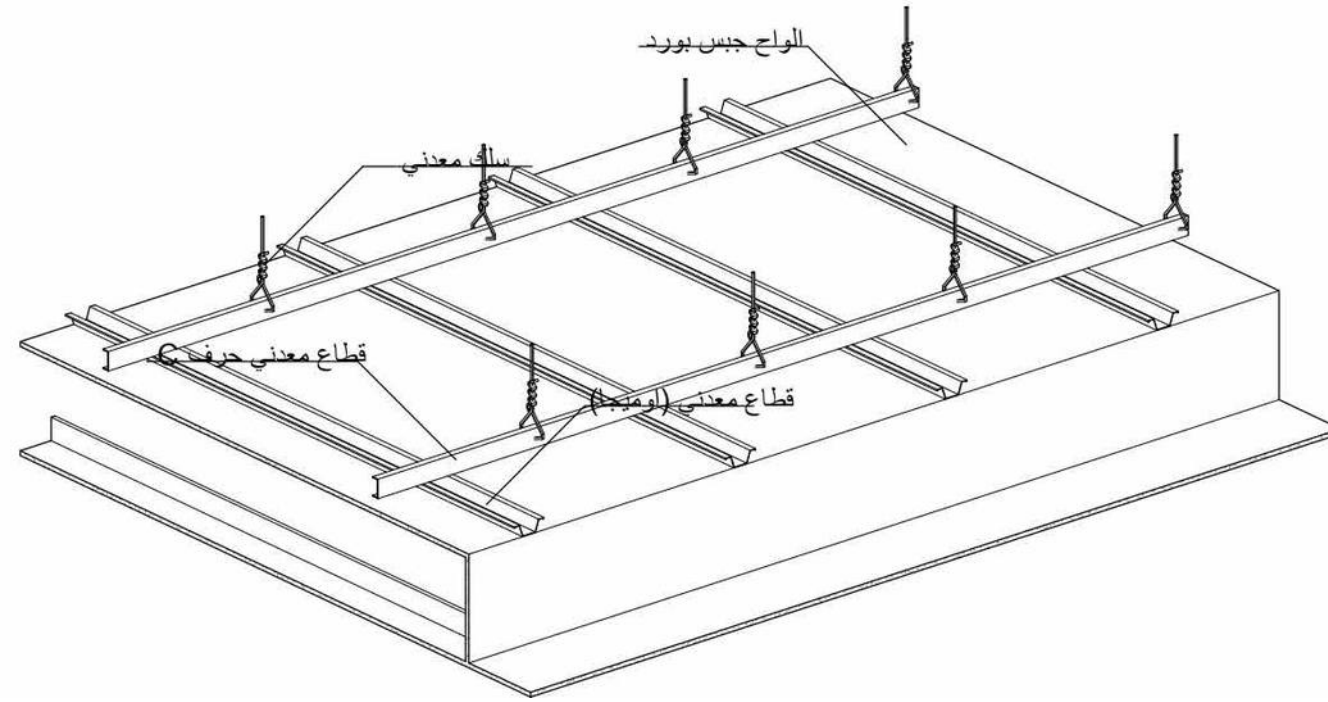
M. Talaat

DISCIPLINE

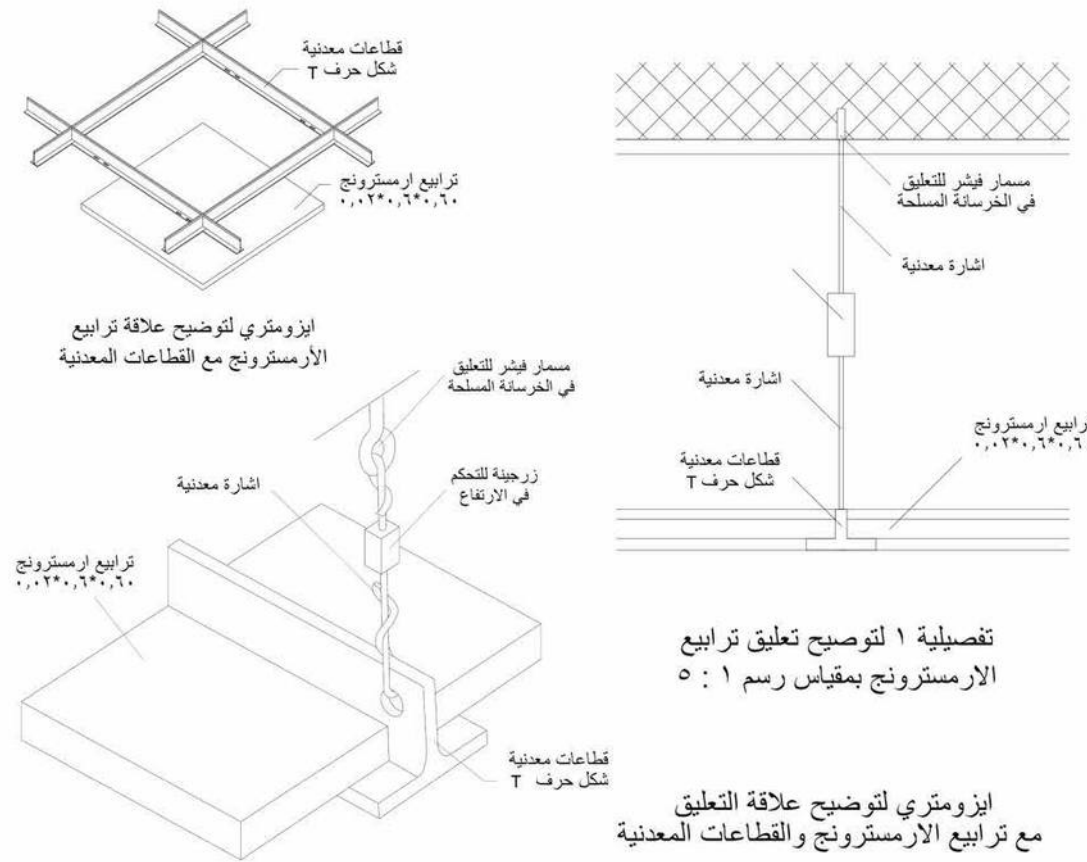
Architectural

SHEET NO.

A-301



تفصيلية تركيب سقف جبس بورد زائف



تفصيلية تركيب سقف ارمسترونج زائف

MAHMOUD TALAAT

ARCHITECTURE | BIM

MULTIPURPOSE HALL

Architectural, Structural & MEP Working Drawings

SHEET TITLE

FALSE CEILING DETAILS

Gypsum board and Armstrong systems

SCALE

N.T.S.

DATE

2022

SHEET NO.

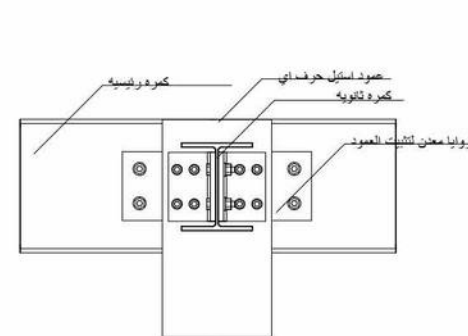
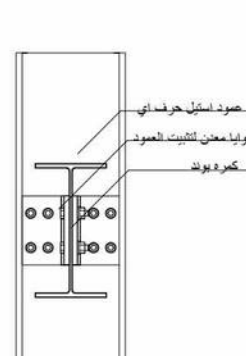
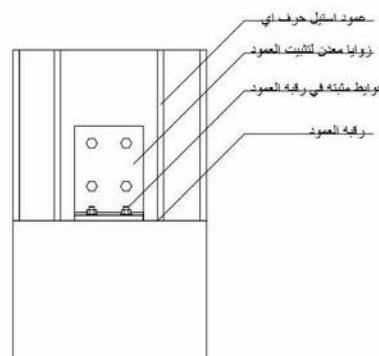
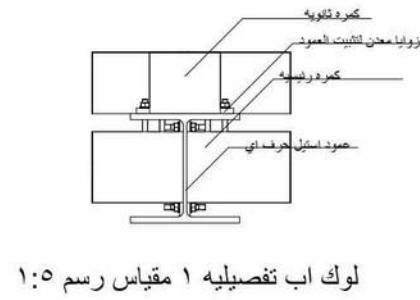
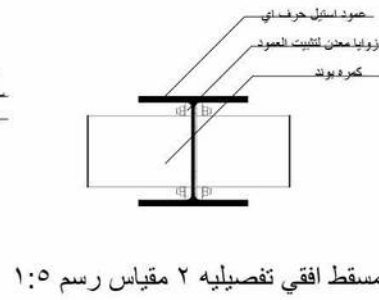
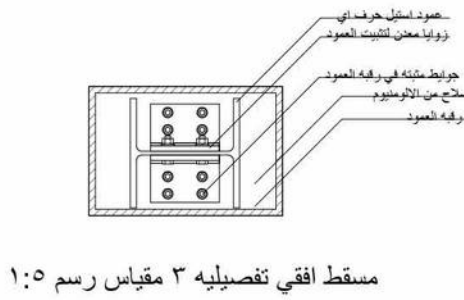
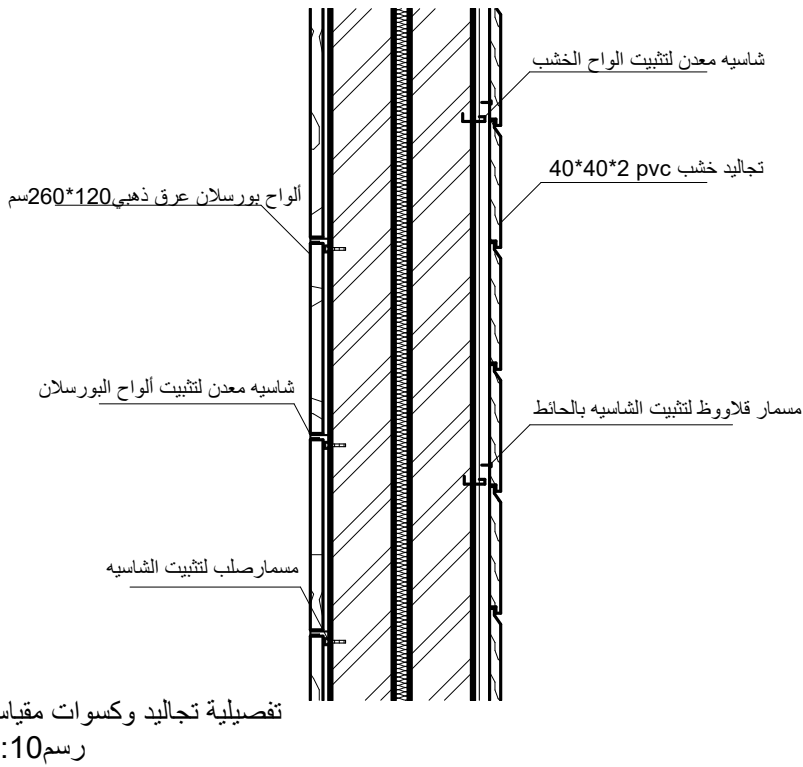
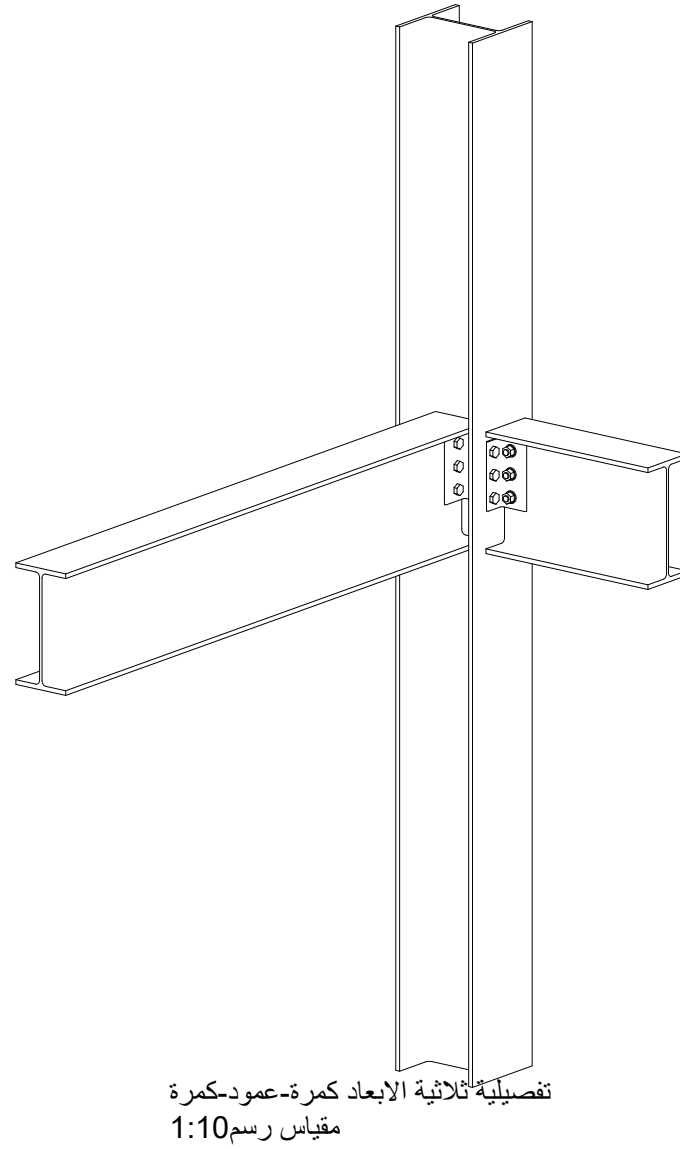
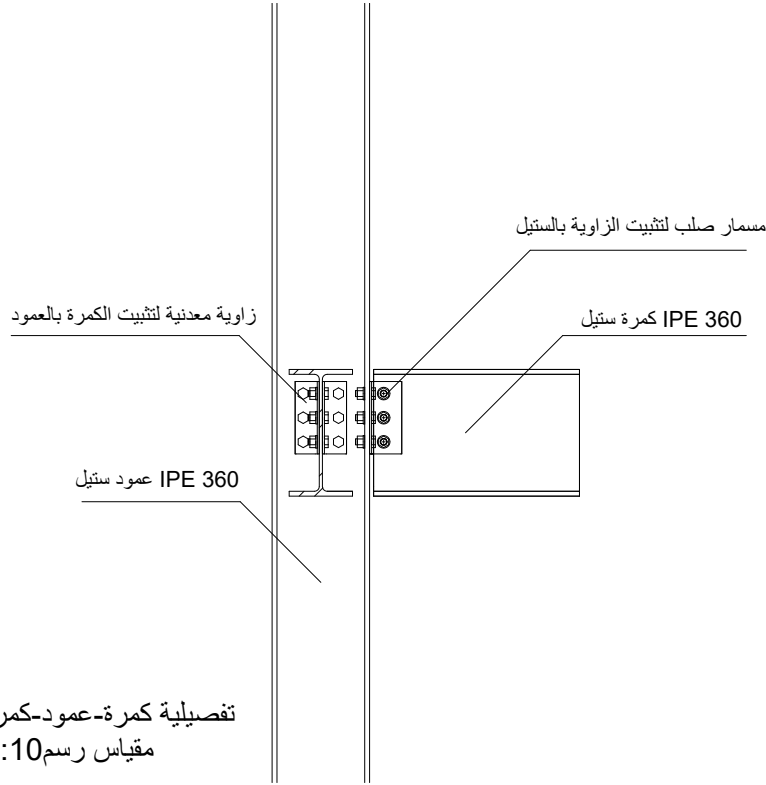
A-302

DRAWN BY

M. Talaat

DISCIPLINE

Architectural



MAHMOUD TALAAT

ARCHITECTURE | BIM

MULTIPURPOSE HALL

Architectural, Structural & MEP Working Drawings

SHEET TITLE

STEEL CONNECTION & CLADDING DETAILS

Beam-column joints and wall cladding

SCALE

1:25, 1:10

DATE

2022

DRAWN BY

M. Talaat

DISCIPLINE

Structural

SHEET NO.

S-301

W3

CAR SHOWROOM

Executive Working Drawings

TYPE	Car showroom with steel roof structure
LEVELS	Ground + first floor
LOCATION	Asyut, Egypt
FOCUS	Steel, glazing & ceiling details
YEAR	2023

PLANS

A01	Ground Floor Plan	32
A02	First Floor Plan	33

SECTIONS & ELEVATIONS

A03	Section A-A	34
A04	Exterior Elevations	35
A05	Interior Elevations	36

INTERIORS

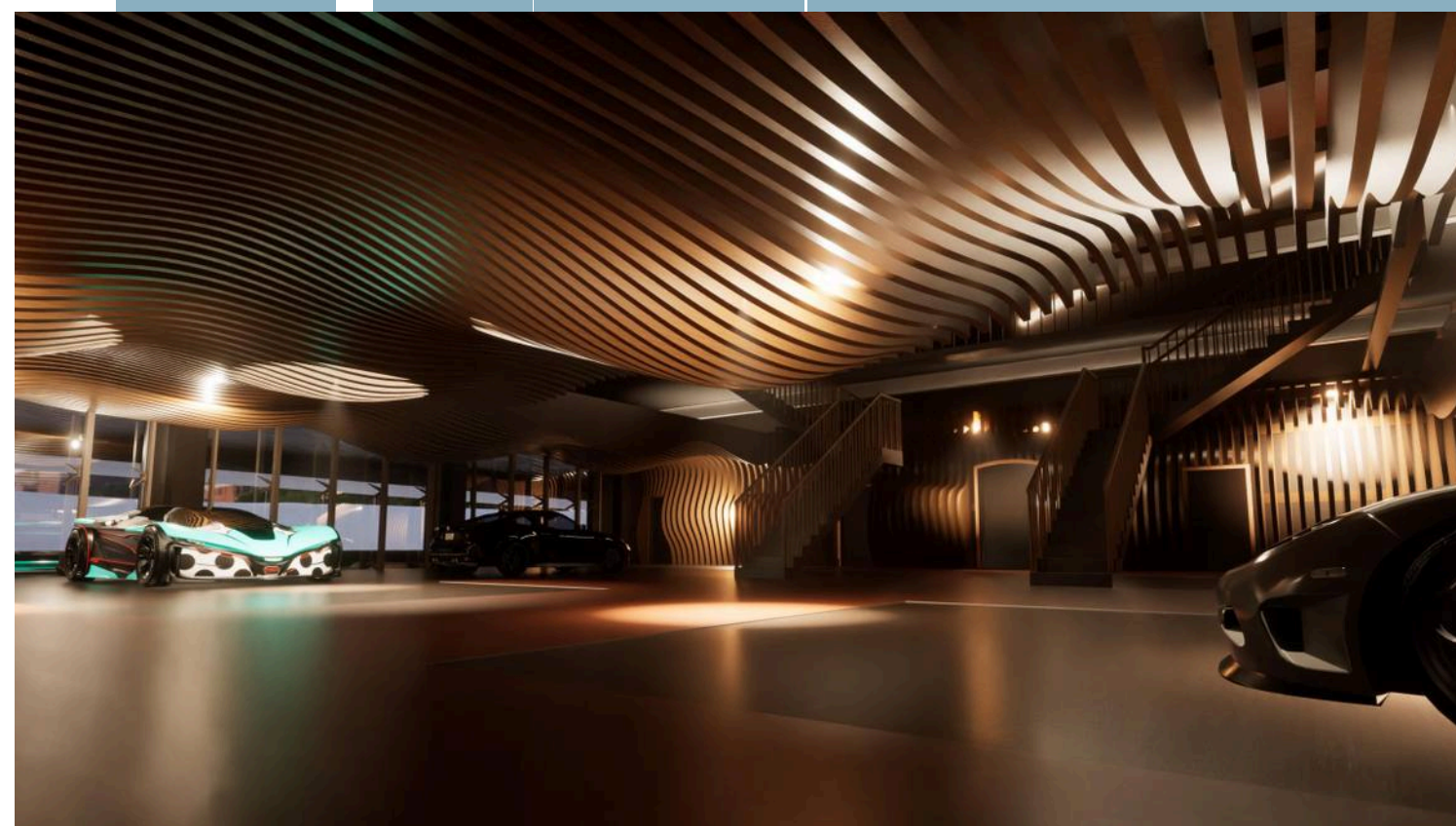
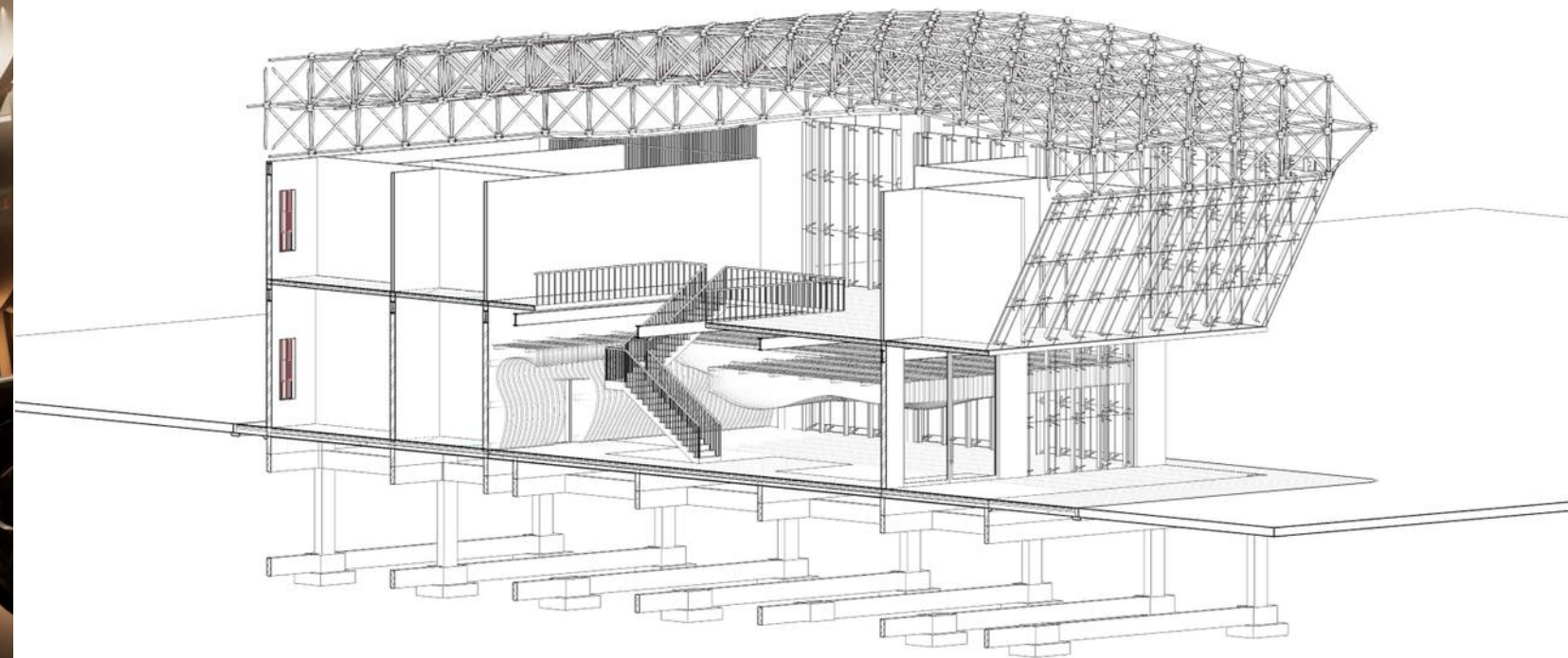
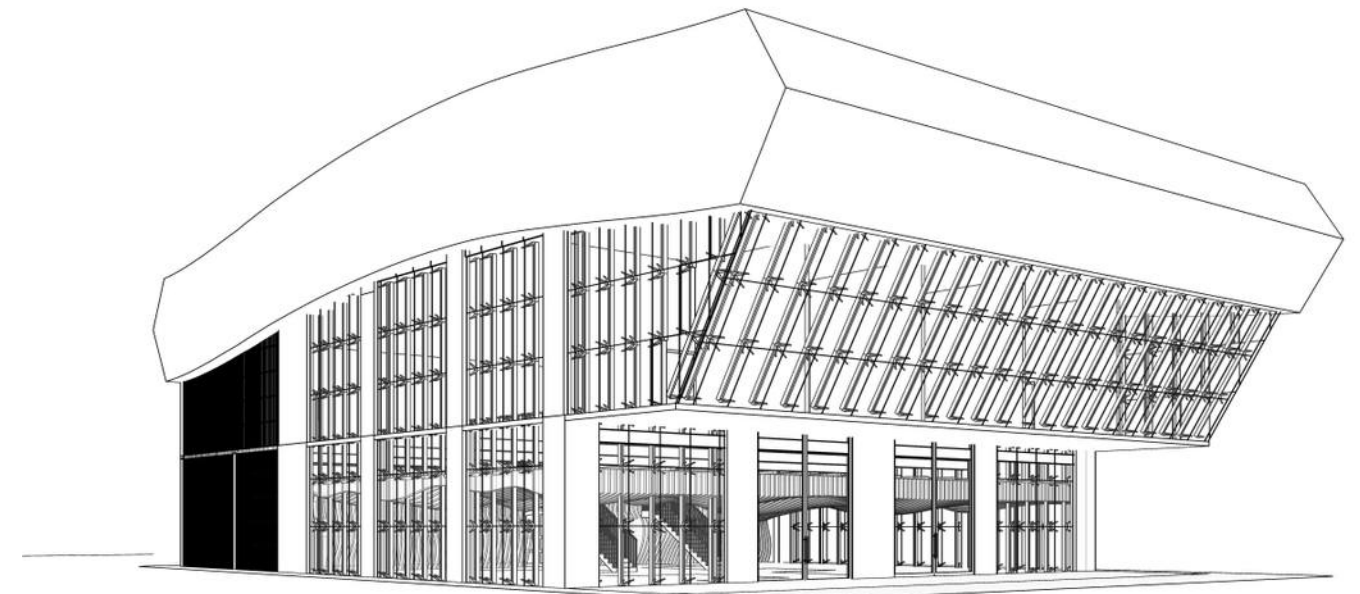
A06	Showroom Ceiling	37
A07	Showroom Flooring	38

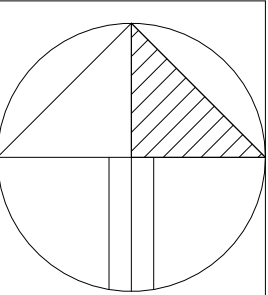
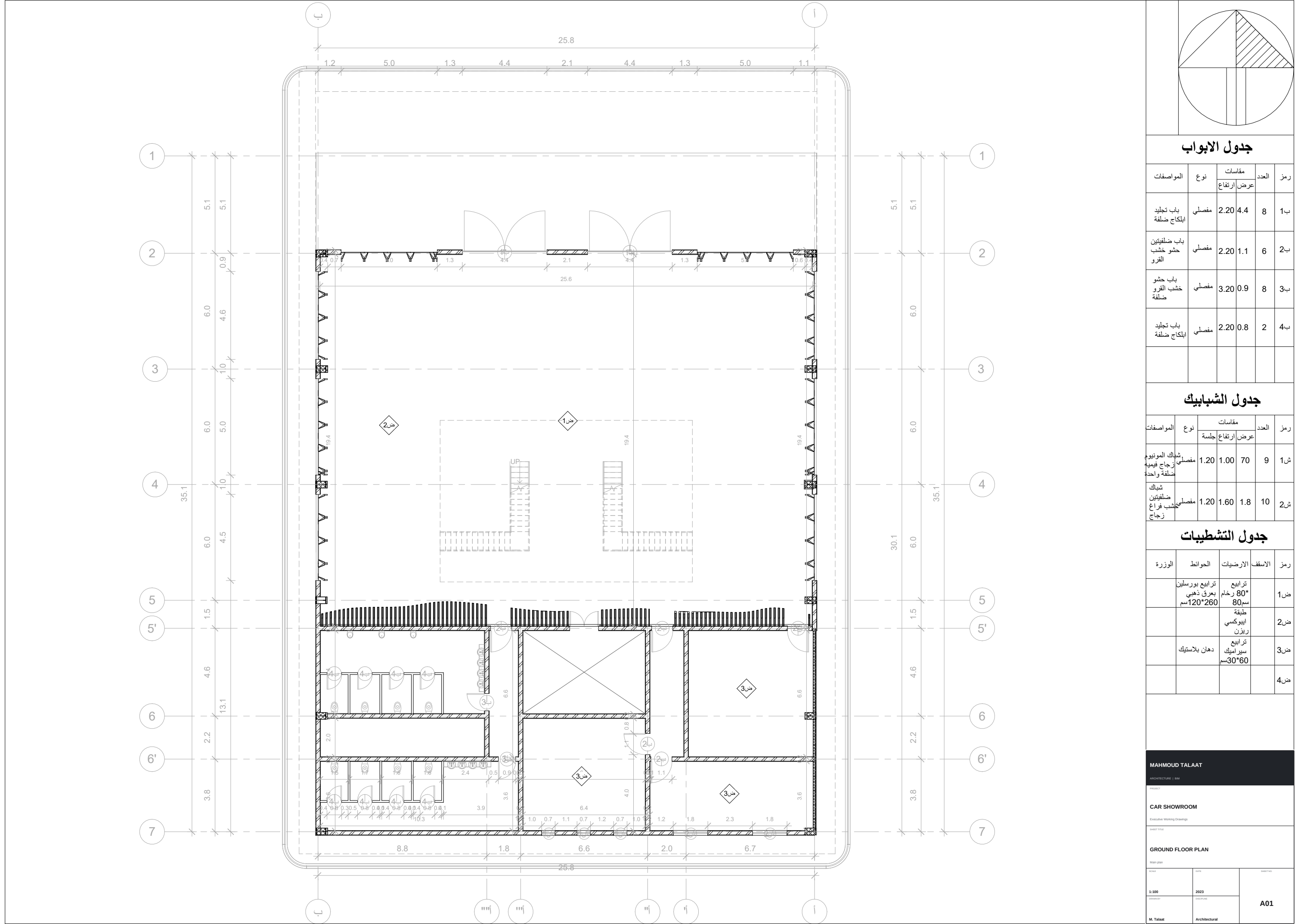
DETAILS

A08	Roof Truss Details	39
A09	Spider Glazing Details	40
A10	Glass Curtain Wall	41
A11	Roof & Wall Details	42

تصميم تنفيذي لمعرض سيارات الفهرس اسم اللوحة ابعاد اللوحة رقم اللوحة مقياس الرسم

1:100	A01	70*50	لوحة المسقط الأفقي الرئيسي
1:100	A02	70*50	لوحة المسقط الأفقي الفرعي
1:100	A03	70*50	لوحة السكشن
1:100	A04	70*50	لوحة الواجهات الخارجية
1:100	A05	70*50	لوحة الواجهات الداخلية
1:50	A06	70*50	لوحة السقف للمعرض
1:50	A07	70*50	لوحة الأرضية للمعرض
-----	A08	70*50	لوحة تفصيليات الجملون
-----	A09	70*50	لوحة تفصيليات الستائر العنكبوتية
-----	A10	70*50	لوحة تفصيليات الستائر الزجاجية
-----	A11	70*50	لوحة التفاصيل للسقف والحائط





جدول الابواب

رمز	العدد	مقاسات		نوع	المواصفات
		عرض	ارتفاع		
ب1	8	4.4	2.20	مفصلي	باب تجليد اللكاج ضلفة
ب2	6	1.1	2.20	مفصلي	باب ضلقتين حشو خشب القرو
ب3	8	0.9	3.20	مفصلي	باب حشو خشب القرو ضلفة
ب4	2	0.8	2.20	مفصلي	باب تجليد اللكاج ضلفة

جدول الشبايك

رمز	العدد	مقاسات		نوع	المواصفات
		عرض	ارتفاع		
ش1	9	70	1.00	مفصلي	شباك المونيوم زجاج فيمييه ضلفة واحدة
ش2	10	1.8	1.60	مفصلي	شباك ضلقتين خشب فراغ زجاج

جدول التشطيبات

رمز	الاسقف	الارضيات	الحوائط	الوزرة
ض1		ترابيع 80*80 رخام سم80	ترابيع بورسليين يعرق ذهبي سم120*260	
ض2		طبقة ايوكسي ريزن		
ض3		ترابيع سيراميك سم30*60	دهان بلاستيك	
ض4				

MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

CAR SHOWROOM

Executive Working Drawings

SHEET TITLE

GROUND FLOOR PLAN

Main plan

SCALE

1:100

DRAWN BY

M. Talaat

DATE

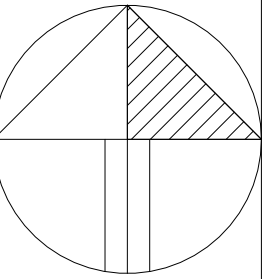
2023

DESIGNED BY

Architectural

SHEET NO.

A01



جدول الابواب

رمز	العدد	مقاييس		نوع	المواصفات
		عرض	ارتفاع		
ب1	8	4.4	2.20	مفصلي	باب تجليد إلکاج ضلفة
ب2	6	1.1	2.20	مفصلي	باب ضلقتين حشو خشب القرو
ب3	8	0.9	3.20	مفصلي	باب حشو خشب القرو ضلفة
ب4	2	0.8	2.20	مفصلي	باب تجليد إلکاج ضلفة

جدول الشبابيك

رمز	العدد	مقاييس		نوع	المواصفات
		عرض	ارتفاع		
ش1	9	70	1.00	1.20	مفصلي شباك زجاج قميبي ضلفة واحد
ش2	10	1.8	1.60	1.20	مفصلي شباك زجاج قميبي ضلفة واحد

جدول التشطيبات

الاسقف	الارضيات	الحوائط	الوزرة
ض1	تر ابيع 80* سم80	تر ابيع بورسلين يعرق ذهبي 260*120سم	
ض2	طيفة ابيوكسي ريزن		
ض3	تر ابيع سيراميك 30*60سم	دهان بلاستيك	
ض4			

MAHMOUD TALAAT

PROJECT

CAR SHOWROOM

Executive Working Drawings

SHEET TITLE

FIRST FLOOR PLAN

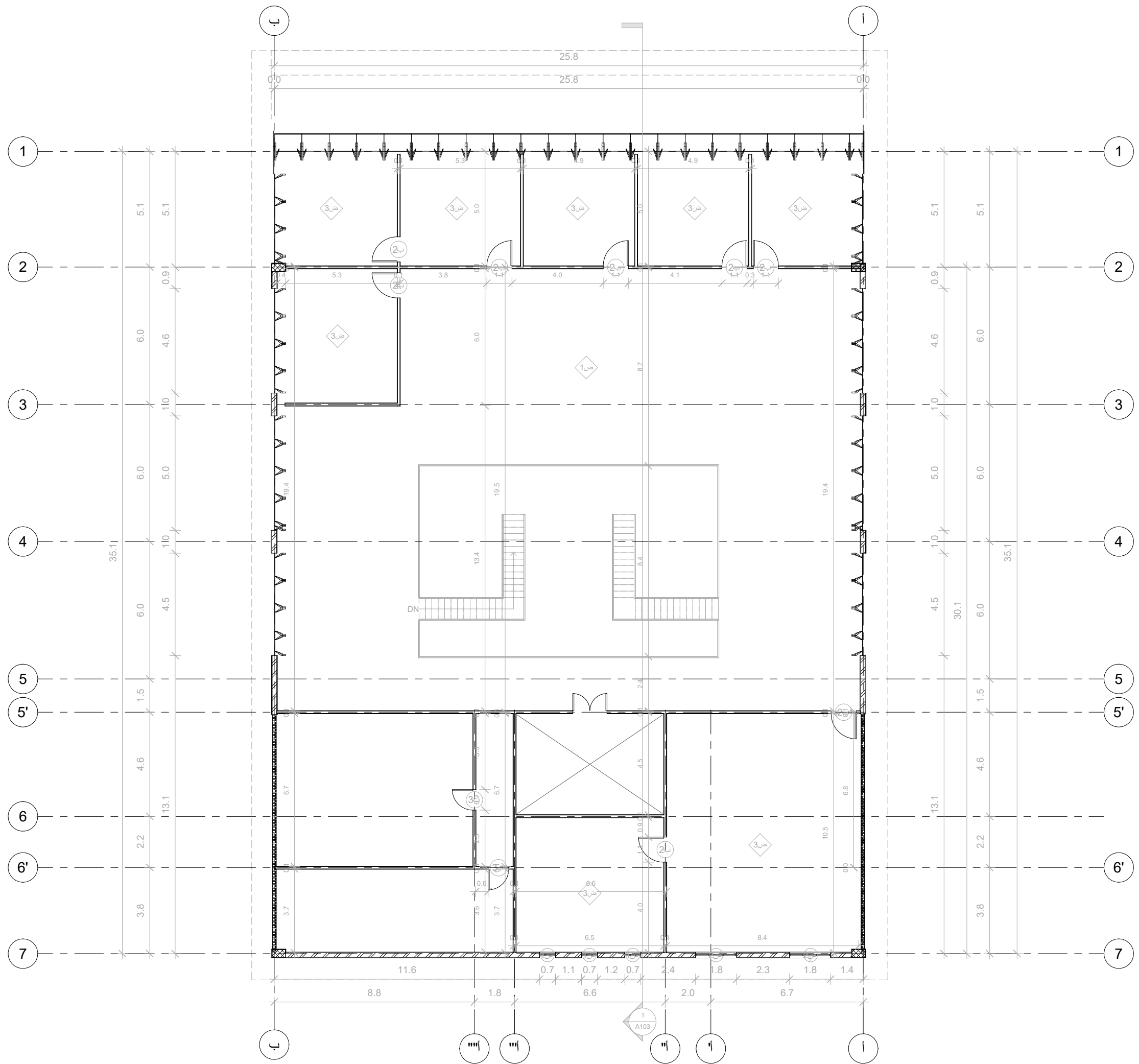
Secondary plan

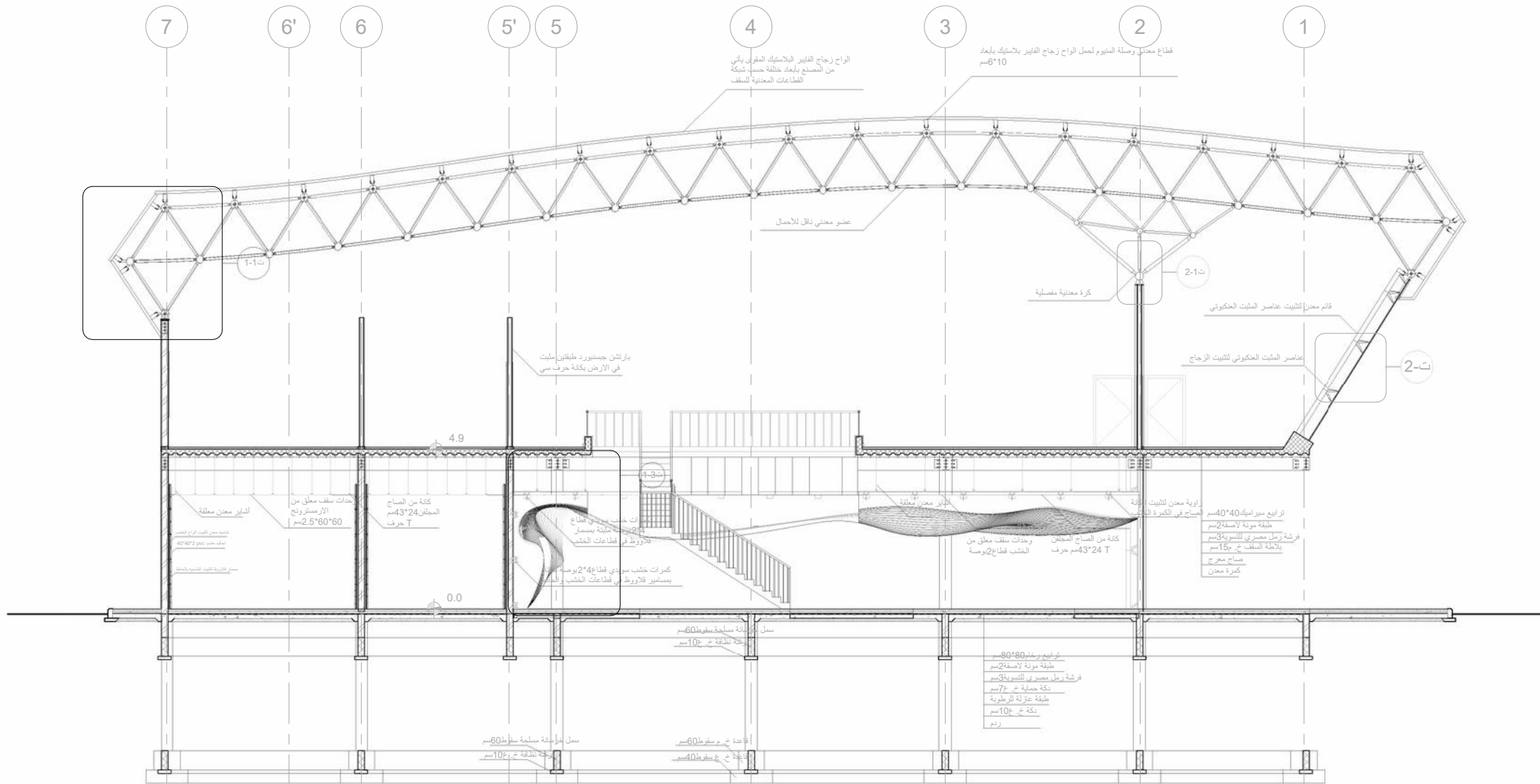
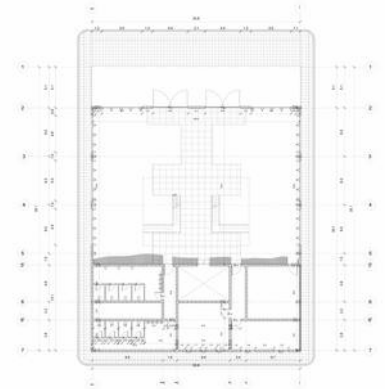
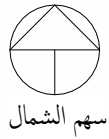
SCALE

1:100

DRAWING

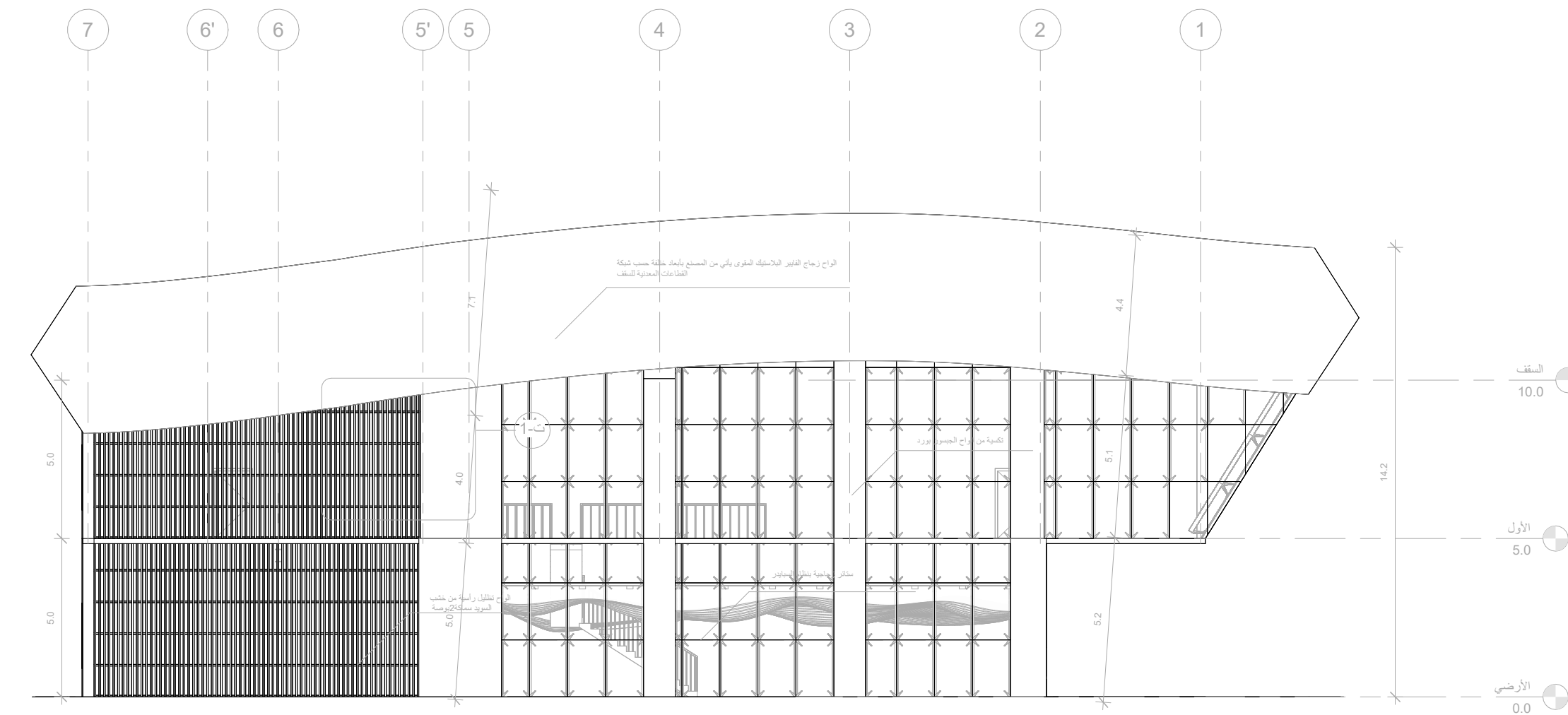
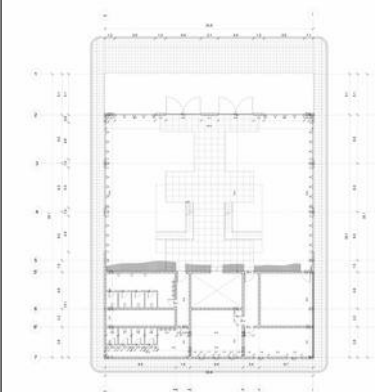
A02



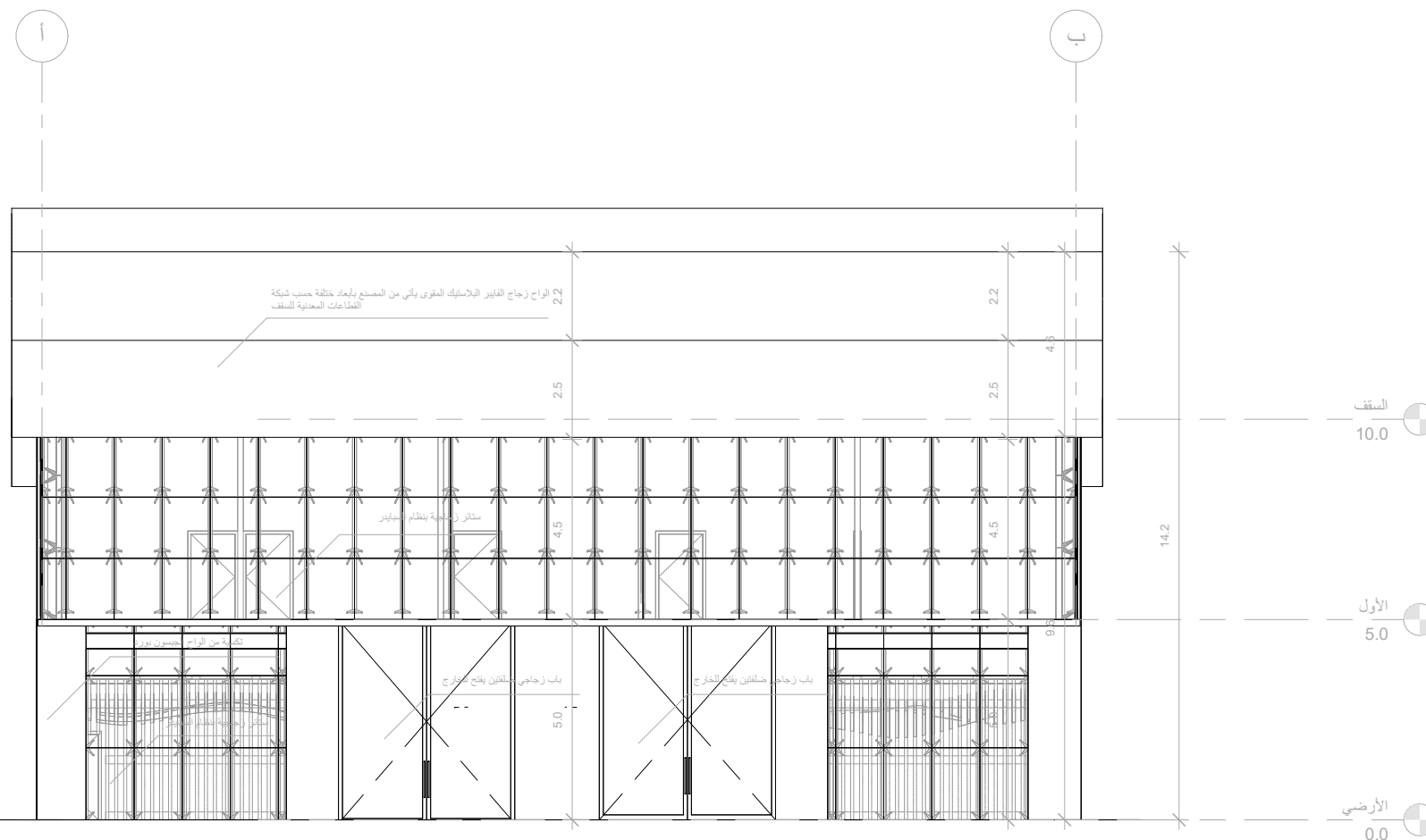


MAHMOUD TALAAT		
ARCHITECTURE BIM		
PROJECT		
CAR SHOWROOM		
Executive Working Drawings		
SHEET TITLE		
SECTION A-A		
Vertical section		
SCALE	DATE	SHEET NO.
1:100	2023	
DRAWN BY	DISCIPLINE	
M. Talaat	Architectural	

A03

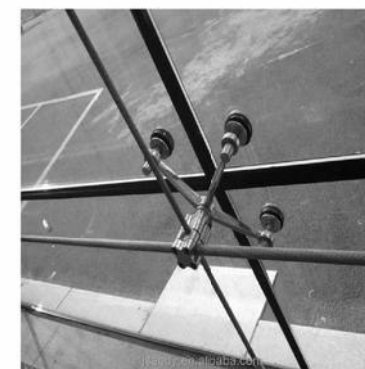
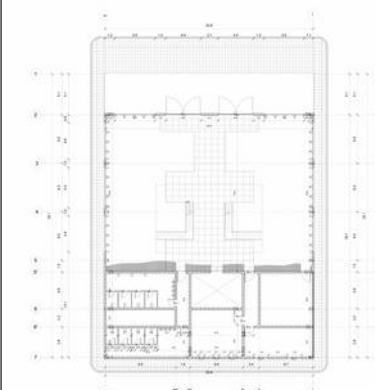


الواجهة الشرقية
مقياس رسم 1:100



الواجهة الشمالية
مقياس رسم 1:100

MAHMOUD TALAAT		
ARCHITECTURE BIM		
PROJECT		
CAR SHOWROOM		
Executive Working Drawings		
SHEET TITLE		
EXTERIOR ELEVATIONS		
SCALE	DATE	SHEET NO.
1:100	2023	A04
DRAWN BY	DISCIPLINE	
M. Talaat	Architectural	



MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

CAR SHOWROOM

Executive Working Drawings

SHEET TITLE

INTERIOR ELEVATIONS

SCALE

1:100

DRAWN BY

M. Talaat

DATE

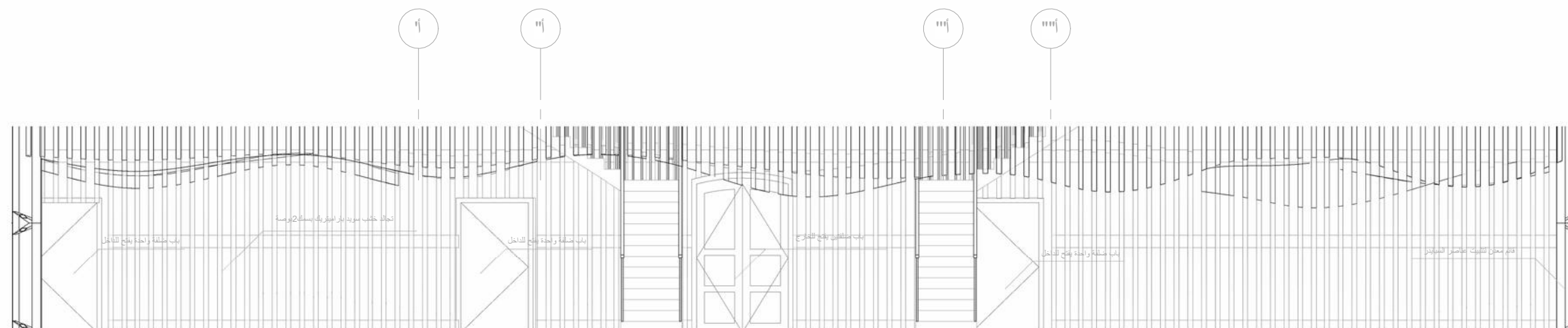
2023

DISCIPLINE

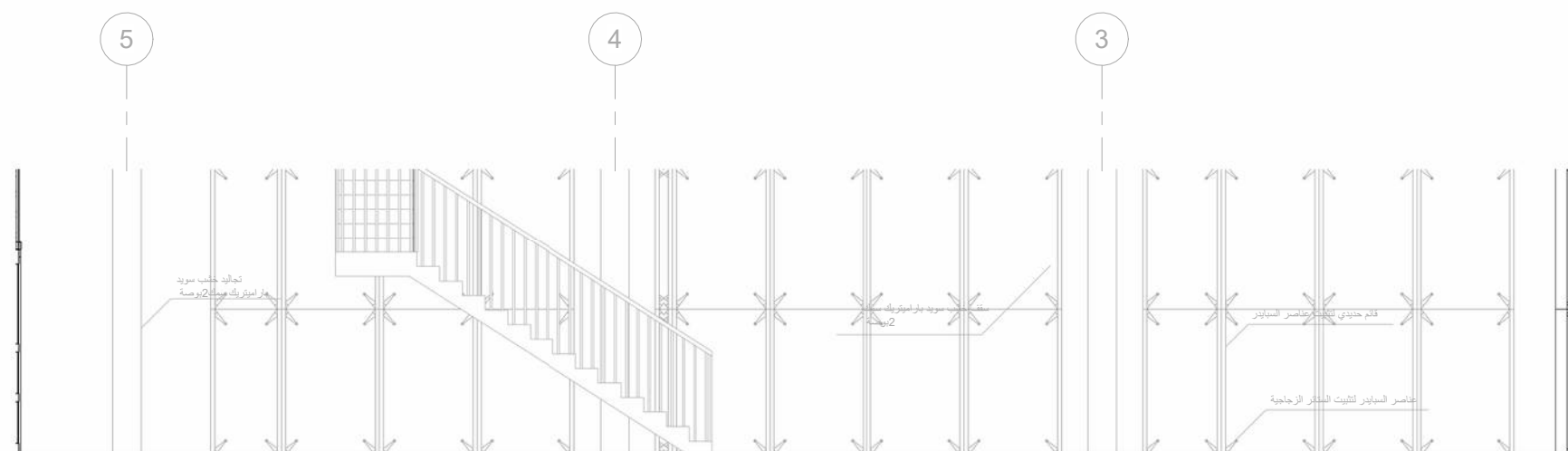
Architectural

SHEET NO.

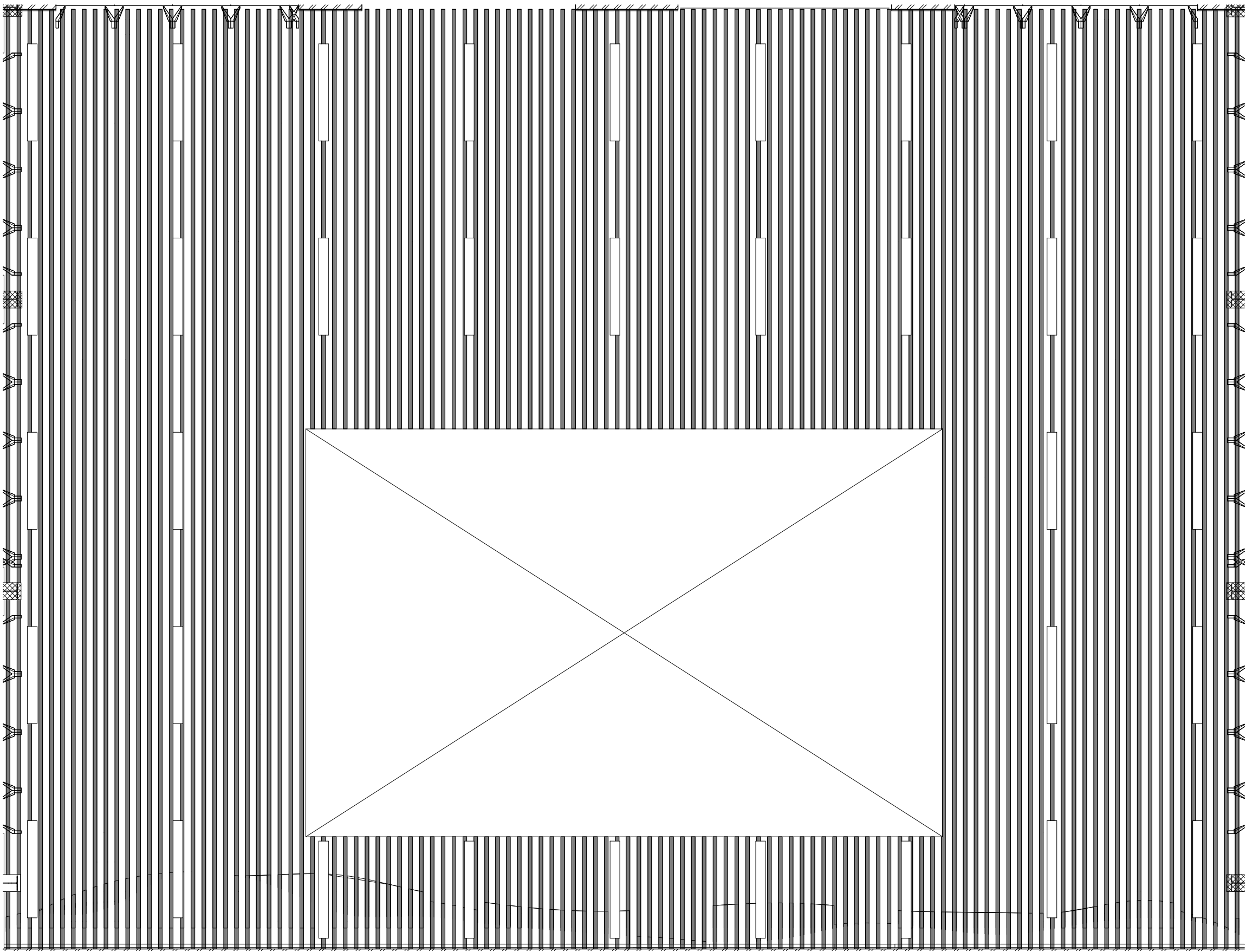
A05



الواجهة الشمالية الداخلية
1:50



الواجهة الشرقية الداخلية
1:50



الرمز	التوصيف
	شريط اضاءة ليد

نموذج



MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

CAR SHOWROOM

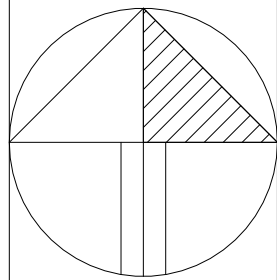
Executive Working Drawings

SHEET TITLE

SHOWROOM CEILING

Display area ceiling

SCALE	DATE	SHEET NO. A06
1:50	2023	
DRAWN BY	DISCIPLINE	
M. Talaat	Architectural	



جدول الابواب

رمز	العدد	مقاسات		نوع	المواصفات
		عرض	ارتفاع		
ب1	8	4.4	2.20	مفصلي	باب تجليد ايلكاج ضلفة
ب2	6	1.1	2.20	مفصلي	باب ضلقتين حشو خشب القرو
ب3	8	0.9	3.20	مفصلي	باب حشو خشب القرو ضلفة
ب4	2	0.8	2.20	مفصلي	باب تجليد ايلكاج ضلفة

جدول الشبايك

رمز	العدد	مقاسات		نوع	المواصفات
		عرض	ارتفاع		
ش1	9	70	1.00	مفصلي	شباك المونيوم ازجاج فيميه ضلفة واحدة
ش2	10	1.8	1.60	مفصلي	شباك ضلقتين خشب فراغ زجاج

جدول التشطيبات

رمز	الاسقف	الارضيات	الحوائط	الوزرة
ض1		ترابيع 80*80 رخام سم80	ترابيع بورسلين يعرق ذهبي سم120*260	
ض2		طبقة اينوكسي ريزن		
ض3		ترابيع سيراميك سم30*60	دهان بلاستيك	
ض4				

MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

CAR SHOWROOM

Executive Working Drawings

SHEET TITLE

SHOWROOM FLOORING

Flooring plan

SCALE

1:50

DRAWN BY

M. Talaat

DATE

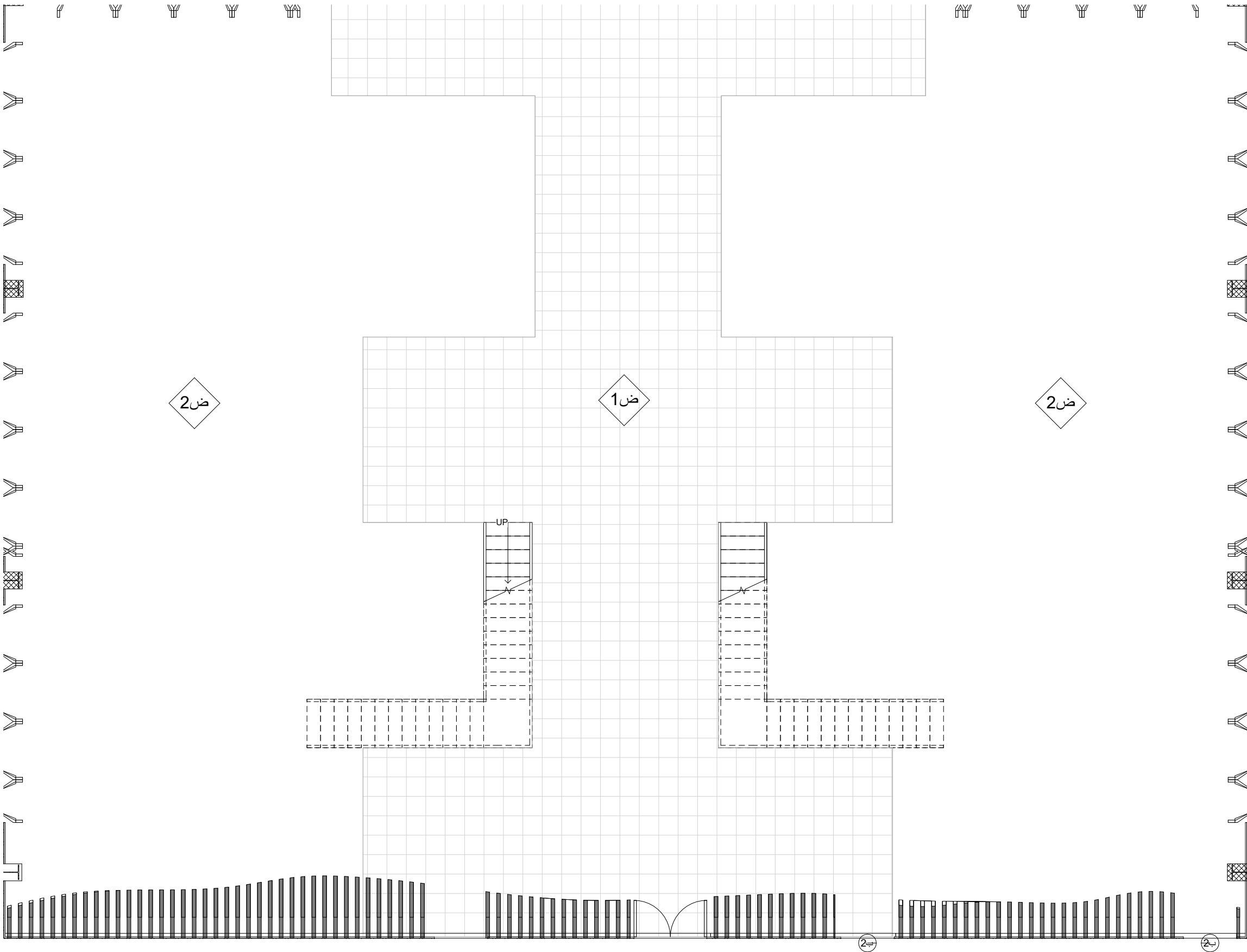
2023

CHECKED BY

Architectural

SHEET NO.

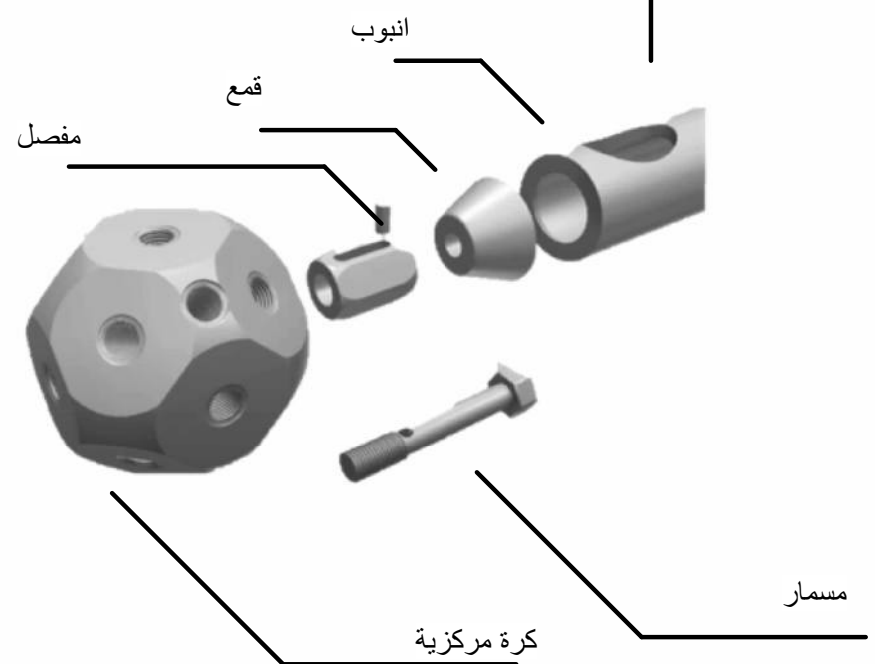
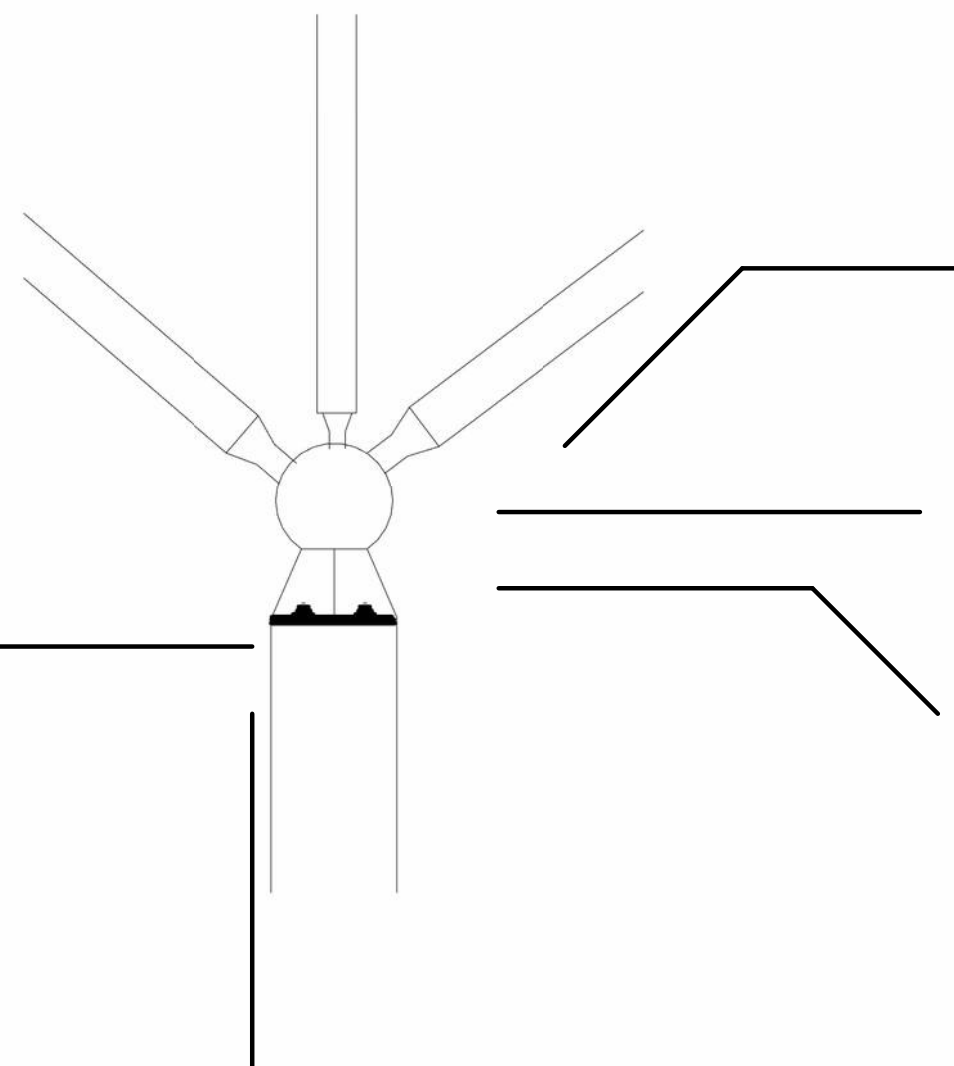
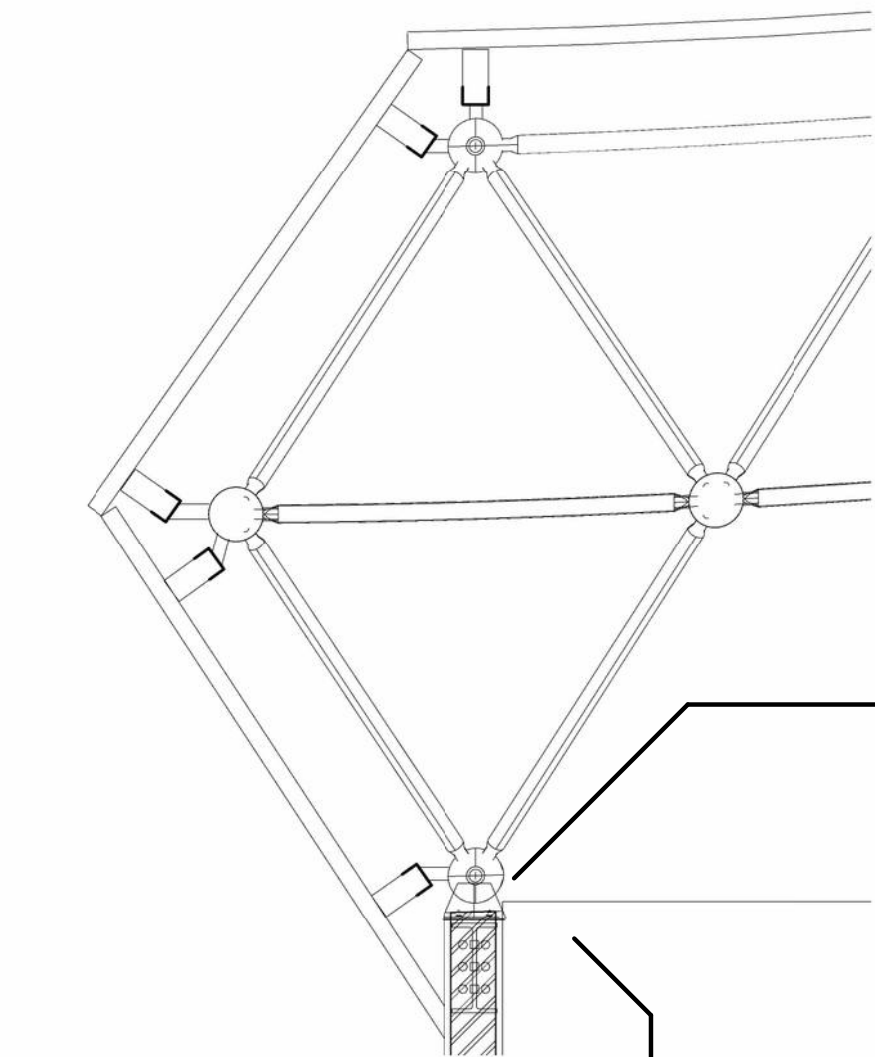
A07



بلان
مقياسه 1:50

1

تفصيلية ت 1-2 لتثبيت الجمالون بالأعمدة الخرسانية



دعامة رأسية

عضو معدني لنقل
الاحمال

مسمار صلب لتثبيت
اللوحة في العمود

قاعدة معدنية لتثبيت
الكرة المركزية

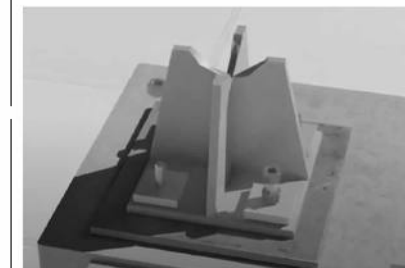
صفحة معدنية لتثبيت
القاعدة بالعمود

جوايط لتثبيت الصفحة
بالعمود الخرساني

رقبة عمود خرساني

مسمار

كرة مركزية



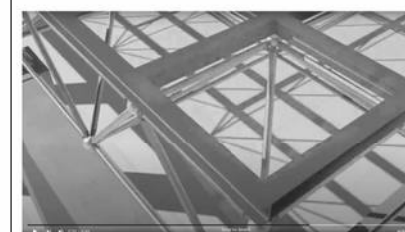
القاعدة المعدنية



الكرة المركزية



مثال لعمود شجري يدعم نظام
جمالوني



اطارات معدنية لتركيب
التجاويد على الجمالون

MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

CAR SHOWROOM

Executive Working Drawings

SHEET TITLE

ROOF TRUSS DETAILS

Truss-to-column connections

SCALE	DATE	SHEET NO.
As indicated	2023	A08
DESIGNED BY	DISCIPLINE	
M. Talaat	Architectural	



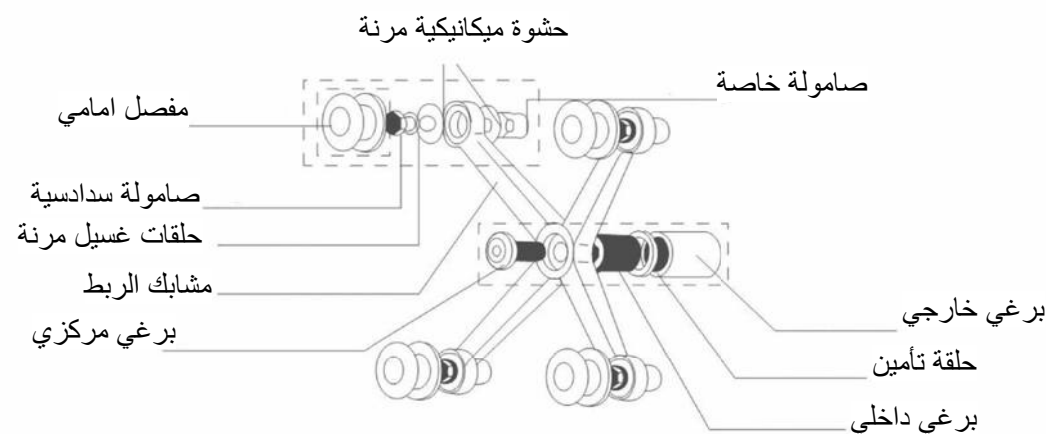
واجهة خارجية للستائر الزجاجية العنكبوتية



واجهة داخلية للستائر الزجاجية العنكبوتية

MAHMOUD TALAAT		
ARCHITECTURE BIM		
PROJECT		
CAR SHOWROOM		
Executive Working Drawings		
SHEET TITLE		
SPIDER GLAZING DETAILS		
Point-fixed glazing fittings		
SCALE	DATE	SHEET NO.
As indicated	2023	
DRAWN BY	DISCIPLINE	
M. Talaat	Architectural	

A09



مكونات العناصر العنكبوتية

أحادي

ثنائي

ثلاثي

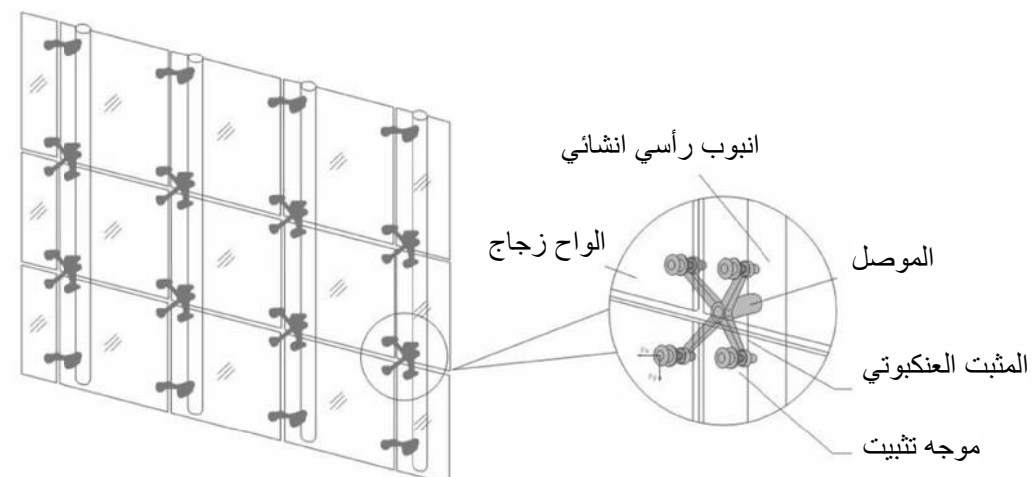
رباعي

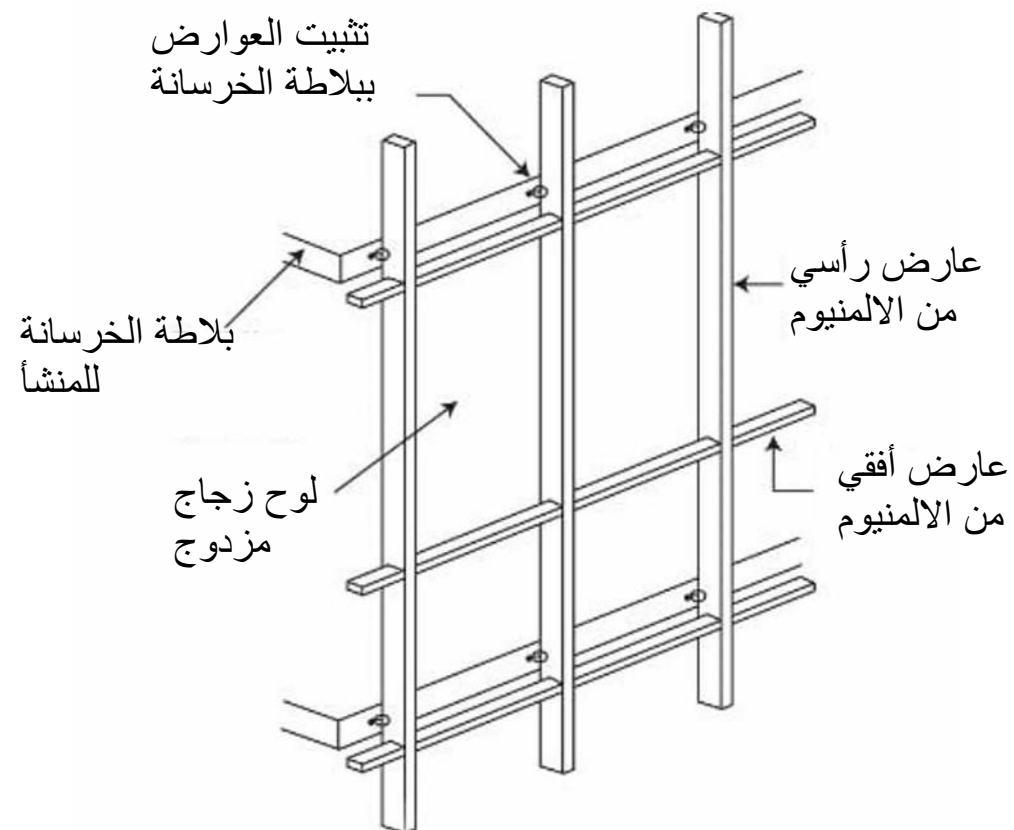
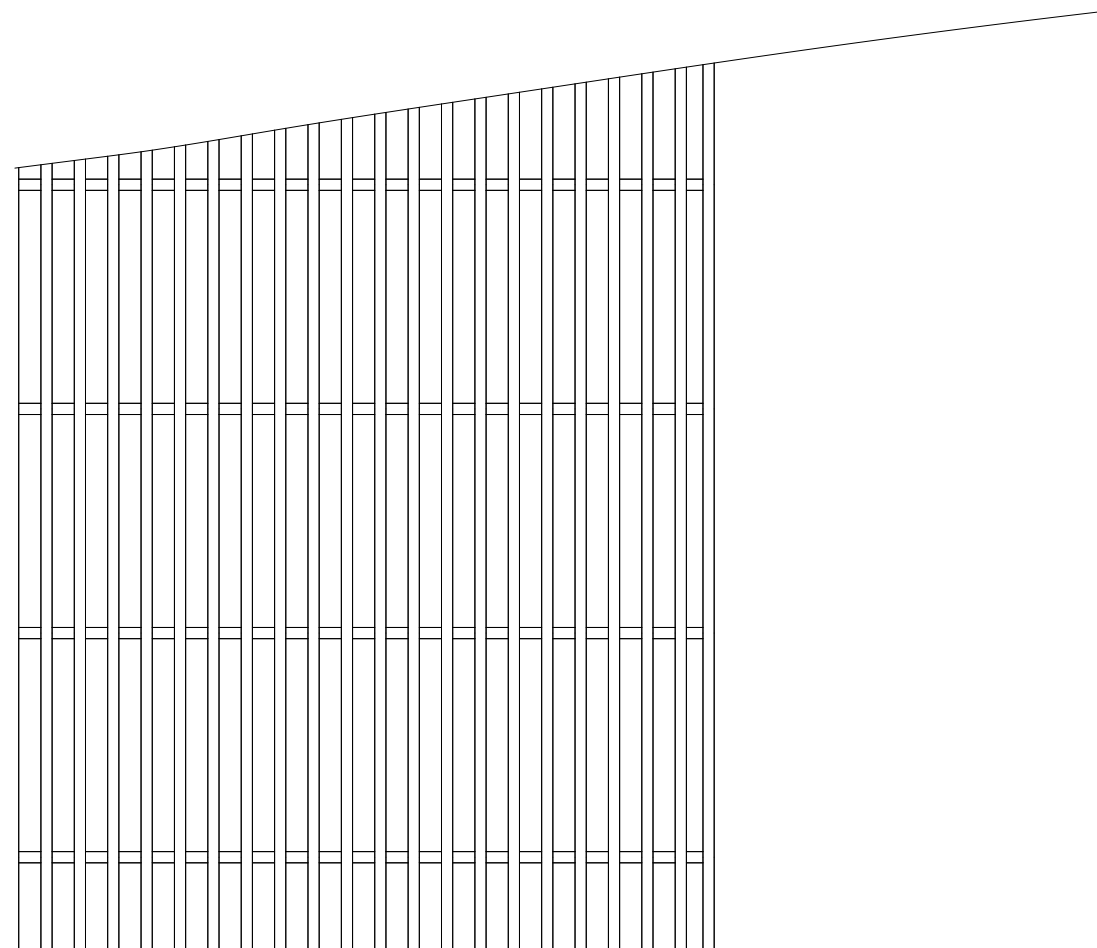


أنواع مفاصل المثبت العنكبوتي

تفصيلية ستائر عنكبوتية

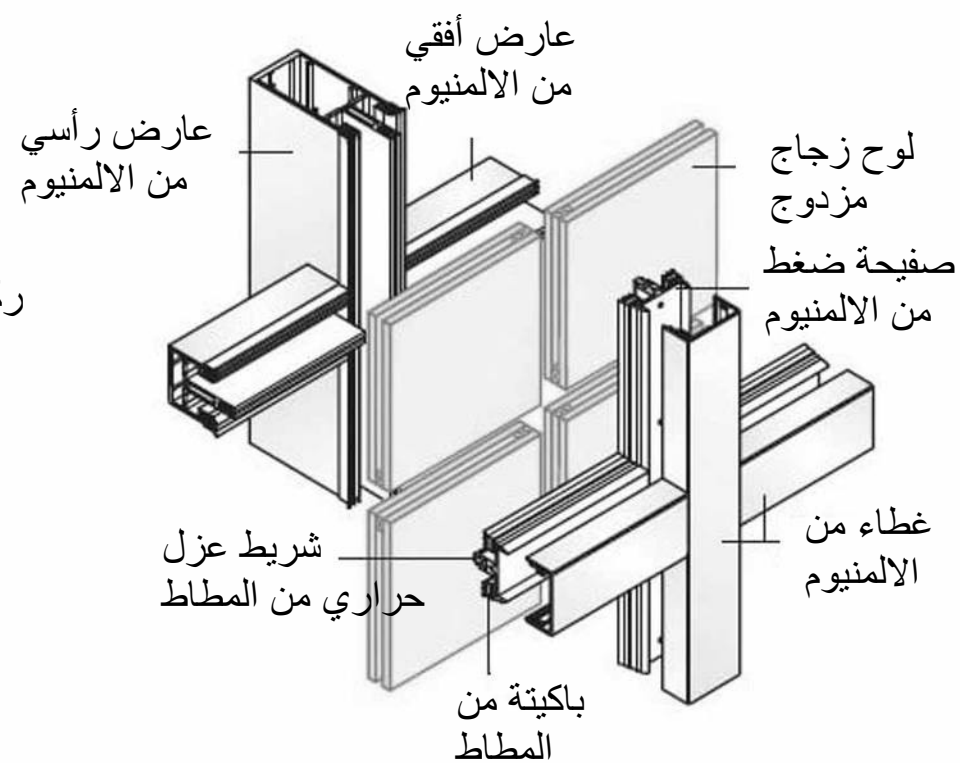
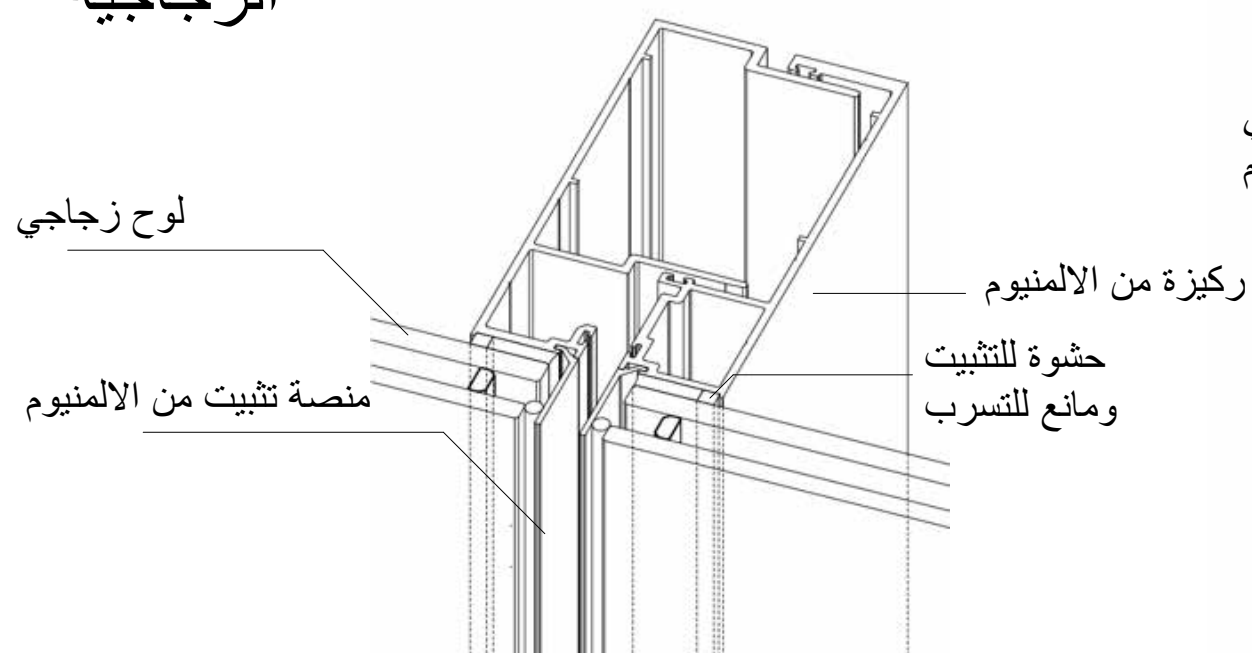
مكونات حائط ستائري عنكبوتي





مثال لواجهة من الستائر الزجاجية

تفصيلية ت-3 الستائر الزجاجية



MAHMOUD TALAAT

ARCHITECTURE | BIM

PROJECT

CAR SHOWROOM

Executive Working Drawings

SHEET TITLE

GLASS CURTAIN WALL

Aluminium mullions & glazing

SCALE

As indicated

DATE

2023

DRAWN BY

M. Talaat

DISCIPLINE

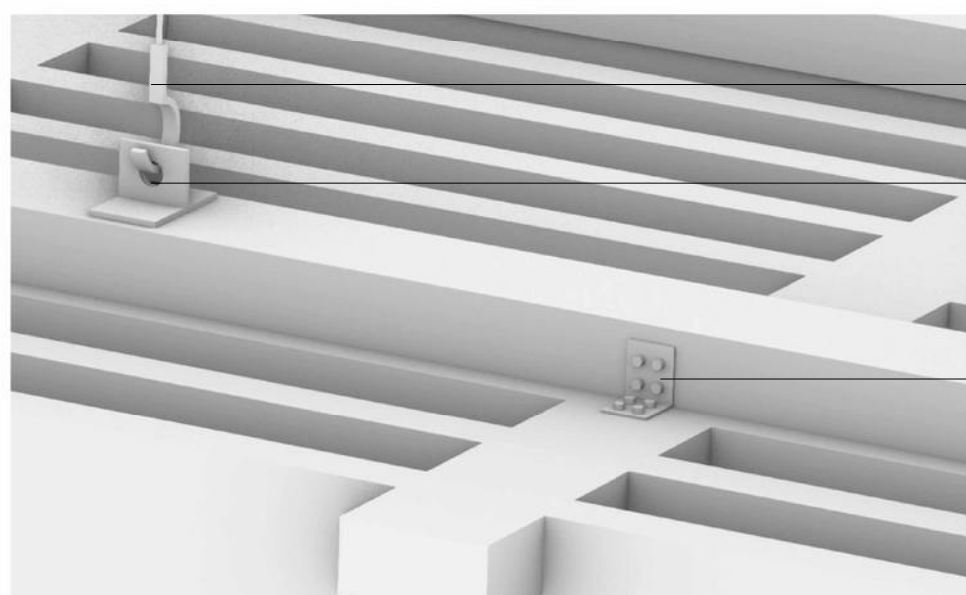
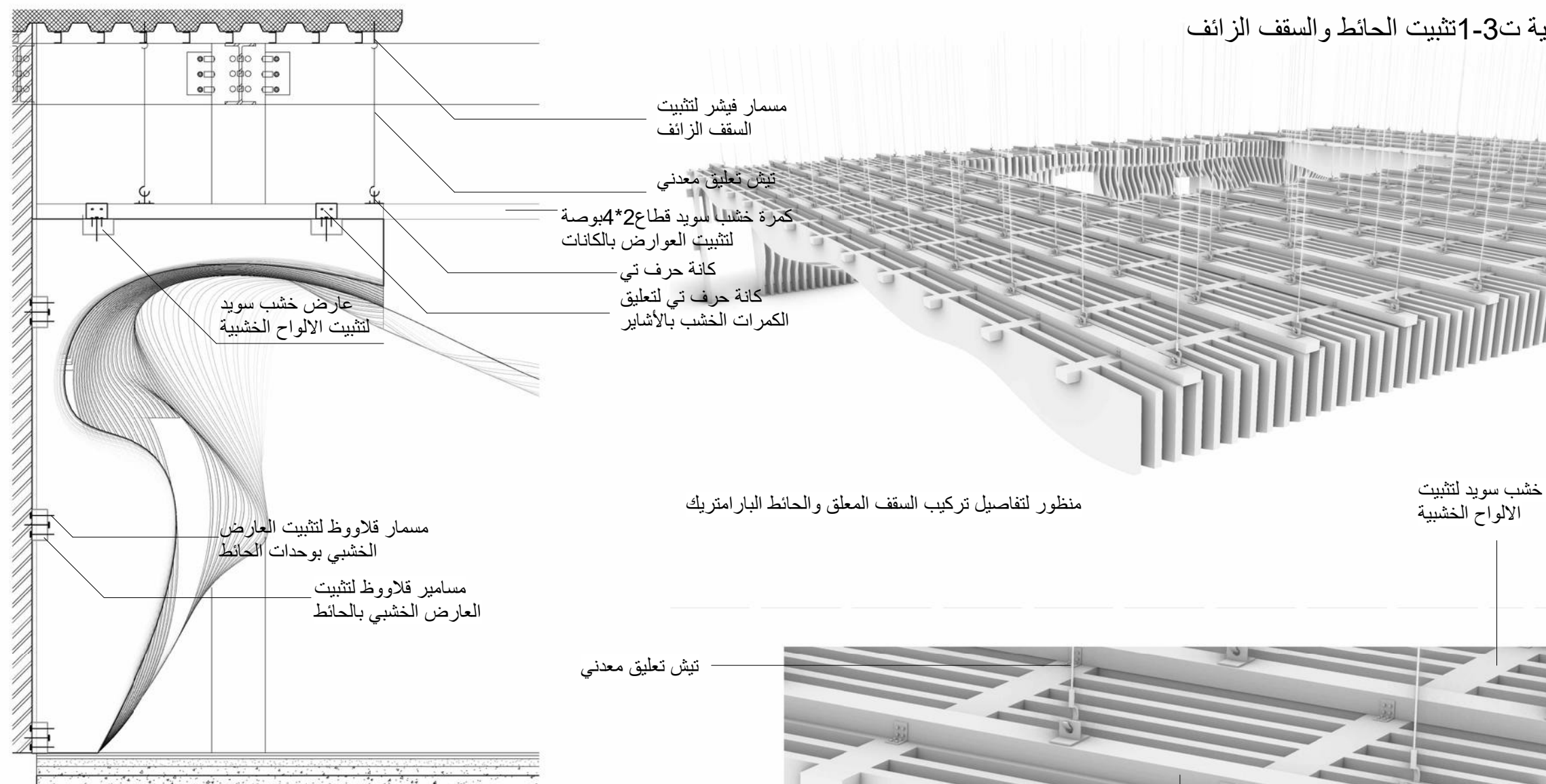
Architectural

SHEET NO.

A10

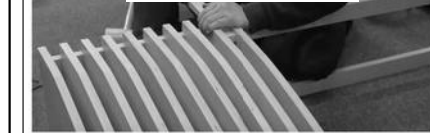


تفصيلية ت3-1 تثبيت الحائط والسقف الزائف



وحدات الواح السقف المعلق من الخشب السويد قطاع 2 بوصة مثبت بمسمار قلاووظ بالعوارض الخشبية

كمرة خشب سويد قطاع 2*4 بوصة لتثبيت العوارض بالكانات



MAHMOUD TALAAT		
ARCHITECTURE BIM		
PROJECT		
CAR SHOWROOM		
Executive Working Drawings		
SHEET TITLE		
ROOF & WALL DETAILS		
Wall & false ceiling fixing		
SCALE	DATE	SHEET NO.
As indicated	2023	A11
DRAWN BY	DISCIPLINE	
M. Talaat	Architectural	

02

BIM AUTOMATION & FAMILIES

Custom tools that remove repetitive modelling work

FOCUS	Repetitive, error-prone modelling tasks in Revit
TOOLS	2 Dynamo workflows · Revit plugin · 1 parametric family
BUILT WITH	Revit · Dynamo · C# Revit API · Revit family editor
INPUTS	Linked models · slab geometry · CAD drawings · named parameters
RESULT	Native Revit elements: cut walls, hosted families, typed beams

DYNAMO SCRIPTS

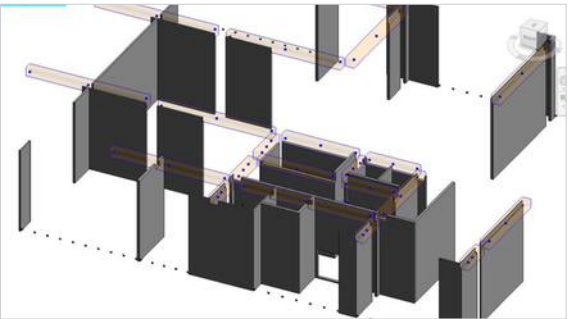
A1	Clash-Driven Wall Openings	44 – 45
A2	Face-Hosted Family Placement	46 – 48

REVIT PLUGIN

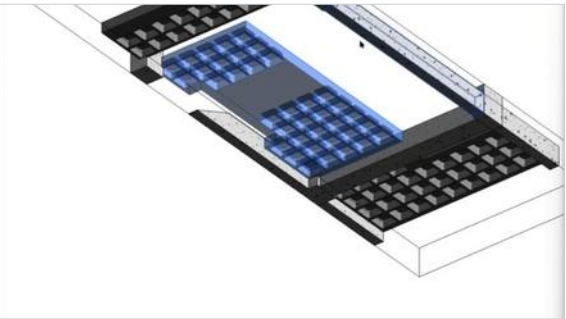
A3	Beam Automation Pipeline	49 – 50
----	--------------------------	---------

REVIT FAMILY

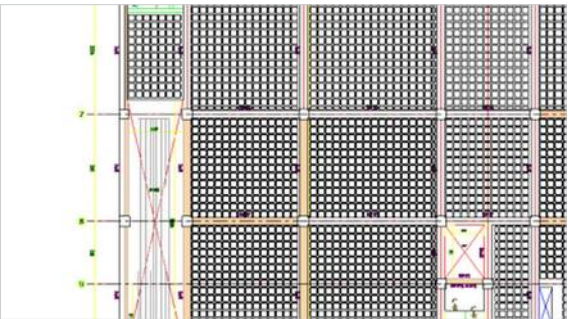
A4	Parametric Door Family	51
----	------------------------	----



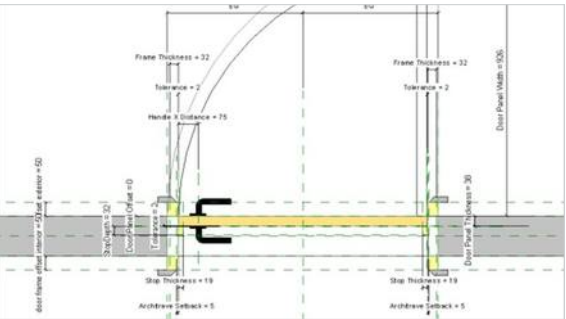
A1 Wall openings across a linked model



A2 Families placed on a slab face



A3 Beams modelled from a CAD drawing



A4 One door driven by 16 parameters

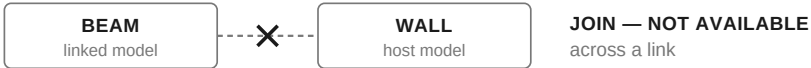
A1

CLASH-DRIVEN WALL OPENINGS

Dynamo for Revit — linked-model clashes, shown on structural beams

THE PROBLEM

Revit cannot join elements of a linked model to elements of the host model. A beam in the linked structural file cannot be joined to, or cut out of, the walls it crosses — the usual manual fix has no equivalent across a link.



THE SOLUTION

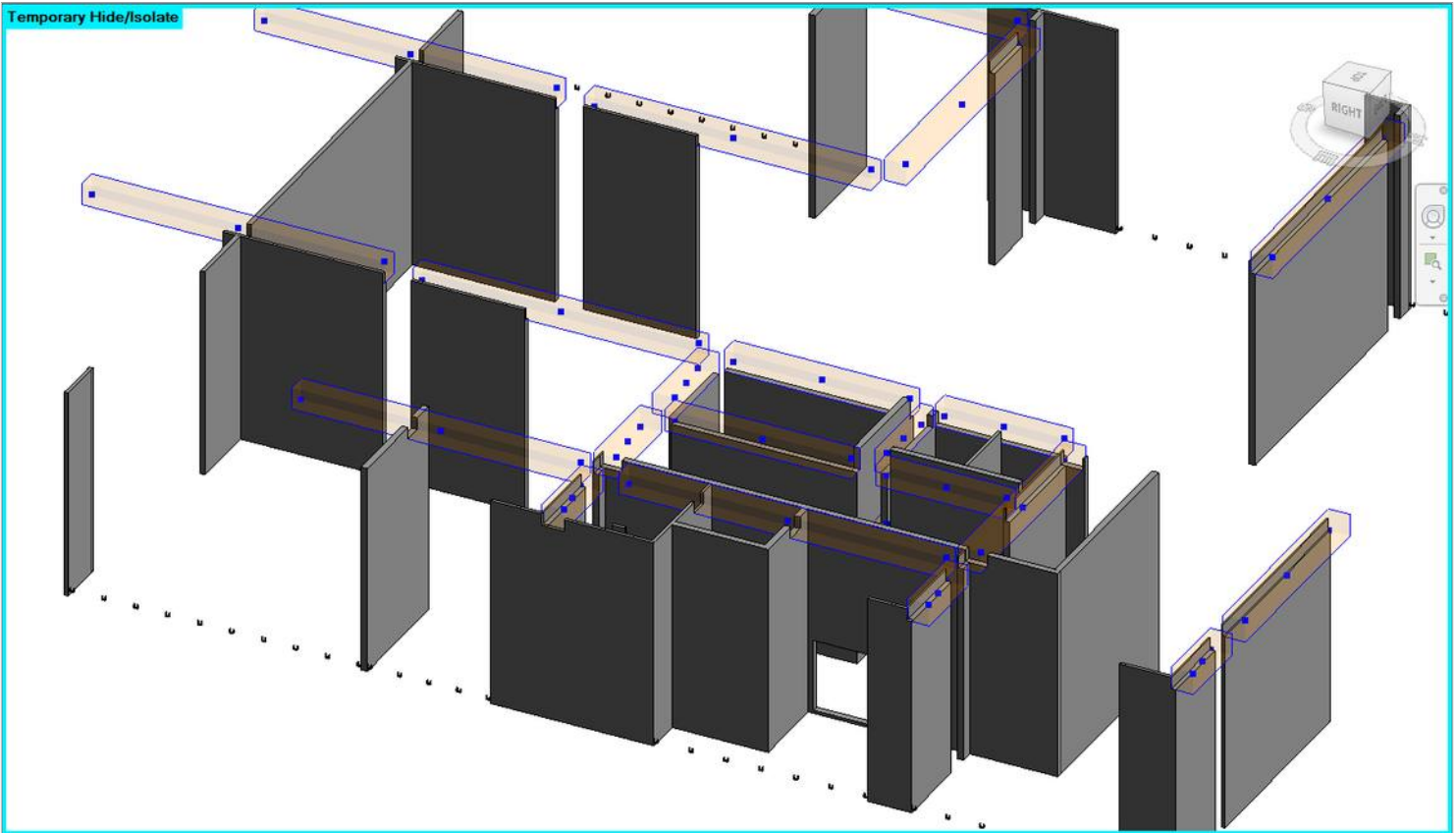
A Dynamo script finds every beam–wall intersection on a chosen level and places one parametric void family per unique beam. Each void is sized from its beam and cuts every wall the beam crosses, so a whole level is resolved in one run.



ACCURATE QUANTITIES

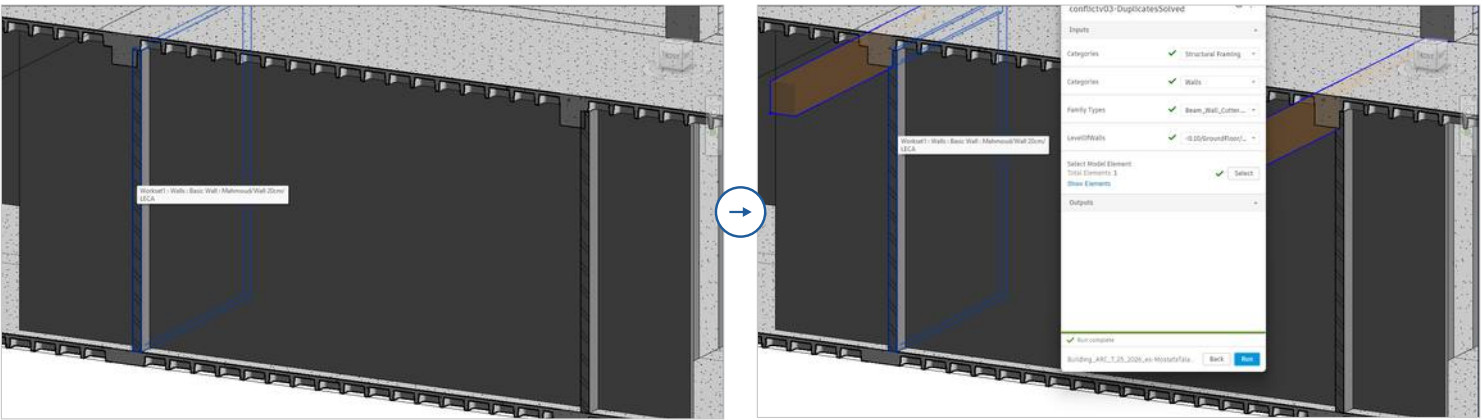
The walls are actually cut by the voids, so material quantities, schedules and takeoffs come out right, not just how the model looks.

TYPE	Dynamo script, run from Dynamo Player
OUTPUT	One void per unique beam, cutting every wall it crosses
SOFTWARE	Autodesk Revit · Dynamo with Python nodes



RESULT — OPENINGS ACROSS THE SELECTED LEVEL

Void families and the walls they cut, isolated from the rest of the model



BEFORE — LINKED STRUCTURAL MODEL

Wall–beam intersection in the linked model

AFTER — RUN COMPLETE

Resulting opening in the wall

DYNAMO WORKFLOW

Five stages, read left to right across the graph. Every group is named for what it does.

- 1

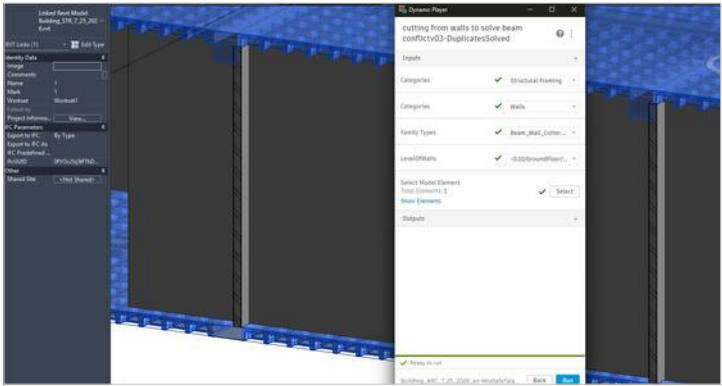
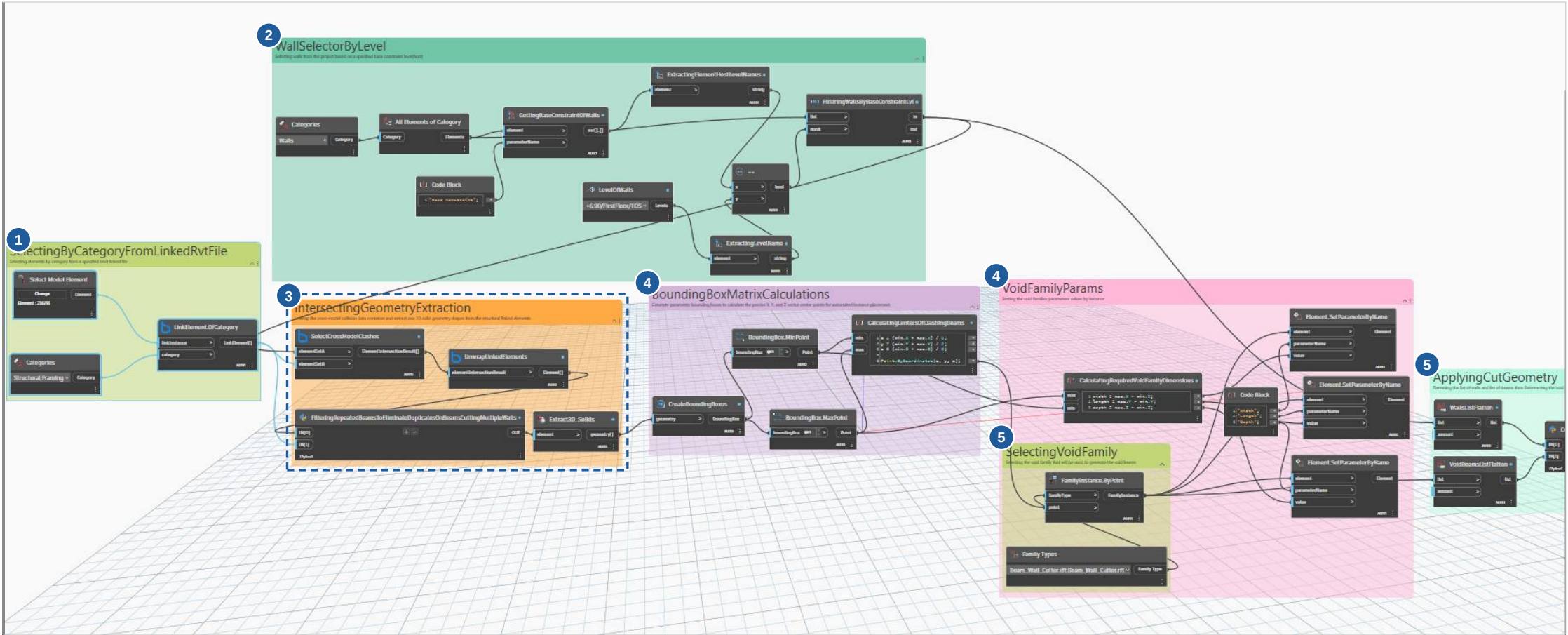
LINKED BEAMS
Pick the link instance and read every Structural Framing element from the linked file.
- 2

WALLS BY LEVEL
Host walls are selected by their base constraint level.
- 3

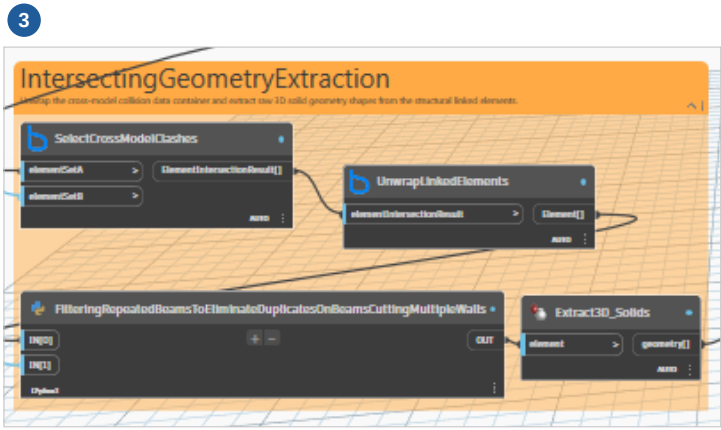
CLASHES + UNIQUE BEAMS
Cross-model clashes between linked beams and walls. Each beam's Element ID is the key, so a beam crossing several walls gives one result.
- 4

SIZE + LOCATE
The beam's bounding box gives the void's centre point and its width, length and depth.
- 5

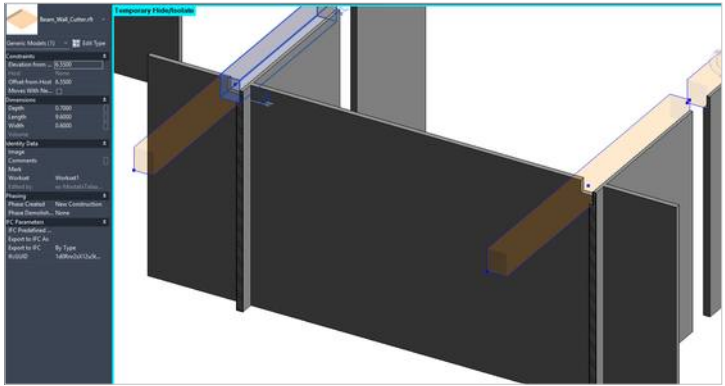
PLACE + CUT
A void family is placed at each point and the cut is applied to the walls.



INPUTS — DYNAMO PLAYER
Linked model, categories, void family and wall level



UNIQUENESS LOGIC
Intersection results are filtered using the underlying beam's Revit Element ID



PARAMETRIC VOID FAMILY — BEAM_WALL_CUTTER.RFT
A Generic Models family. Depth, Length and Width are the parameters the script sets for every instance.

BEYOND BEAMS

The element categories and the void family are inputs. The same logic — **place a family of set dimensions wherever a condition is met** — can be pointed at other categories.

A2

FACE-HOSTED
FAMILY PLACEMENT

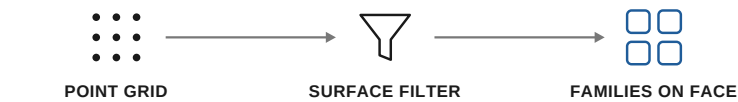
Dynamo for Revit — configurable grid placement, shown on a waffle slab

THE PROBLEM

A waffle slab carries hundreds of pans on a regular grid. Each one is a face-hosted family that must sit on the slab surface, clear of beams and edges — and every change to spacing or layout means redoing the placement.

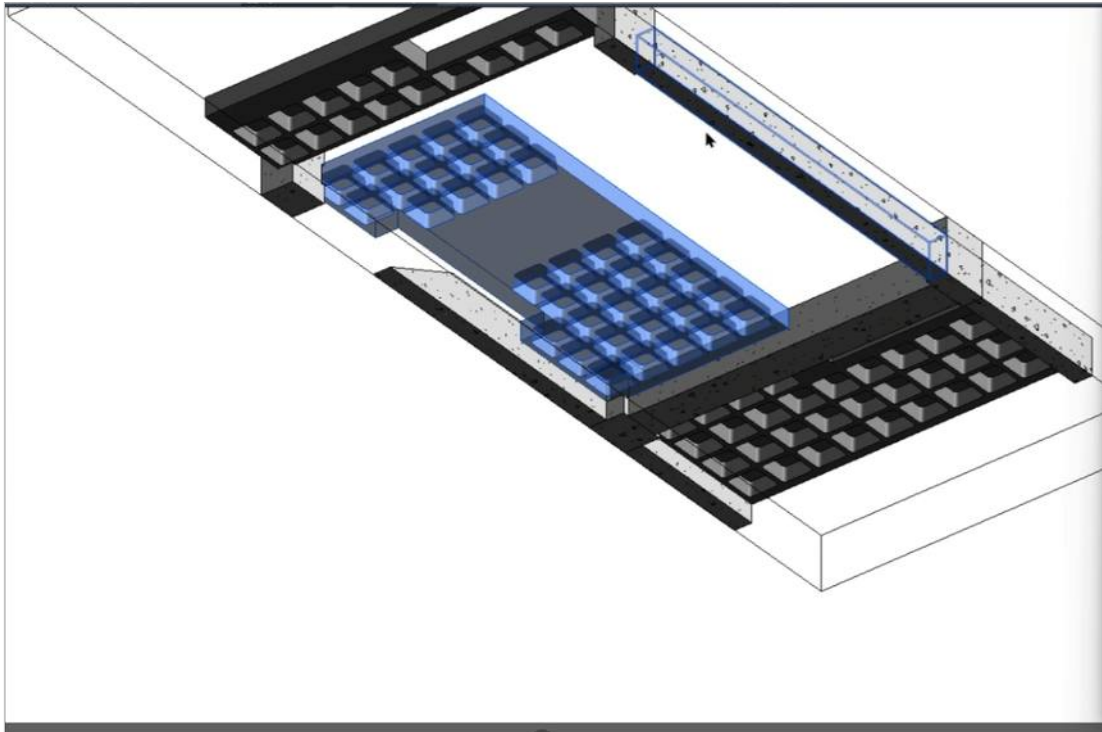
THE SOLUTION

A Dynamo script builds a point grid from the slab geometry, keeps only the points that land on its surface and places the chosen family on the face at each one. Offsets, spacing, rotation and excluded rows or columns are set in Dynamo Player.



ACCURATE QUANTITIES
The pans are actually cut out of the slab, so material quantities, schedules and takeoffs come out right, not just how the model looks.

TYPE	Dynamo script, run from Dynamo Player
HOSTING	Face of the slab (face-hosted families)
CONTROLS	Offsets · spacing · rotation · row and column exclusion · A / B



RESULT — FAMILIES PLACED ON THE SLAB FACE
Void families cutting the waffle slab; placed families shown selected



INPUTS
Dynamo Player

GRID FROM THE SLAB
Points are generated inside the slab's bounding box, then filtered to the surface.

OFFSETS
Left, right and bottom offsets position the grid within the bay.

SPACING + ROTATION
Pan spacing and a 0–360° rotation are set at run time.

ROW / COLUMN EXCLUSION
Numbers entered in Dynamo Player are removed from the point list by index.

TWO FAMILY TYPES
Family B, with its own X / Y offsets, can be assigned to chosen columns.

ANY FACE-HOSTED FAMILY
Not limited to pans: any face-hosted family can be chosen as type A or B.

DYNAMO WORKFLOW

Eight named groups take the slab geometry to placed families.

- 1

BOUNDING BOX
Min and max points of the slab geometry.
- 2

OFFSETS
The grid location is controlled with offsets.
- 3

GRID
A grid of points is created from the box and the spacing inputs.
- 4

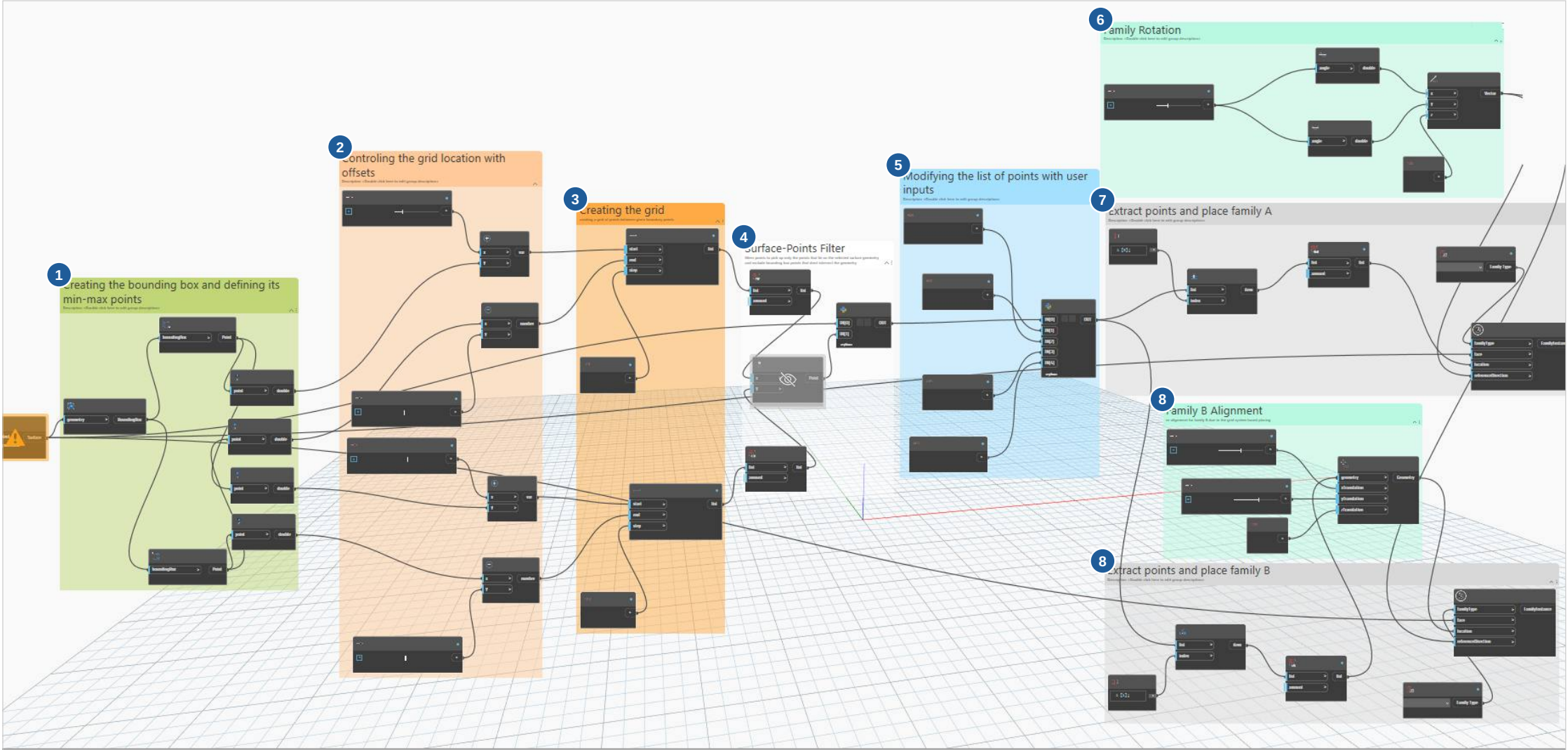
SURFACE-POINTS FILTER
Keeps only points on the selected surface; box points that miss the geometry are dropped.
- 5

USER INPUTS
Excluded rows and columns are removed from the point list by index.
- 6

FAMILY ROTATION
The rotation angle becomes the reference direction.
- 7

PLACE FAMILY A
Points are extracted from the list and the family is placed on the face.
- 8

FAMILY B
Alignment offsets for the second family, placed on its own columns.



SURFACE-POINTS FILTER
Grid points come from the slab's bounding box; only the points that land on the slab surface are kept.



INDEXED EXCLUSION
Row and column numbers entered in Dynamo Player are removed from the point list, leaving gaps where no family is wanted.

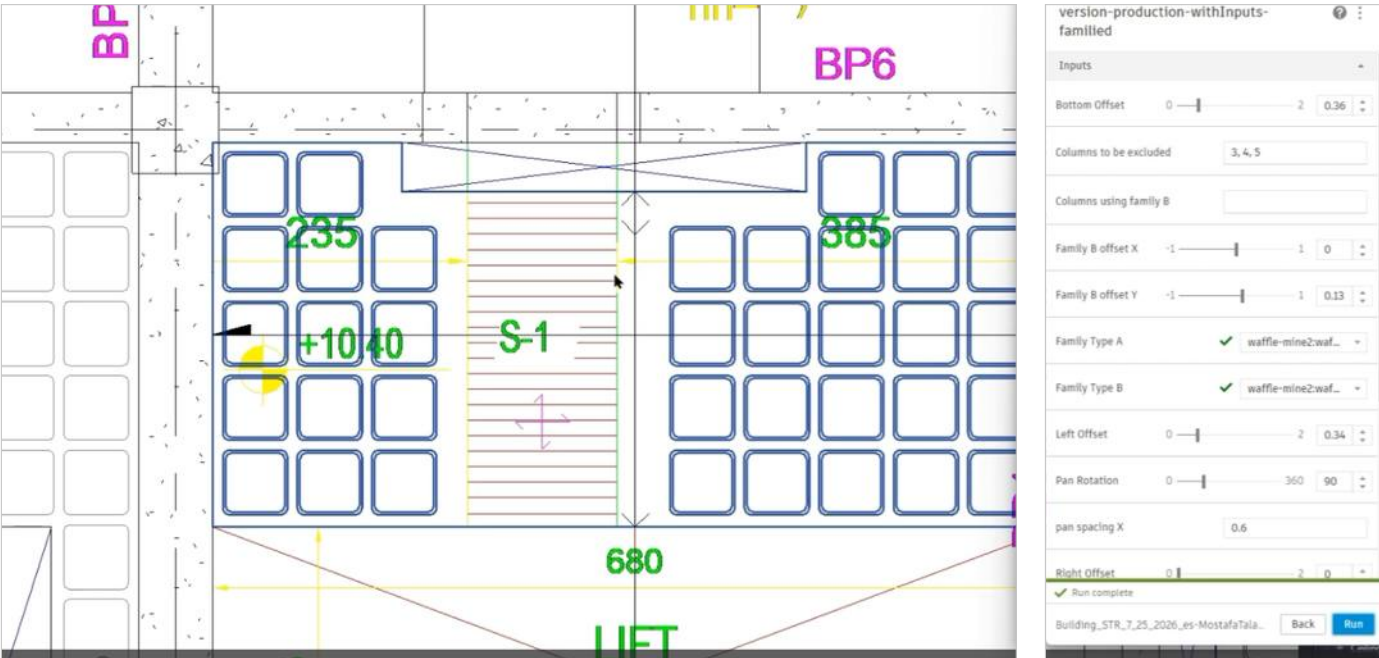


TWO FAMILY TYPES
Family A fills the grid. Family B, with its own X / Y offsets, can be assigned to chosen columns.

SAME WORKFLOW, DIFFERENT CONFIGURATION

Two real runs of the same script with different inputs. Each plan is the Revit result; each panel shows the Dynamo Player inputs used.

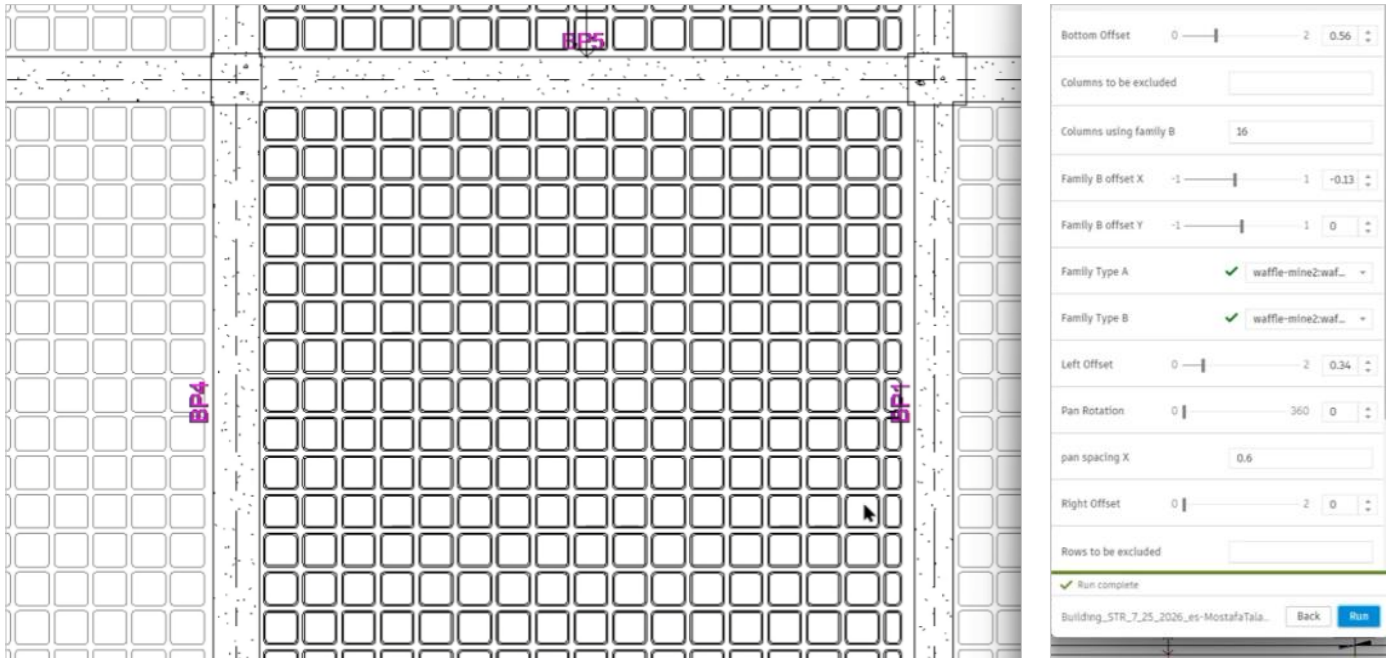
A CONFIGURATION A



CONFIGURATION A

Columns 3, 4, 5 excluded — pans rotated 90°

B CONFIGURATION B



CONFIGURATION B

No exclusions — family B on column 16, rotation 0°

WHAT CHANGED BETWEEN THE TWO RUNS

BOTTOM OFFSET	PAN ROTATION	COLUMNS EXCLUDED	COLUMNS USING FAMILY B	FAMILY B OFFSET X	FAMILY B OFFSET Y
0.36 → 0.56	90 → 0	3, 4, 5 → none	none → 16	0 → -0.13	0.13 → 0

Inputs listed are those that differ in the two Dynamo Player panels shown above.

A3

BEAM AUTOMATION PIPELINE

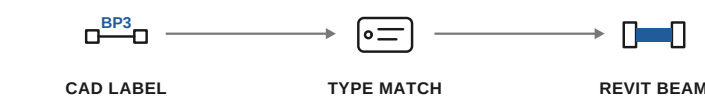
Revit plugin — beams modelled from a CAD drawing

THE PROBLEM

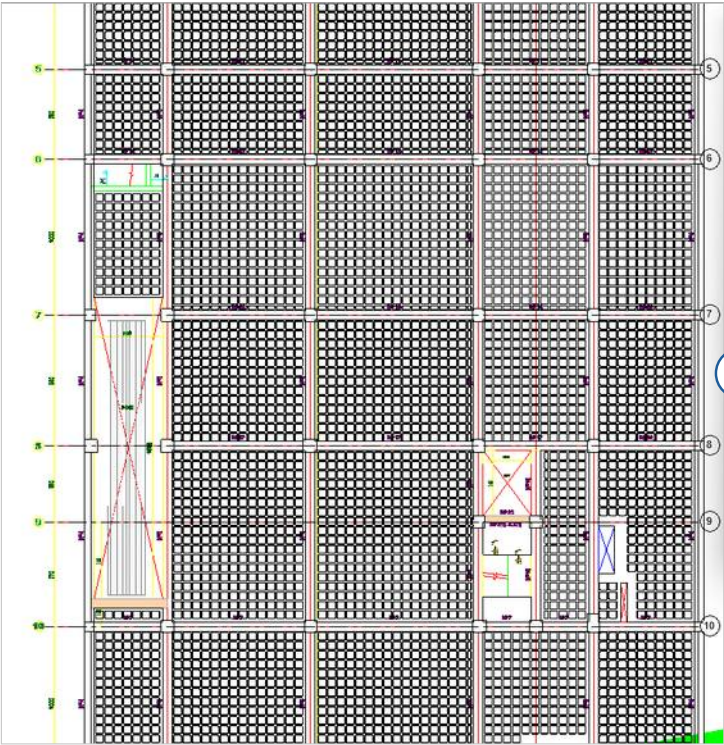
Structural beams are usually drawn in CAD as lines between columns, each tagged with a text label such as BP3. Rebuilding that layout in Revit means reading every label, choosing the matching beam type and placing each beam by hand.

THE SOLUTION

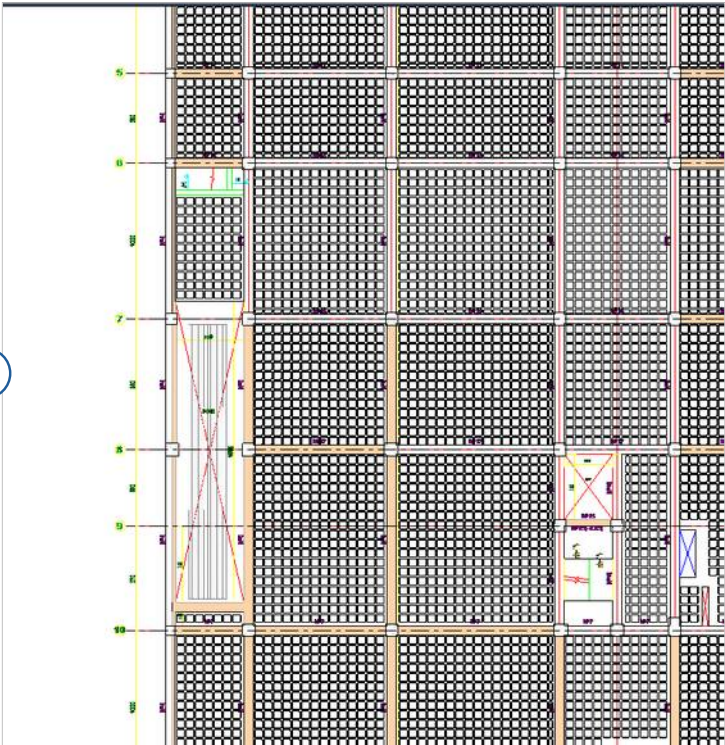
The plugin reads the CAD file in Revit, finds the beam spans between columns and the label — prefix and name — aligned with each, and maps that name to a beam type already in the project. Beams whose label cannot be resolved, or whose type does not exist, are filtered out before anything is created.



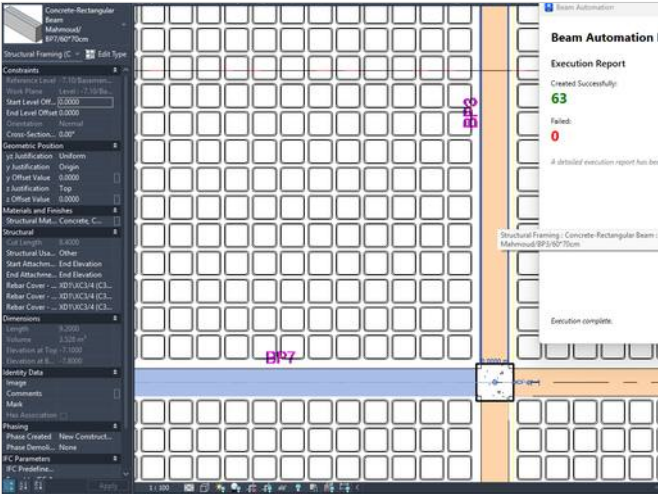
TYPE	Revit plugin with its own dialog
INPUT	CAD file in Revit · beam prefix · tolerances · geometry rules
THIS RUN	63 beams created, 0 failed — 59 unresolved, excluded



CAD FILE IN REVIT
Beam layout and labels, before the run



BEAMS CREATED
Resolved beams placed along the CAD layout



TYPE TAKEN FROM THE CAD LABEL
Label BP3 → Concrete-Rectangular Beam : Mahmoud/BP3/60*70cm

63	BEAMS RESOLVED AND CREATED
59	UNRESOLVED — EXCLUDED, NO MATCHING TYPE IN THE PROJECT
0	FAILED

1

CONFIGURE

Scan settings

Beam Automation

Beam Automation Pipeline

Configuration

Beam Prefix:BP

☒ Map Beam Types Dynamically (match CAD text)

Beam Family (Fallback):W Shapes-Wide Flange

Beam Type (Fallback):W310X32.7

Advanced Tolerances (meters)

☒ Use Metric Units (m)

Min Span:0.3Max Span:15

CAD & Geometry Rules

☒ Orthogonal Beams Only (0° and 90°)

Check this if the project ONLY contains horizontal and vertical beams. The algorithm will strictly reject any slanted candidate spans. Leave unchecked if your project contains boundary beams or slanted grids.

☒ Text Aligns with Beam Direction

Check this if the CAD drafter rotated the beam text tags to match the angle of the beams. Leave unchecked if all text is just stamped horizontally across the page (0°).

ReadyCancelScan CAD

Beam Prefix	text prefix that marks a beam label (BP)
Map Beam Types Dynamically	the CAD text decides the beam type
Min / Max Span	span limits in metres (0.3 – 15)
Orthogonal Beams Only	reject slanted candidate spans
Text Aligns with Beam Direction	for labels rotated with the beam

2

REVIEW

Nothing is created yet

Beam Automation

Beam Automation Pipeline

Review Plan

Please review the proposed changes. Nothing has been created yet.

Resolved (Ready):63

Ambiguous:0

Unresolved:59

New Beams:63

Moved (Flagged):0

Orphaned (Flagged):69

Review pending changes.CancelCreate Elements

Resolved (Ready)	63
Ambiguous	0
Unresolved	59
New Beams	63
Moved (Flagged)	0
Orphaned (Flagged)	69

3

CREATE

Execution report

Beam Automation

Beam Automation Pipeline

Execution Report

Created Successfully:63

Failed:0

A detailed execution report has been written to disk.

Execution complete.CancelFinish

Create Elements places the resolved beams. Beams whose label has no matching beam type in the project are excluded. The report shows 63 created, 0 failed; a detailed report is written to disk.

HOW A LABEL BECOMES A BEAM



Revit family — parametric design

A door driven by named parameters instead of fixed geometry: overall size, frame, stop, panel, handle position and the tolerances around the opening. Equal-constrained dimensions keep it symmetric about its centre as the size changes.

The same idea as the Beam_Wall_Cutter void family in A1, whose Depth, Length and Width are set from outside: geometry that is driven by named parameters, not fixed numbers.

The image contains two technical drawings of a door assembly, showing front and side elevations with dimensions and tolerances.

Front Elevation (Left):

- Overall Width: 1010
- Overall Height: 2110
- Door Panel Width: 926
- Door Panel Thickness: 38
- Frame Thickness: 32
- Tolerance: 2
- Handle X Distance: 75
- Stop Thickness: 19
- Architrave Setback: 5
- Structural Tolerance: 8
- Door Panel Offset: 0
- Stop Depth: 32
- Door frame offset interior = 50, exterior = 50

Side Elevation (Right):

- Overall Height: 2110
- Overall Width: 1010
- Door Panel Width: 926
- Door Panel Thickness: 38
- Frame Thickness: 32
- Tolerance: 2
- Handle X Distance: 75
- Stop Thickness: 19
- Architrave Setback: 5
- Structural Tolerance: 8
- Door Panel Offset: 0
- Stop Depth: 32
- Door frame offset interior = 50, exterior = 50

Frame, stop, panel and handle parameters at the wall cut

Height, handle height and undercut

OVERALL		FRAME AND OPENING		DOOR PANEL		STOP AND HARDWARE	
Width	1010	Frame Thickness	32	Door Panel Width	926	Stop Thickness	
Height	2110	Door frame offset exterior	50	Door Panel Thickness	38	StopDepth	
		Door frame offset interior	50	Door Panel Offset	0	Handle X Distance	
		Architrave Setback	5	UnderCut	8	Handle Z Distance	
		Structural Tolerance	8				
		Tolerance	2				

EQ Equal-constrained dimensions keep the door centred about its reference plane.

FLIP Flip controls in plan and elevation, shown as blue arrows in the family editor.

03

ARCHITECTURAL DESIGN

Five design projects and a graduation project

PROJECTS	5 design projects + 1 graduation project
YEARS	2022 – 2024
SOFTWARE	Revit · Rhino · Grasshopper · Twinmotion · Enscape

DESIGN PROJECTS

D1	The Sun That Sets in the South	Train station	53
D2	Audi Showroom	Car showroom	57
D3	Boutique Hotel of the Arabian Nights	Boutique hotel	60
D4	Siwa Discovery Center	Discovery center	63
D5	Alexandria Community Center	Community center	65

GRADUATION PROJECT

D6	Oblivion: The Sin of Time	Museum, graduation project	68
----	---------------------------	----------------------------	----



D1 Train station



D2 Car showroom



D3 Boutique hotel



D4 Discovery center



D5 Community center



D6 Museum



D1

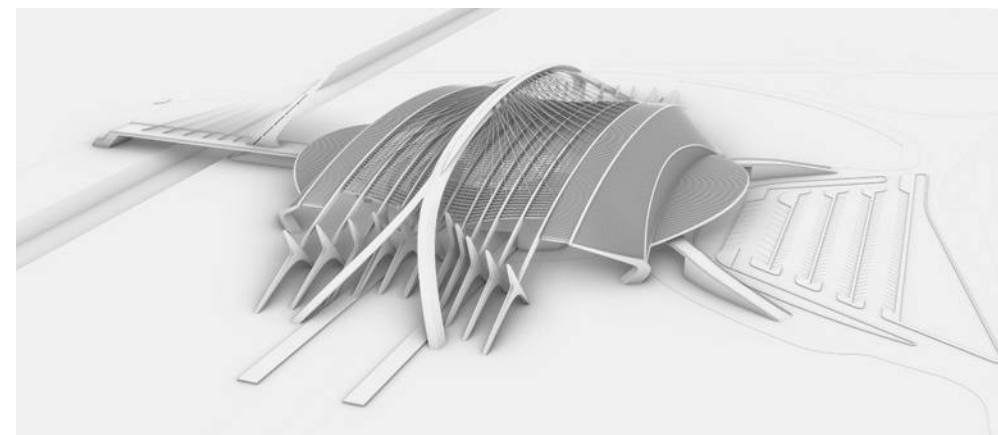
THE SUN THAT SETS IN THE SOUTH

Train station, Assiut, Egypt

A train station in Assiut, Egypt inspired by the sunsets of southern Egypt. A structural sea and a cable-supported framing system support the building and mark the entrance, with large windows and skylights open to the sky. Arrivals and departures are separated onto different floors to ease passenger flow.



WESTERN ENTRANCE VIEW

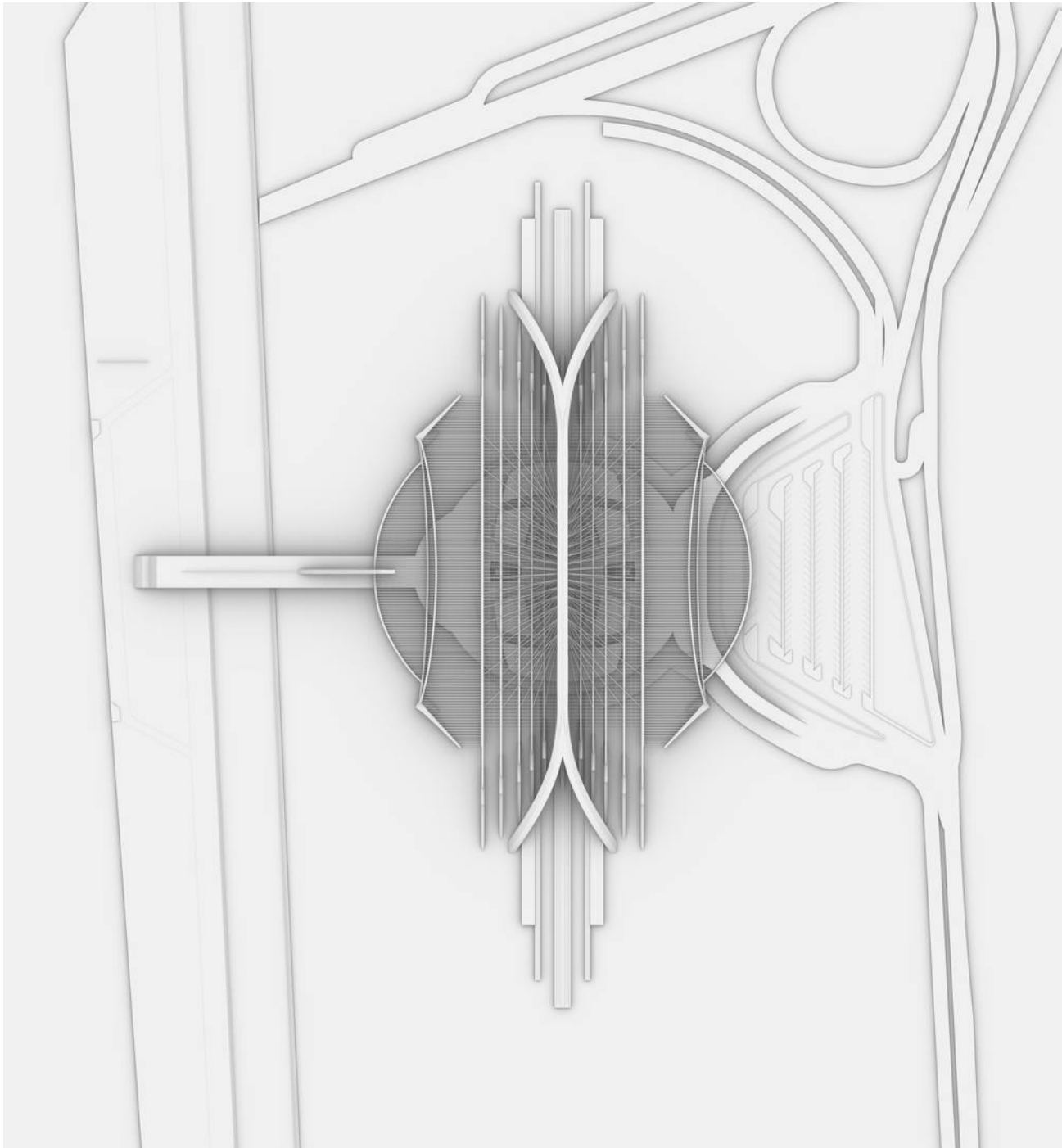


MASSING VIEW

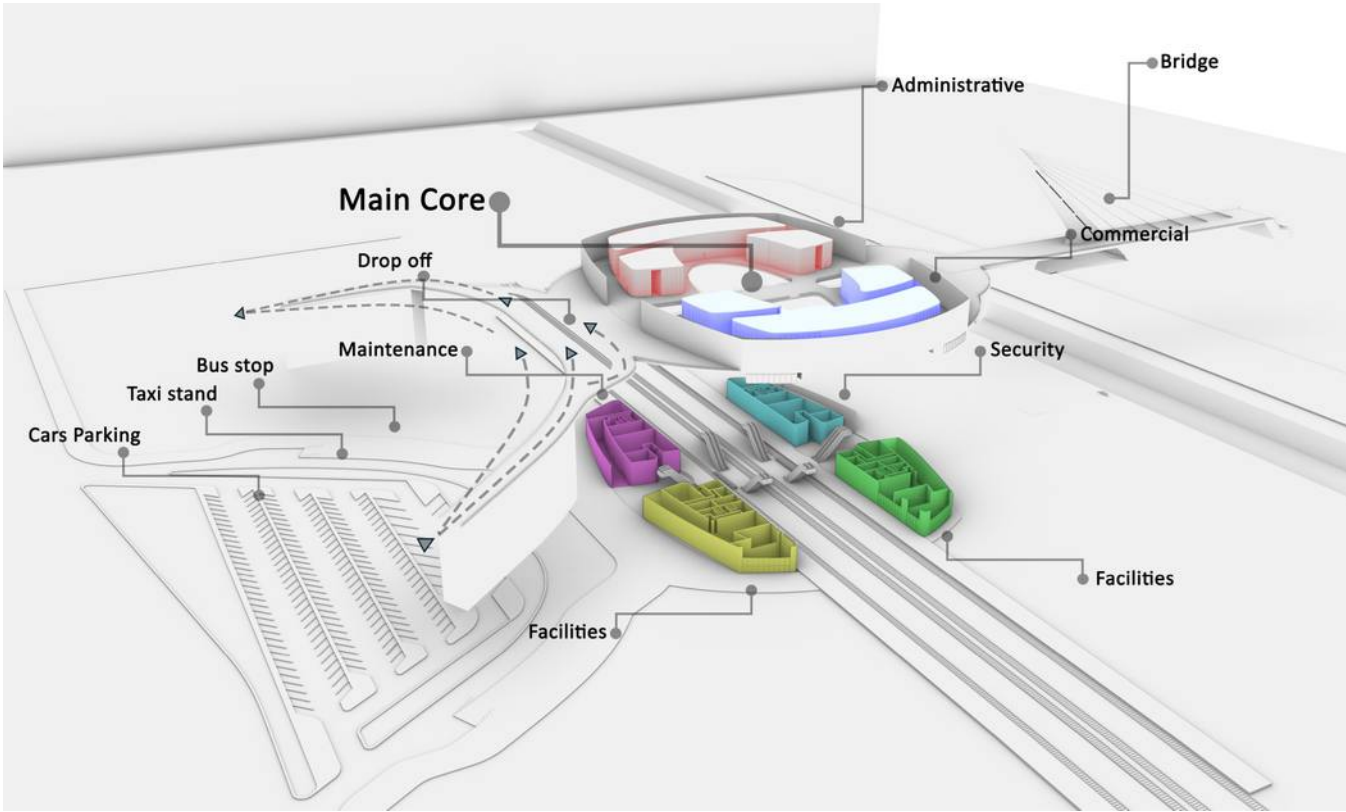
STRUCTURE CONCEPT

The design incorporates an expansive structural sea and a cable-supported framing system. The structural sea provides a visually striking entrance and support for the building, while the cables offer flexibility and open spaces within.

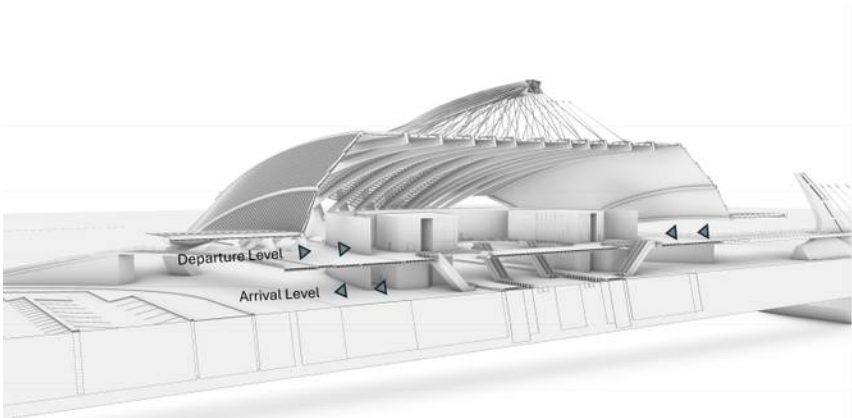
TYPE	Train station
LOCATION	Assiut, Egypt
YEAR	2024
SOFTWARE	Revit, Rhino, Grasshopper, Twinmotion
FOCUS	Structure system and separated passenger circulation



SITE PLAN
Station between the road and the parking



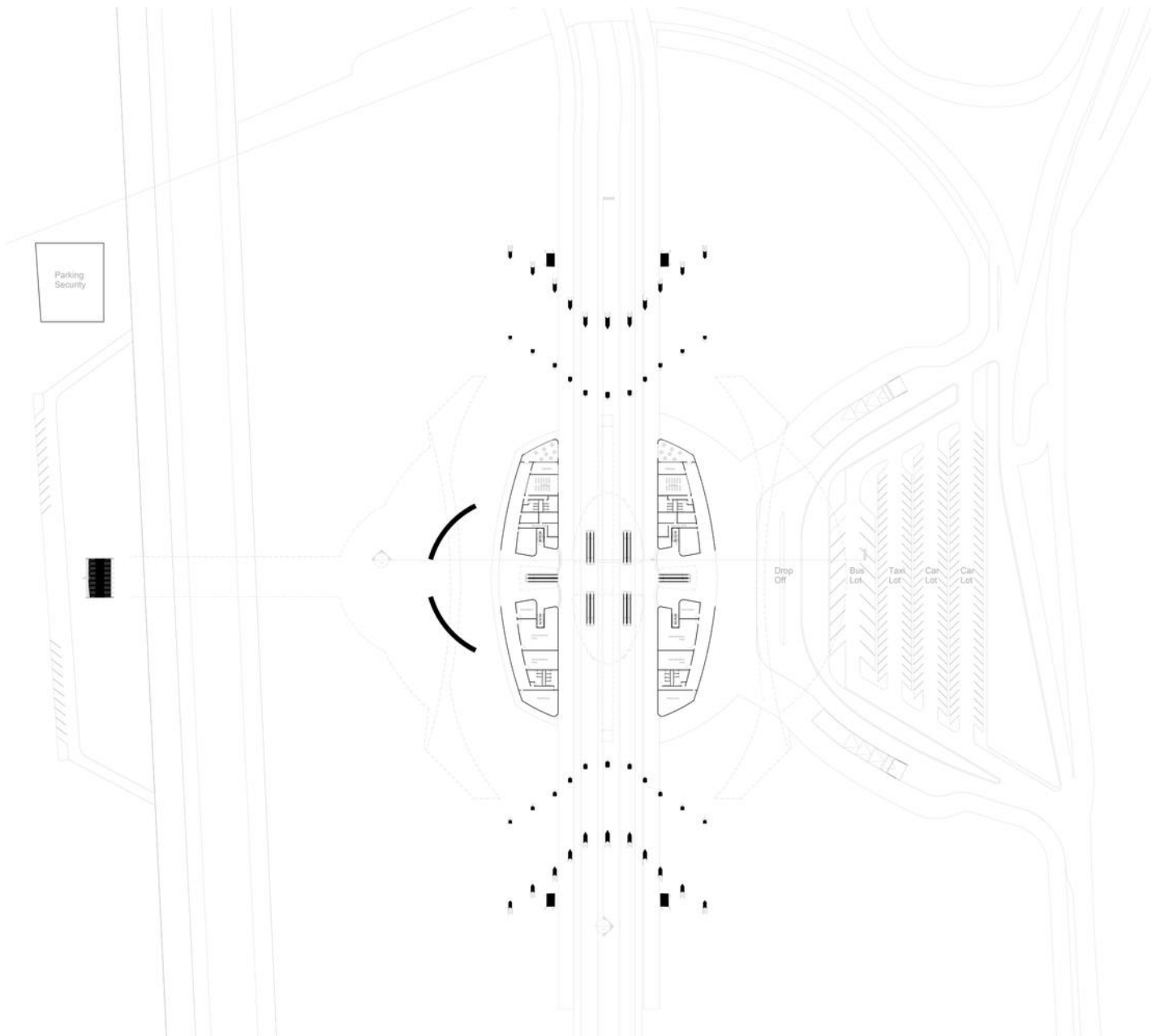
CIRCULATION CONCEPT
Arrivals and departures separated onto different floors



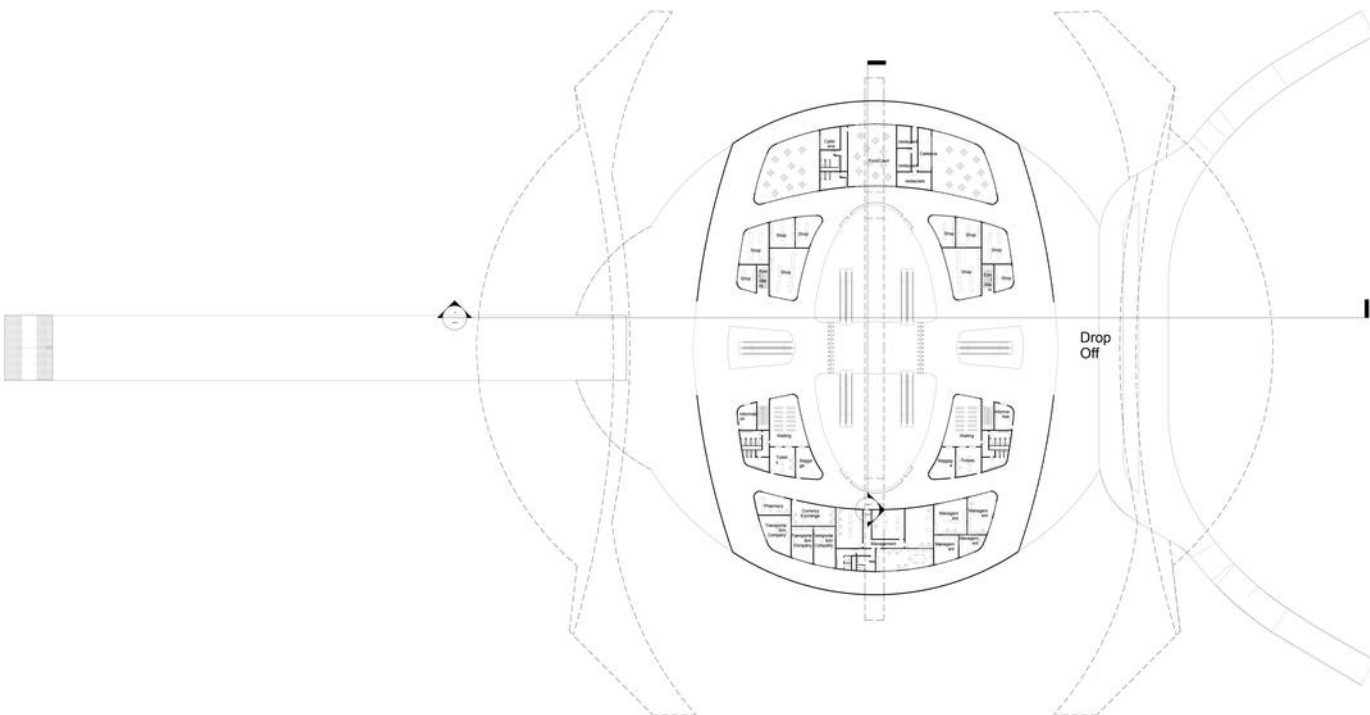
PASSENGER LEVELS
Departure and arrival levels

PASSENGER FLOW

Passenger areas are divided and segregated to optimize efficiency and organization. Functional zones, physical barriers, platform segregation, queue management and accessibility are key elements. This smooths passenger flow, reduces confusion and enhances the travel experience.



GROUND FLOOR PLAN
Station building and platforms between the road and the parking lots



FIRST FLOOR PLAN
Food court, shops, waiting, tickets, baggage, management

FIRST FLOOR, AS LABELED ON PLAN

Food court and restaurants

Cafeteria

Shops

Waiting

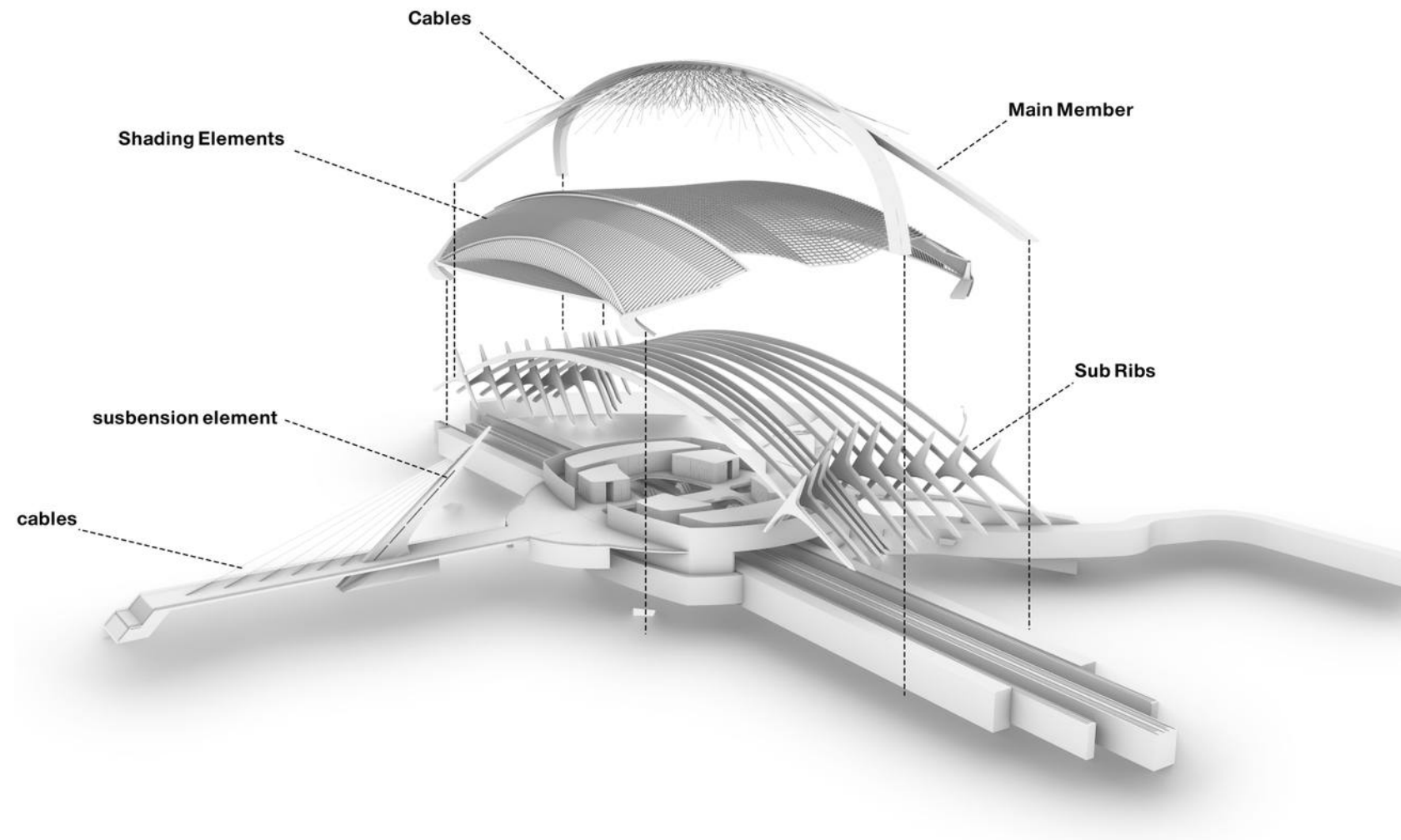
Tickets and baggage

Information

Pharmacy and currency exchange

Transportation companies

Management

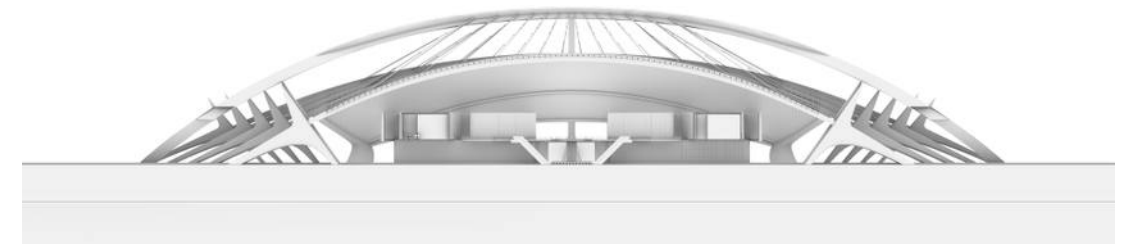


STRUCTURE SYSTEM

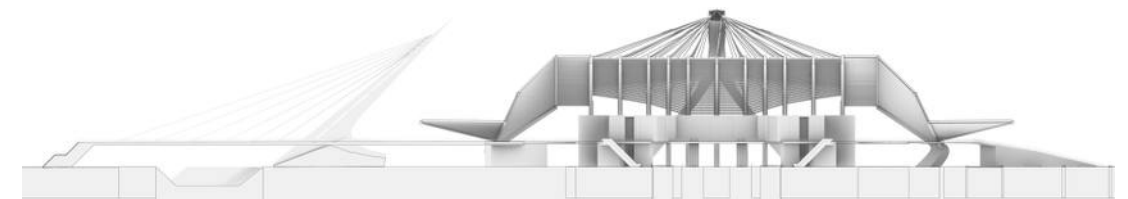
Cables, shading elements, main member, sub ribs, suspension element

STRUCTURE CONCEPT

The design incorporates an expansive structural sea and a cable-supported framing system. The structural sea provides a visually striking entrance and support for the building, while the cables offer flexibility and open spaces within. The design emphasizes transparency, natural light and sustainability.



SECTION A-A



SECTION B-B



SOUTHERN ELEVATION



EASTERN ELEVATION



D2

AUDI SHOWROOM

Brand showroom shaped by the Audi logo

A showroom for Audi whose form and levels of space are derived from the four circles of the brand's logo.

TYPE	Car showroom
LEVELS	Ground + first floor
YEAR	2023
SOFTWARE	Revit, Rhino, Grasshopper, Twinmotion
FOCUS	Logo-derived form; stacked levels of space



EXTERIOR PERSPECTIVE

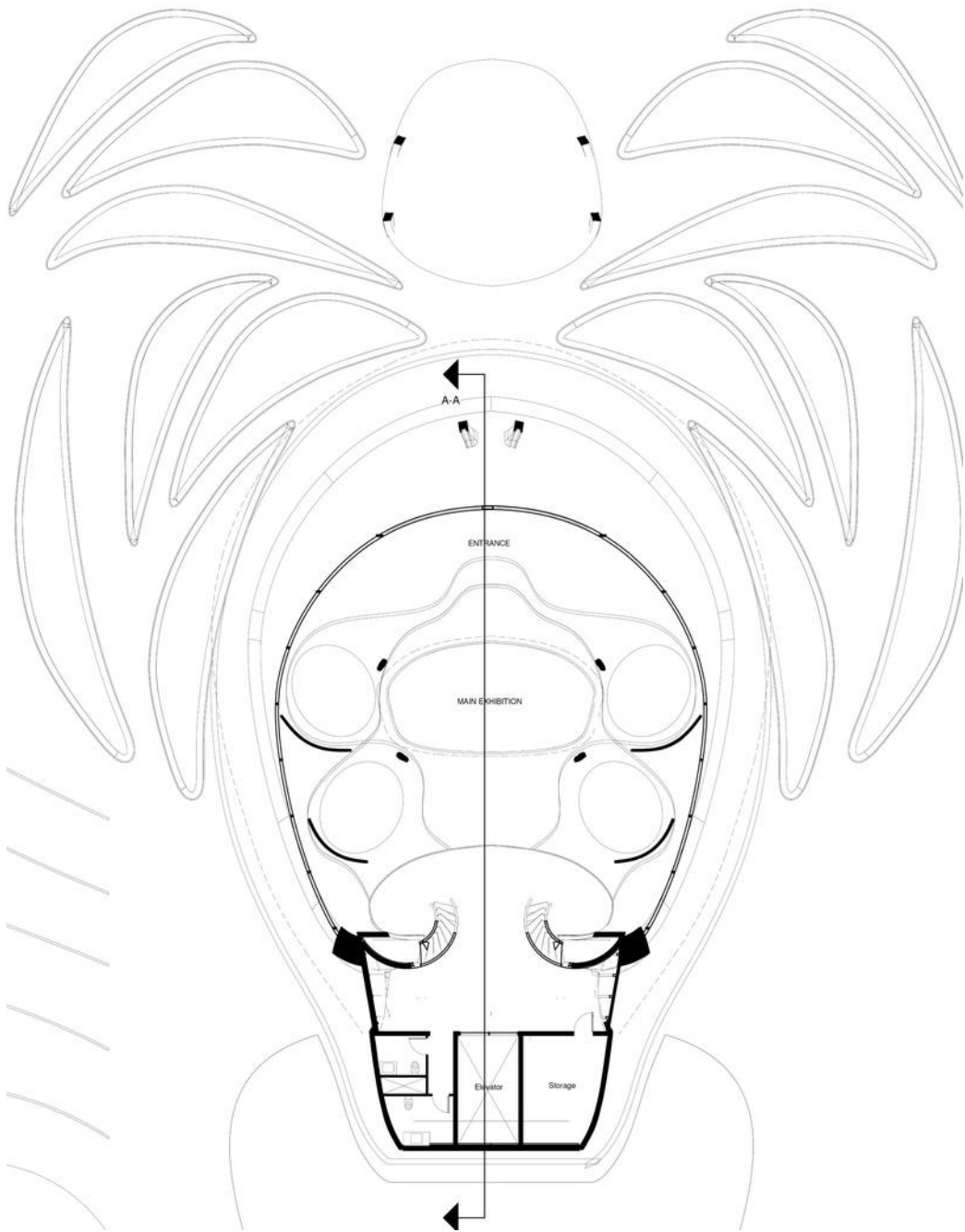
Glazed volume under a dark curved roof, with landscaped forecourt

Logo-derived form

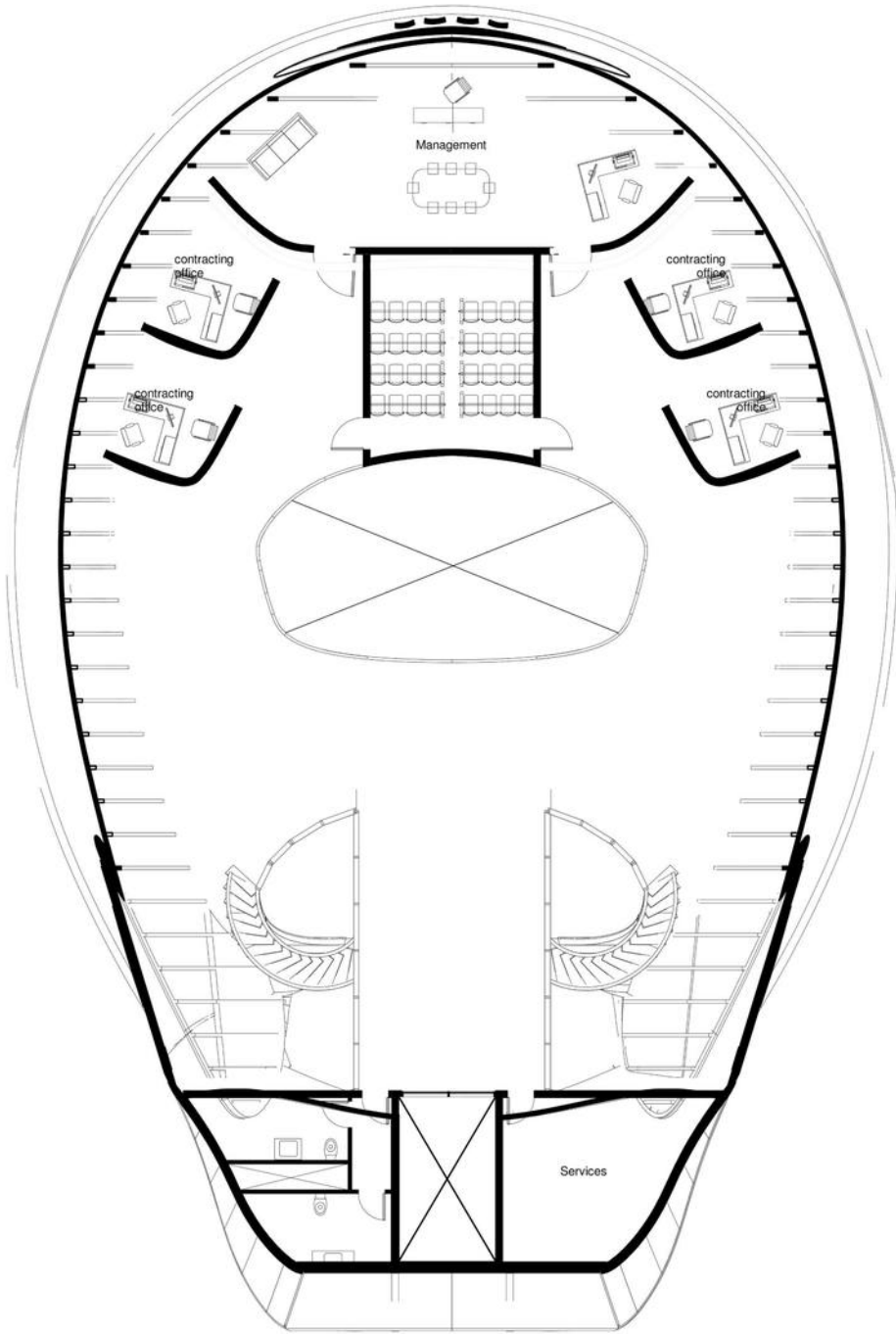
The four circles of the Audi logo are stacked to define different levels of space, each with its own function.

Interior space

The interior follows the curvatures and slopes of Audi's automotive design, as an alliance between architecture and automotive design.



GROUND FLOOR PLAN
Entrance, main exhibition, elevator, storage



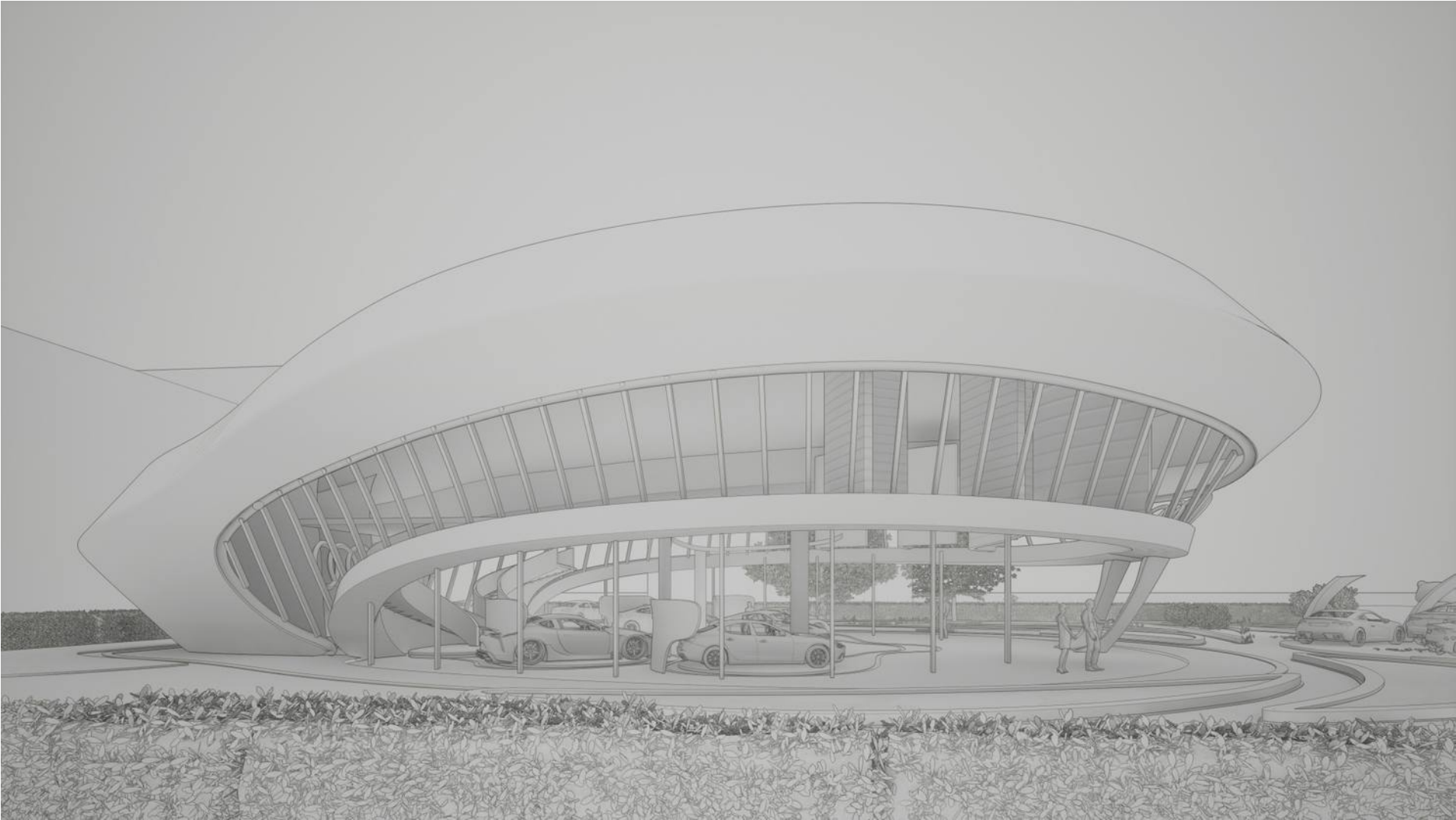
FIRST FLOOR PLAN
Management, contracting offices, services



SITE VIEW
Building and forecourt from above

PROGRAM, AS LABELED ON PLANS

GROUND	Entrance
	Main exhibition
	Elevator
	Storage
FIRST	Management
	Contracting offices (4)
	Services



EXTERIOR LINE VIEW

Curved roof and glazed ribs

STRUCTURE SYSTEM

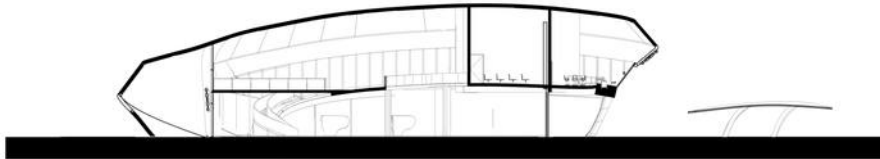
A frame truss structure system carries the building. In the line view and Section A-A, the curved roof sits over a glazed volume raised above the circular display platforms on slender supports.

SYSTEM	Frame truss structure
ROOF	Curved roof over a glazed volume
DISPLAY	Cars on circular platforms



INTERIOR PERSPECTIVE

Showroom floor on circular platforms



SECTION A-A

Stacked levels under the curved roof



D3

BOUTIQUE HOTEL OF THE ARABIAN NIGHTS

Courtyard hotel in Alexandria

A boutique hotel in Alexandria with an Islamic Mediterranean theme drawn from the Islamic Golden Age. An open-air court is the centre of the plan; guests move between private sea-view rooms, a floor for social interaction and the court.

TYPE	Boutique hotel
LOCATION	Alexandria
YEAR	2023
SOFTWARE	Rhino, Revit, Enscape
FOCUS	Open-air court; privacy and interaction



EXTERIOR PERSPECTIVE



COURTYARD PERSPECTIVE



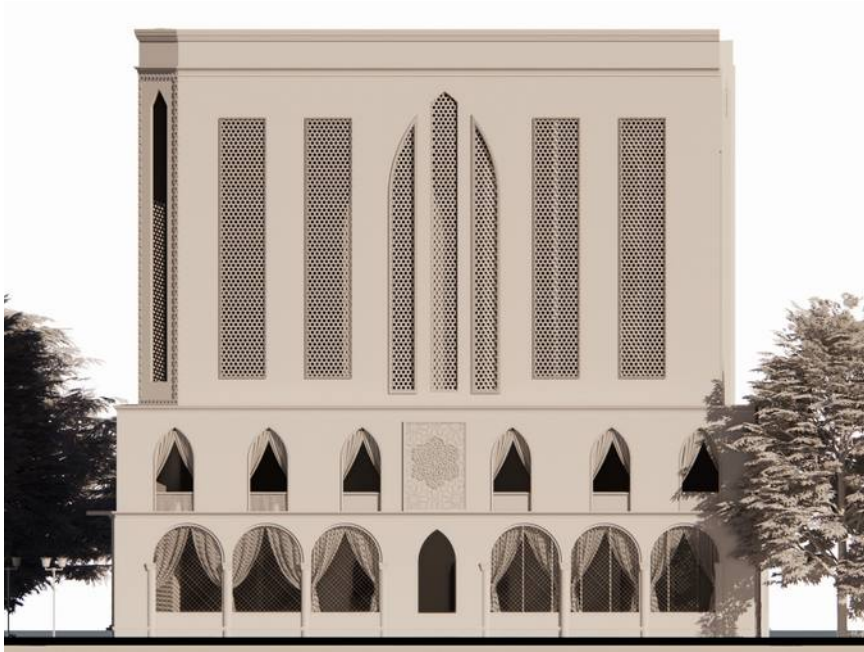
GROUND FLOOR PLAN

Reception court, salon, cafeteria, events hall, laundry, loading dock

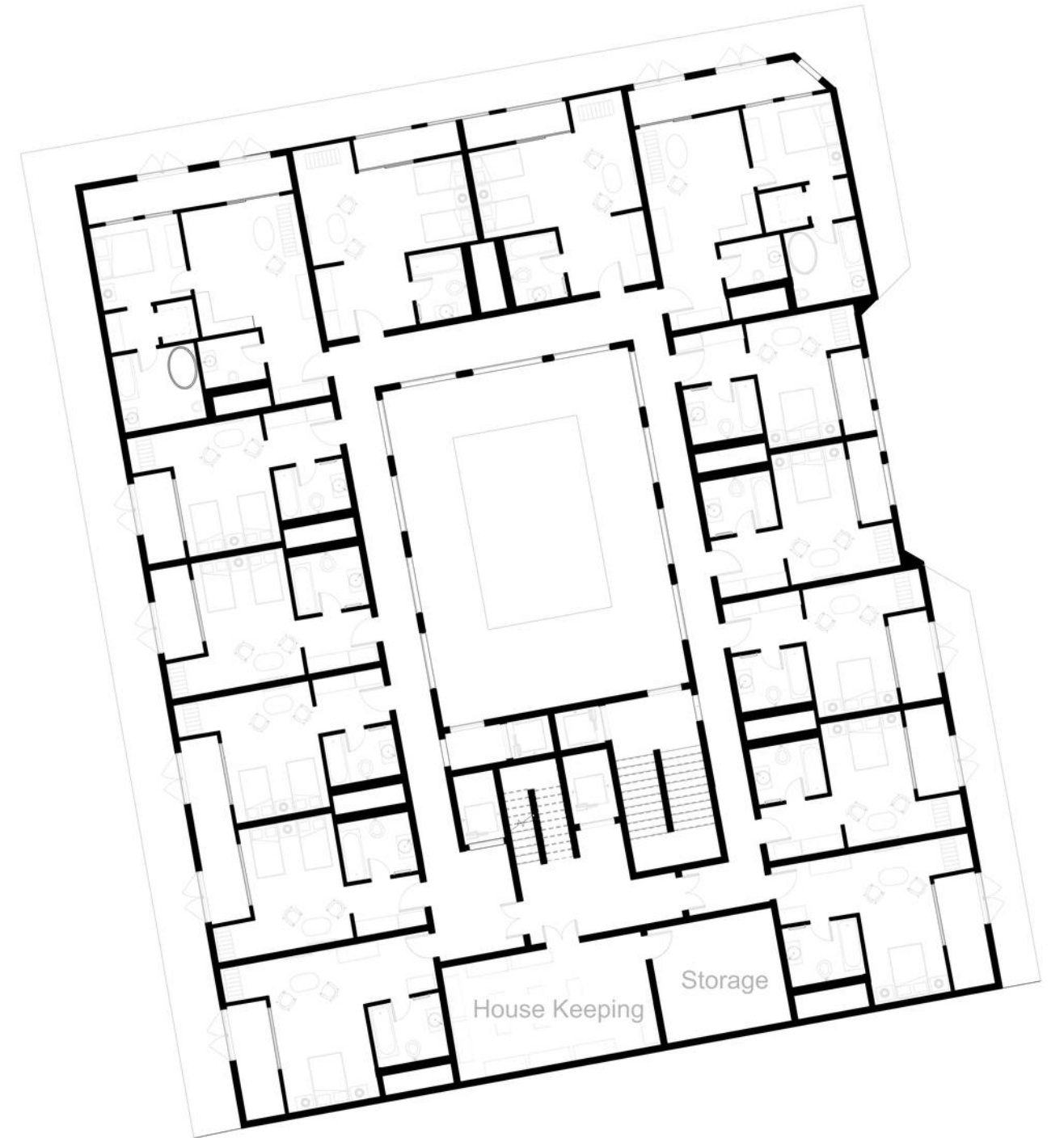


FIRST FLOOR PLAN

Restaurant, kitchen, spa, sauna, jacuzzi, gym, health station

**FRONT ELEVATION**

Arched openings and screened facade

**MAIN HOTEL LOBBY****GUEST FLOOR PLAN**

Guest rooms around the court; housekeeping and storage



D4

SIWA DISCOVERY CENTER

Dune-inspired discovery center in Siwa Oasis, Egypt

A discovery center at Siwa Oasis whose form follows the surrounding dunes: one sun-facing surface and one shaded surface. The building responds to heat, wind and low humidity with thick rammed-earth walls, a south-facing dune profile with roof openings, and evaporative holes. Its plan combines multipurpose halls, research areas and an observatory.

TYPE	Discovery center
LOCATION	Siwa Oasis, Egypt
YEAR	2022
SOFTWARE	Rhino, Revit, Enscape
FOCUS	Climate response: heat, wind, low humidity



COURTYARD PERSPECTIVE
Palms and water basins; dune-profile roofs and arched entrances



AERIAL VIEW
Dune-profile roofs seen from the entrance side



FORM CONCEPT

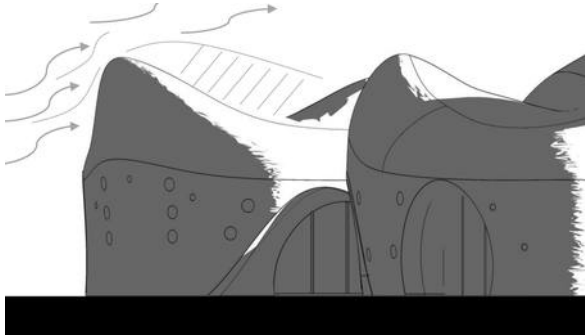


CONCEPT REFERENCE

Dunes (internet image)

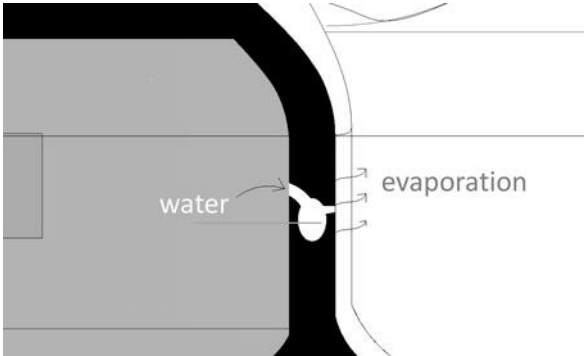
Dunes have two surfaces. One roasts the feet of travelers; the other stays cool in the dune's shadow for most of the day. The form echoes the surrounding fabric of dunes.

WIND STRATEGY



Siwa is exposed to hot, dusty wind from the south for most of the year. The dune shape is oriented south, facing that air and redirecting it upward to create a cool zone above the building, so the roof surface is not exposed to it. Roof openings bring in natural light and ventilation to save energy.

HUMIDITY STRATEGY



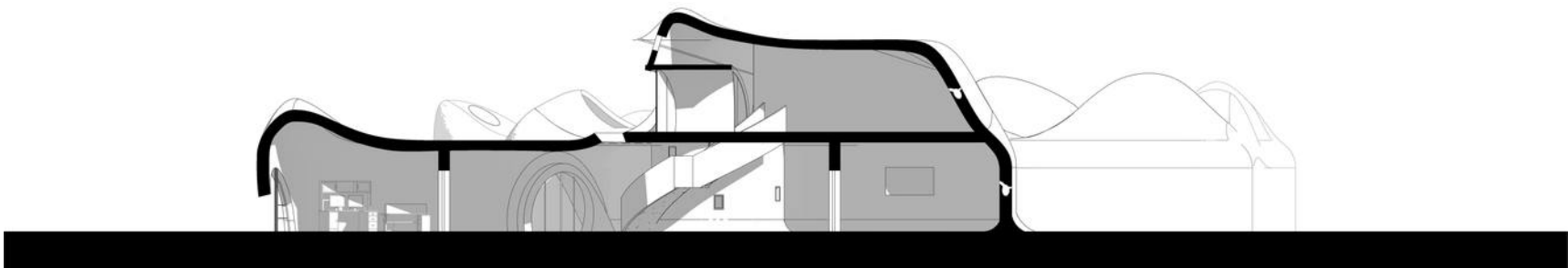
Siwa has low humidity. Walls facing the sun path carry built-in water containers and evaporative holes, which let the sun evaporate the water to add humidity and cool the building.

HEAT STRATEGY

0.8 m

rammed-earth wall thickness

Siwa has high temperatures. The walls are 0.8 m thick rammed earth with high thermal mass: it absorbs heat in the morning, keeps it from the interior spaces, and releases it at night when temperatures fall.



SECTION A-A

Dune-profile roof and roof openings

GROUND FLOOR PROGRAM, AS LABELED ON PLAN

Multipurpose halls (2)

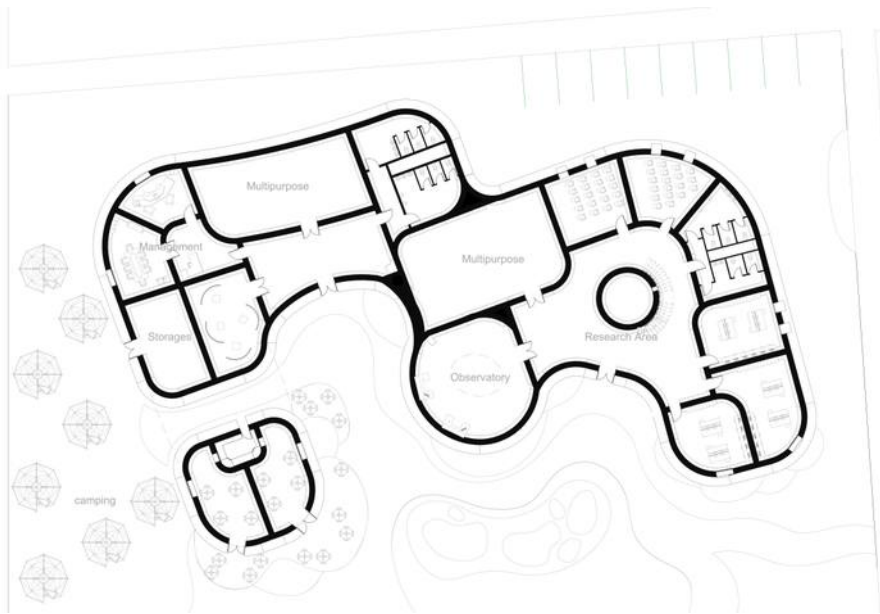
Observatory

Management

Research area

Storage

Camping area (outside the building)



GROUND FLOOR PLAN

Multipurpose halls, management, storage, observatory and research area; camping area outside



D5

ALEXANDRIA COMMUNITY CENTER

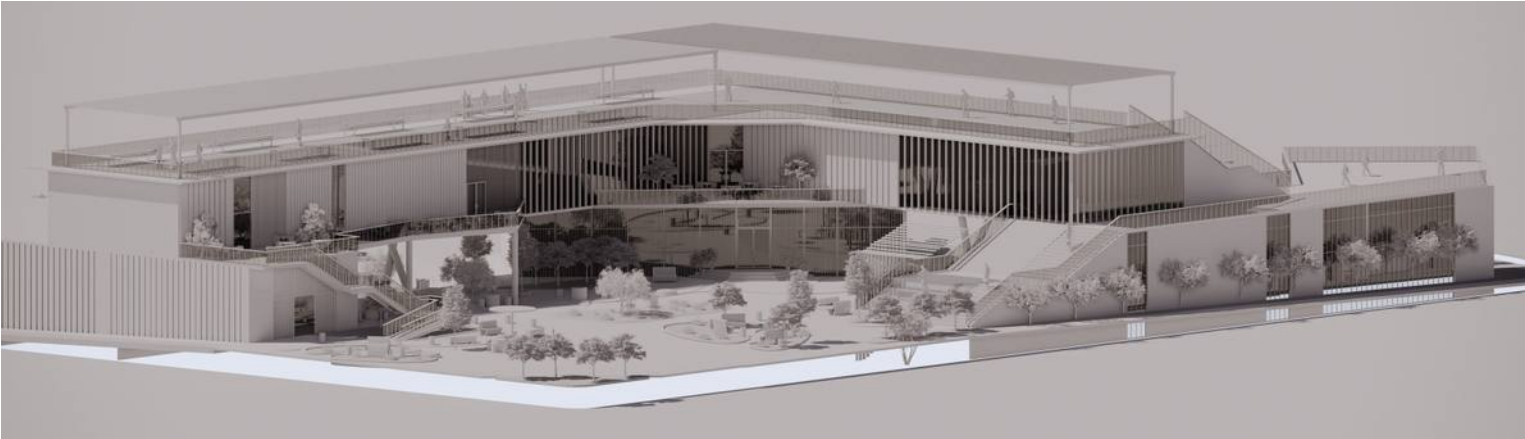
Community center in Alexandria

A community center in Alexandria organised around three visitor groups: adults, children and craftspeople. The plaza is divided into three areas, one per group, and the building into three blocks, each with a private courtyard. The roof serves as a summer market for local producers; its cover is removed in winter.

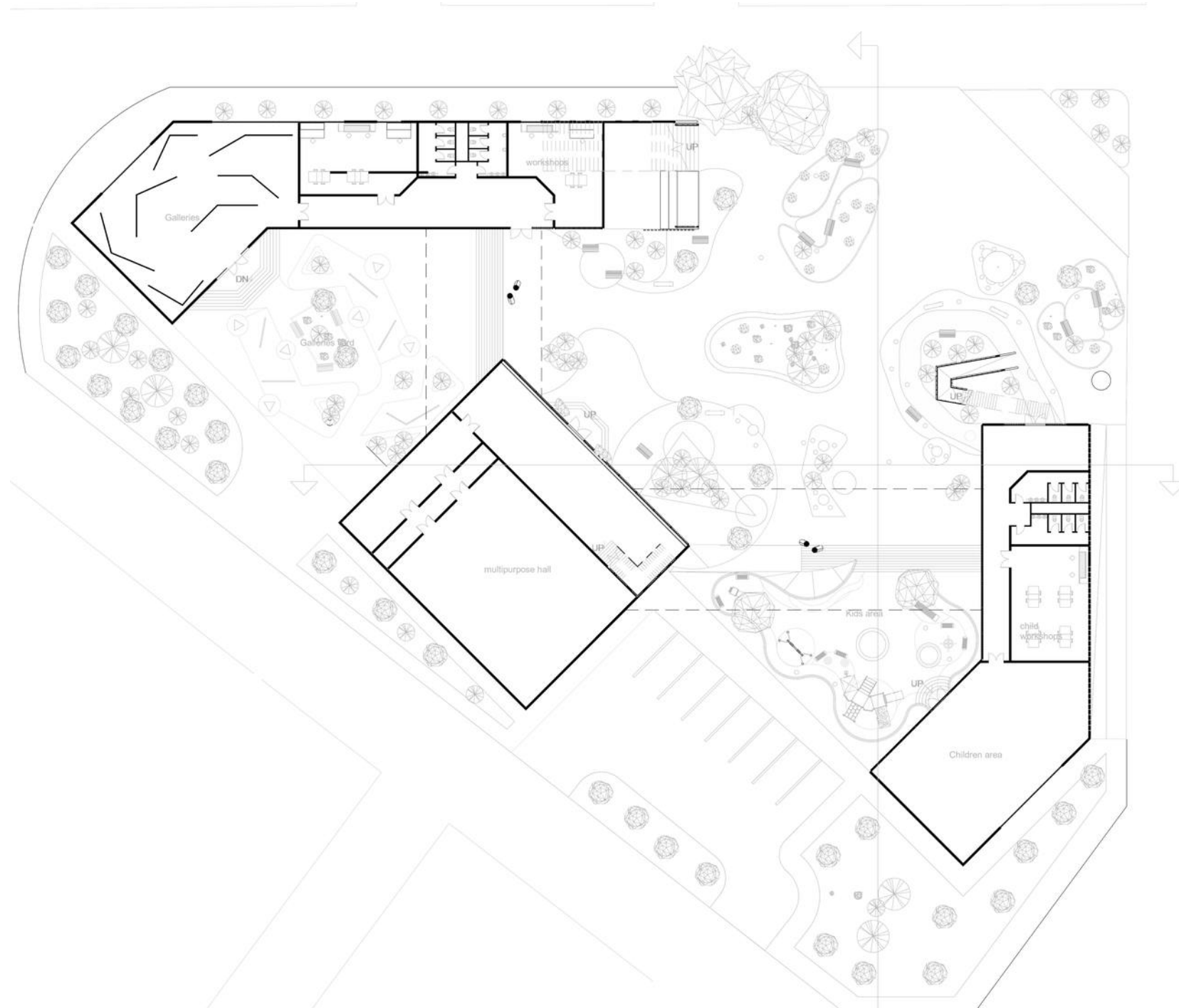
TYPE	Community center
LOCATION	Alexandria
YEAR	2023
SOFTWARE	Rhino, Revit, Twinmotion
FOCUS	Three visitor groups, three blocks, three courtyards



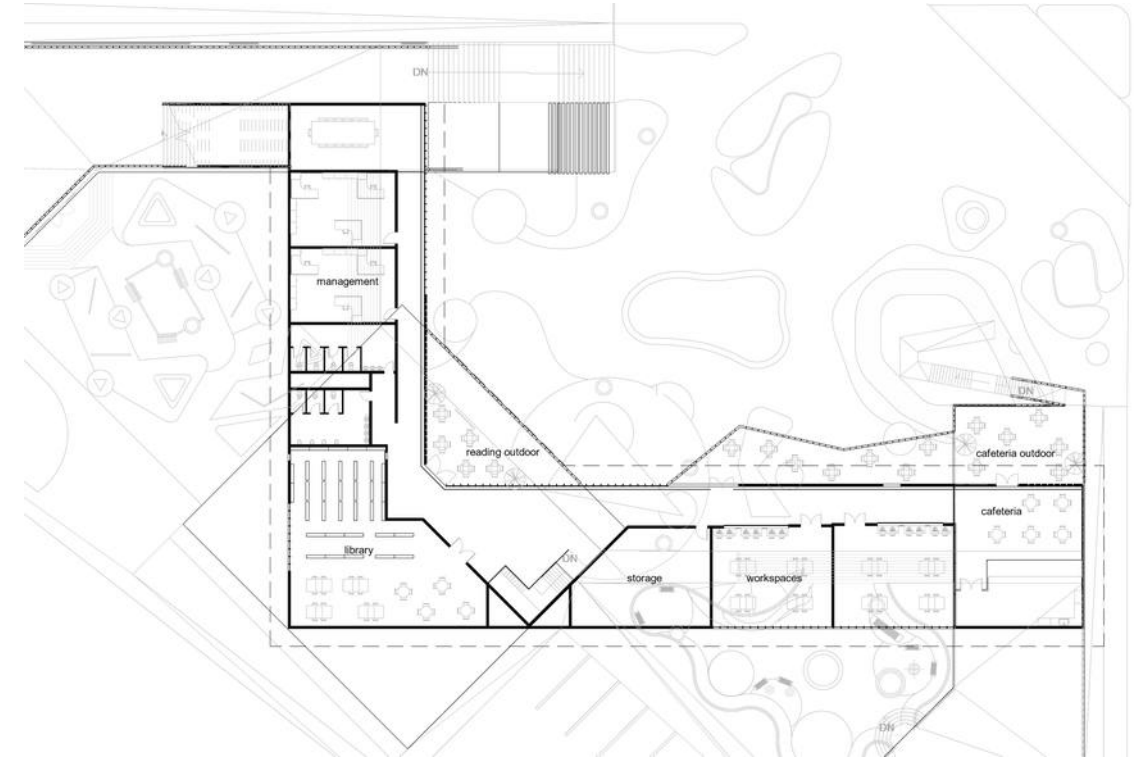
COURTYARD PERSPECTIVE
Courtyard with seating, stair and terrace



AERIAL CUTAWAY
Roof terrace and courtyard

**GROUND FLOOR PLAN**

Galleries, workshops, multipurpose hall, children's area

**FIRST FLOOR PLAN**

Library, workspaces, cafeteria, management, outdoor reading

DESIGN STRATEGY**Three visitor groups**

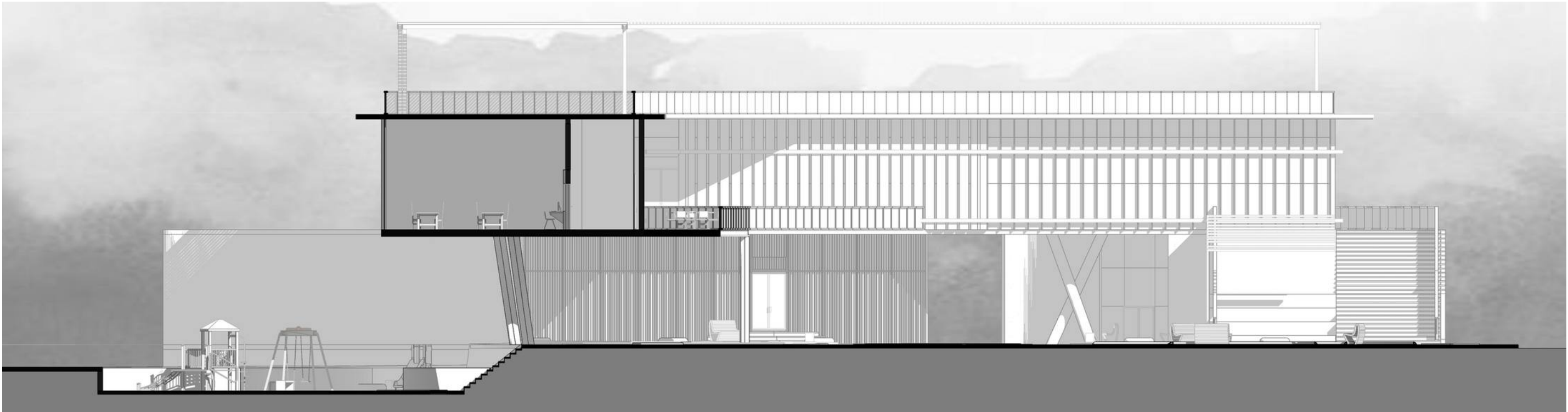
Adults, children and craftspeople each have a designated area of the plaza.

Three blocks, three courtyards

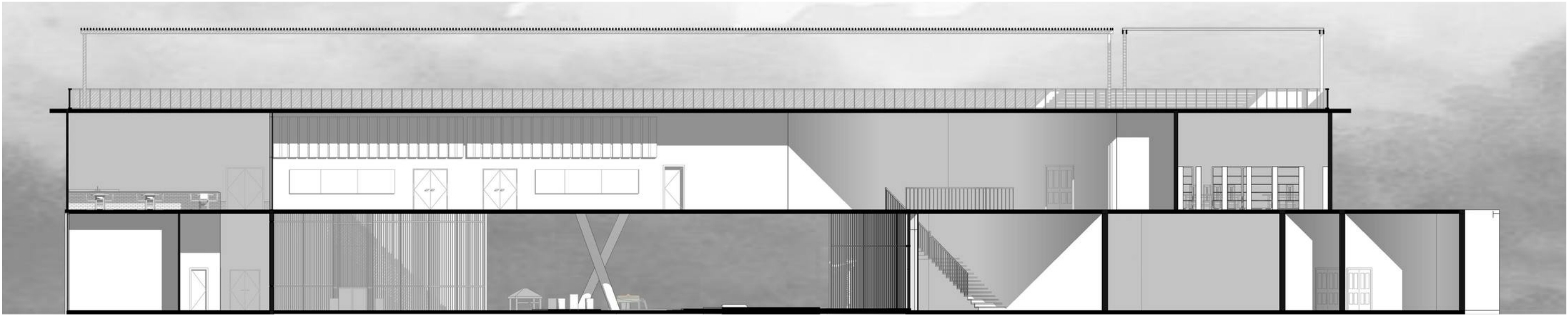
Each block serves one category and opens onto its own private courtyard, giving direct movement paths.

Roof as summer market

The roof extends the usable area as a market for local producers; the cover is removed in winter.



SECTION A-A



SECTION B-B

D6

GRADUATION PROJECT

Oblivion: The Sin of Time

TYPE	Museum, graduation project
LOCATION	Asyut, Egypt
YEAR	2024
SOFTWARE	Revit · Rhino · Twinmotion
FOCUS	Three civilizations, ring of time, underground ruins

IN THIS PROJECT

OBLIVION: THE SIN OF TIME

D6.1	Overview	69
D6.2	Three phases and form concept	70
D6.3	Site and arrival	71
D6.4	Exploded axonometric and ground floor plan	72
D6.5	Floor plans and ruins chamber	73
D6.6	Narrative and interior views	74



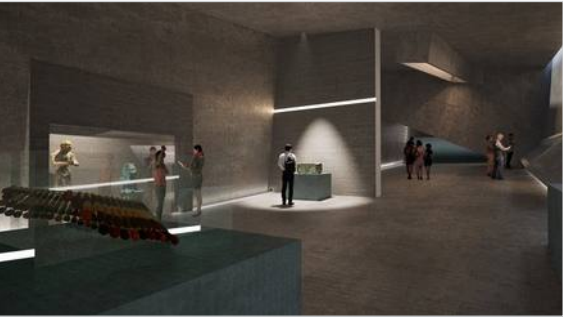
D6 Front view at night



D6 Ruins hall



D6 Entrance plaza



D6 Exhibition gallery

D6

OBLIVION:
THE SIN OF TIME

Museum of three civilizations in Asyut

A museum in Asyut, Upper Egypt, set into a rocky hillside. Its form comes from three civilizations, Pharaonic, Coptic and Islamic, crossing as lines of time, combined with a ring of time that becomes a circular underground chamber of ruins. A marked main route links the phase galleries.

TYPE	Museum, graduation project
LOCATION	Asyut, Egypt
YEAR	2024
SOFTWARE	Revit, Rhino, Twinmotion
FOCUS	Three civilizations, ring of time, underground ruins
GRADE	Excellent



FRONT VIEW AT NIGHT

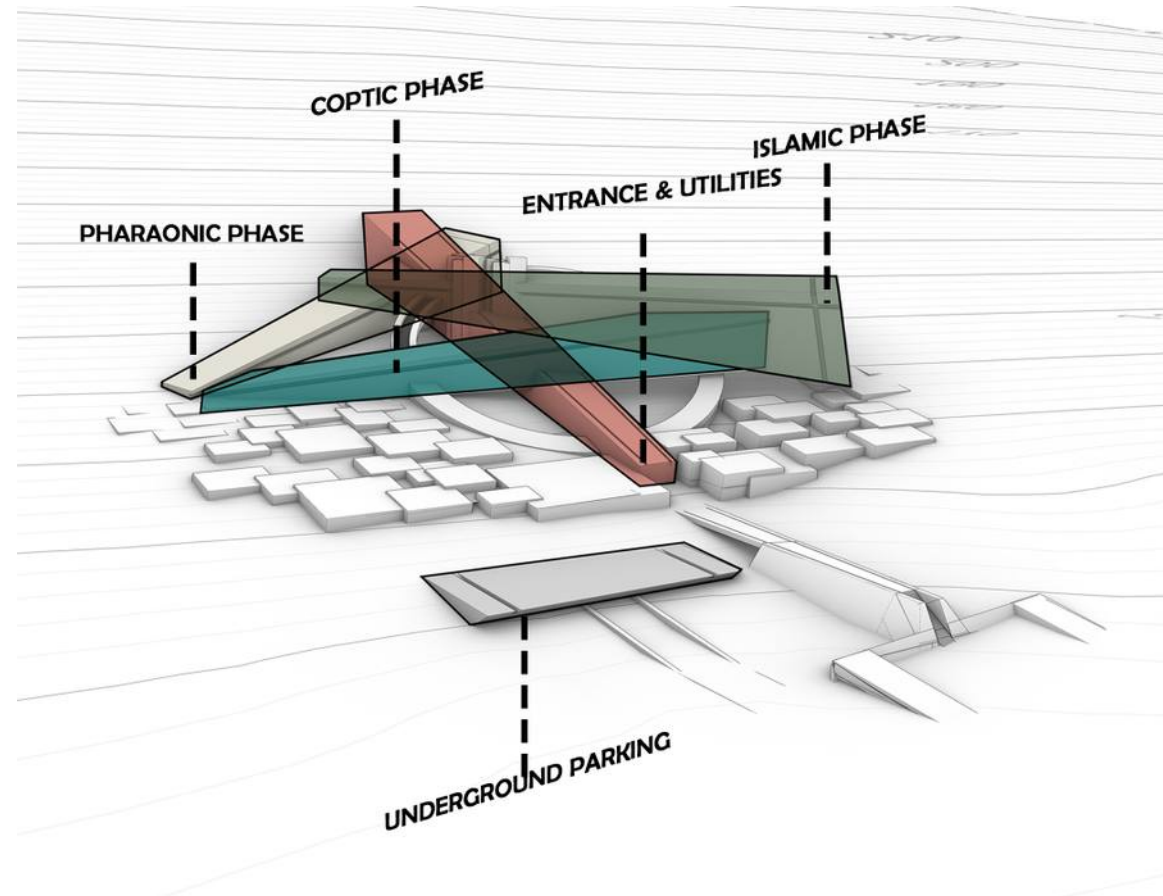
Arrival plaza, building and road



WESTERN SIDE

The building set into the hillside

THREE PHASES



PHASE MASSING

Entrance and utilities, underground parking

Three crossing masses

The Pharaonic, Coptic and Islamic phases cross one another. Each crossing ends one civilization and starts the next.

The ring of time

A ring, the tomb of all that was made and born, becomes a circular chamber below ground that holds the ruins of the three phases.

Route and services

A marked main route links the phase galleries. Entrance and utilities and the underground parking sit beside them.

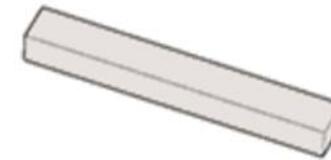
FORM CONCEPT

01



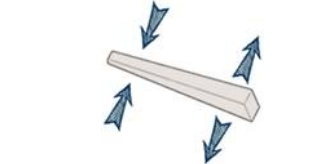
The line of time that never slows down, devouring worlds and civilizations.

02



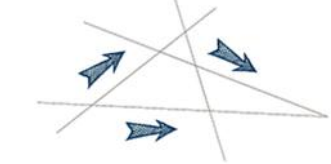
The mass of decay that put all to destruction.

03



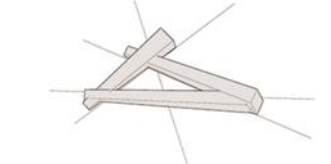
As civilization grows from less to more, its falter grows towards vanishing.

04



Pharaonic, Coptic and Islamic civilizations, all three turned into ruins in the land of time.

05



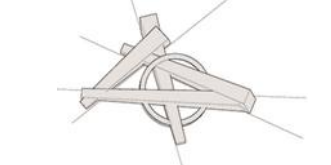
The three civilizations crossed each other at different points. Each point was the start for the new civilization, and the end to the old one.

06



The ring of time that resembles the tomb for everything that was made and everything that was born.

07



The final building. A building that resembles the fate of civilizations, laws, people, and us.

SITE AND ARRIVAL

The building is set into a rocky hillside outside Asyut. Visitors arrive from the road to an entrance plaza with a water basin and a stair.

The maps narrow the location from the Nile Valley to Asyut, the nearby village and the site.

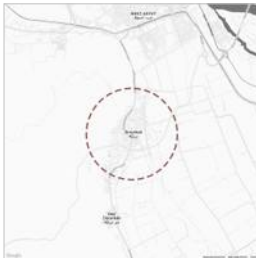
SITE LOCATION



REGION



CITY



VILLAGE

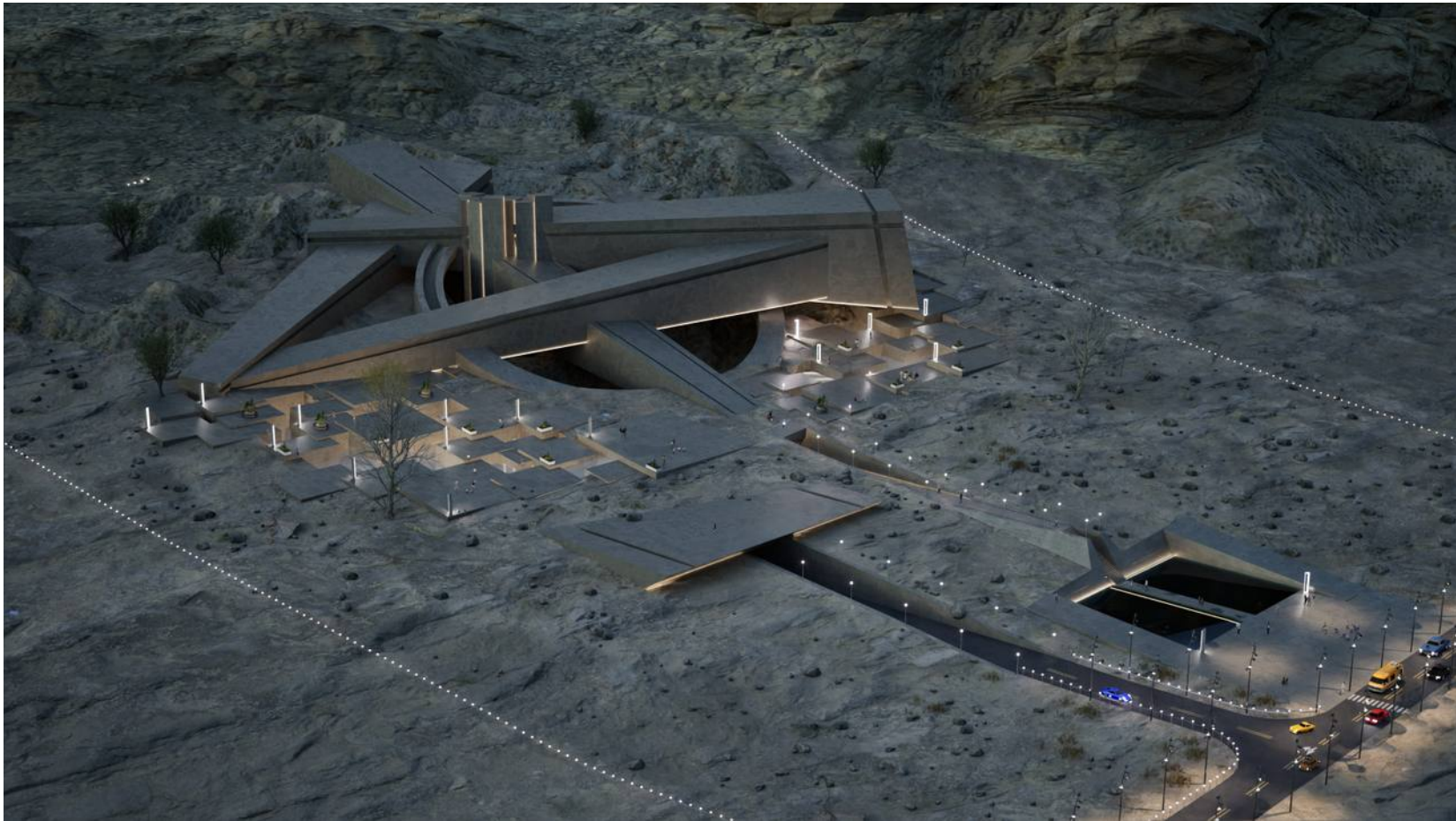


SITE

WHY HERE

The site matches the environment of the old civilizations, so the visitor feels like travelling through time. Visitors cross open rock, go below ground and follow the main route through the three phases to the ruins.

SITE PLAN

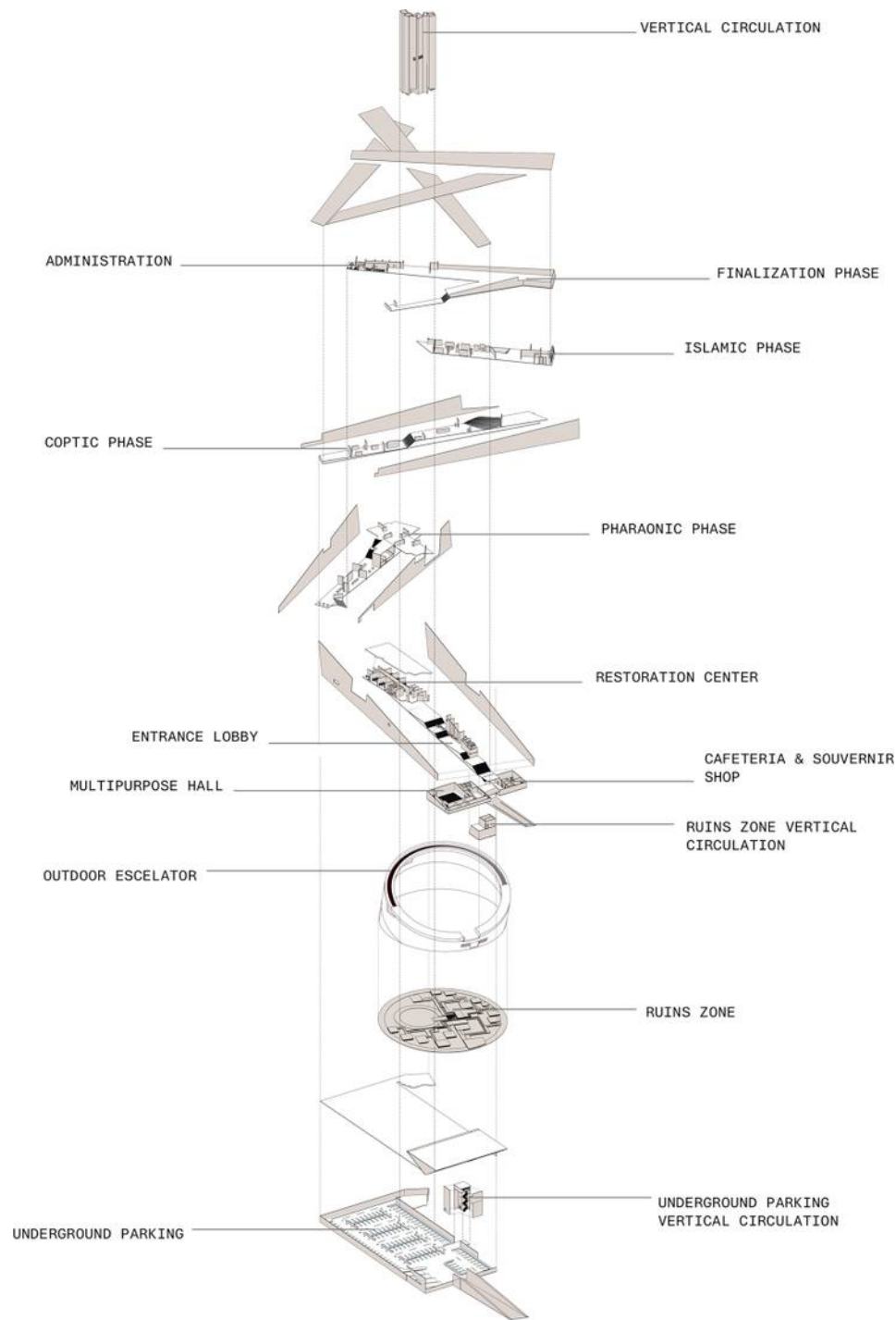


AERIAL VIEW AT NIGHT

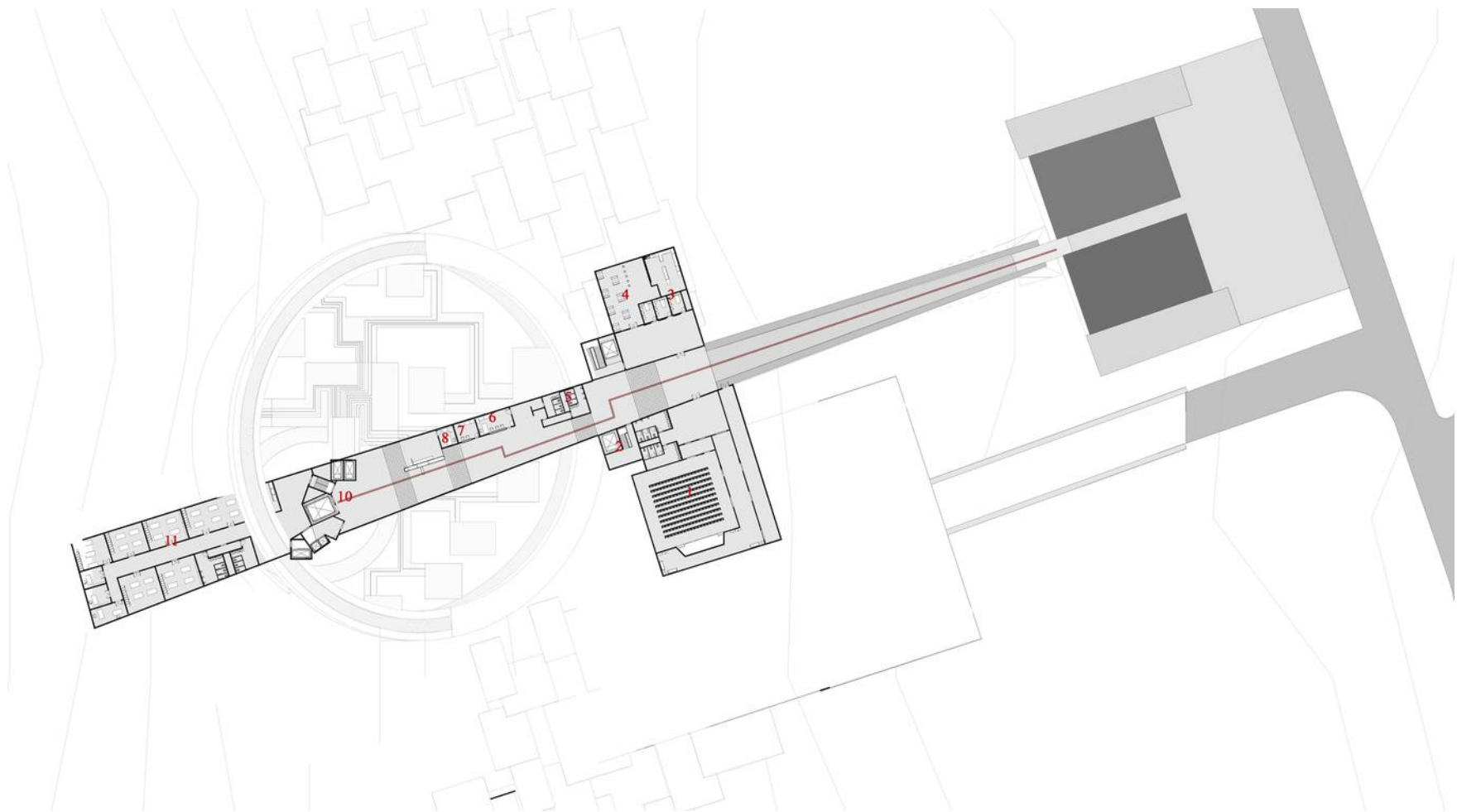
THE LAND OF TIME



ENTRANCE PLAZA



EXPLODED AXONOMETRIC



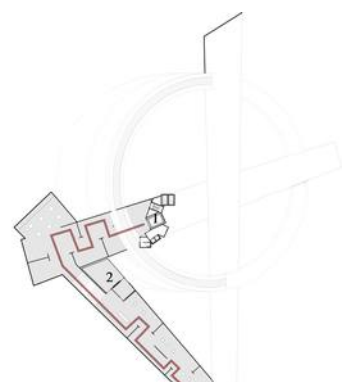
GROUND FLOOR PLAN P1

Main route in red

LEGEND

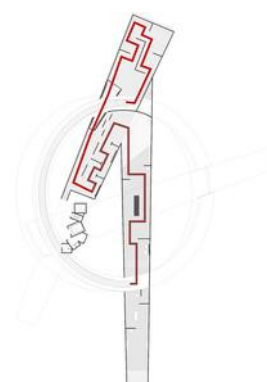
- 1 Multipurpose hall
- 2 Parking vertical circulation
- 3 Souvenir shop
- 4 Cafe
- 5 WC

- 6 Tickets
- 7 Clock room
- 8 Security check
- 10 Main vertical circulation
- 11 Restoration center



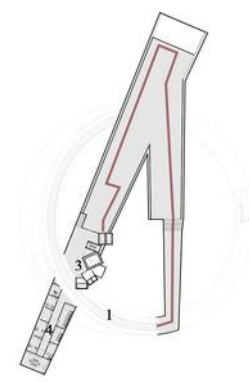
B2 FLOOR PLAN
Main route

- 1 Vertical circulation
- 2 Hologram optional room



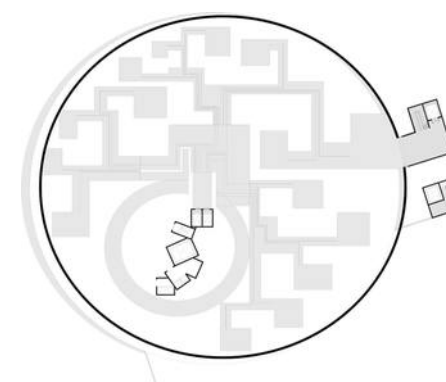
B3 FLOOR PLAN
Main route

- 1 Coptic phase
- 2 Islamic phase
- 3 Outdoor escalator circulation



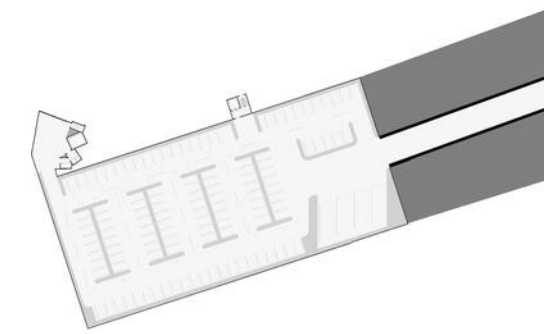
B4 FLOOR PLAN
Finalization phase, main route

- 1 Outdoor escalator
- 2 Digital finalization phase
- 3 Vertical circulation
- 4 Administration



B5 RUINS FLOOR PLAN
Inside the circular ring

- 1 Vertical circulation (entrance to ruins)
- 2 Pharaonic phase ruins
- 3 Coptic phase ruins
- 4 Islamic phase ruins
- 5 Vertical circulation (exit of ruins)



UNDERGROUND PARKING PLAN

- 1 Parking
- 2 Visitor vertical circulation
- 3 Service vertical circulation



RUINS CHAMBER FROM ABOVE



LIT ROUTE THROUGH THE RUINS



RUINS CHAMBER, ELEVATED VIEW

*In the triangle of civilization, a story unfolds,
Where time's relentless ring devours and molds,
I had a dream, the dunes once said,
Of endless ruins where all hearts tread.*

*The fate of all that is made and all that is born,
A tomb that conceals no joy, no motion, no pain, no mourn.
Where only dust shall exist, and only dust is forlorn*

NARRATIVE

The museum reads the rise and fall of three civilizations as lines of time. The Pharaonic, Coptic and Islamic phases cross one another, each crossing an end and a beginning. A ring of time, the tomb of all that was made and born, gathers their ruins in a circular chamber below ground.



RUINS HALL



CHAPEL RUIN



EXHIBITION GALLERY

CONTACT

Mahmoud Talaat

EMAIL	mahmoud.talaat605@gmail.com
PHONE	+20 115 674 1997
LINKEDIN	linkedin.com/in/mahmoud-talaat605
BASED IN	Alexandria, Egypt

01	Working Drawings Residential Villa · Multipurpose Hall · Car Showroom	4
02	BIM Automation & Families Dynamo scripts · Revit plugin · Parametric family	43
03	Architectural Design Five design projects · Graduation project	52